

Chapter 8 :

FET Biasing Circuits

8.1 Introduction

In Chapter 6, we have discussed constructions and operation of field-effect transistor (FET). Characteristics of different types of FET in different configurations have also been described. In this chapter, we will describe the biasing process of FETs. Like the BJTs, FETs are also used in three configurations – **common source**, **common drain**, and **common gate**. Though common source circuits are widely used, common drain and common gate circuits are also used in some special cases. In this chapter, different biasing circuits for different types of FETs in three configurations will be described. The main difference of FETs circuits is, being the current controlled device, there is no gate current. Thus, the input of the circuit has very high impedance. As there is no gate current, the drain current (I_D) is equal to source current (I_S). As we have seen in Chapter 6, the transfer characteristic curve of FET is much non-linear at the lower portion. So, in FET biasing circuits, we try to set the DC operating point at upper portion, i.e., we set the drain current as much as possible. The gain determining parameter, trans-conductance (g_m) is also higher at the upper portion of the transfer characteristic curves.

8.2 DC Load-Line and Q-Point

The concept of load-line and Q-point (quiescent point) of FET circuits is same as those of the BJT. However, to recapture the concept, load-line and Q-point are described again here.

8.2.1 DC Load Line

As we have described for BJT, the load-line is drawn for the output circuit. To draw the load-line let us consider the N-channel FET circuit of Fig.8-1 (a common source configuration). Here, the FET is biased as a common source amplifier. The gate-source circuit is the input circuit, and the drain-source circuit is the output circuit. Like the NPN transistor drain is connected to the positive terminal of supply V_{DD} and the gate is connected to the negative terminal of another supply V_{GG} that reverse biases the gate-source junctions. The source is directly connected to ground. We know that load-line is a line drawn at the output characteristics that gives the variation of output current I_D with the variation of output voltage V_{DS} . To draw the load line, we can redraw the circuit of Fig.8-1 as in Fig.8-2. Now, applying KVL in the output loop of this circuit we get,

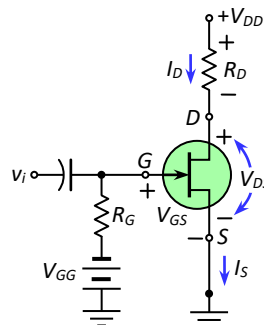


Fig.8-1: N-channel JFET circuit

Now, applying KVL in the output loop of this circuit we get,

$$V_{DD} - I_D R_D - V_{DS} = 0 \quad (8-1)$$

Rearranging above equation,

$$I_D = -\frac{1}{R_D} V_{DS} + \frac{V_{DD}}{R_D} \quad (8-2)$$

Equ.(8-1) is the equation of an straight line that gives the variation of I_D with the variation of V_{DS} , i.e., the load line of the circuit of Fig.8-2. This equation is comparable to the equation of load-line for BJT. The line has a negative slope ($-\frac{1}{R_D}$) and an y-intercept ($\frac{V_{DD}}{R_D}$). To draw this line on the output characteristics of the transistor we have to determine two points. Let's do that.

Point A:

When, $I_D = 0$ from Equ.(8-1) we get, $V_{DS} = V_{DD}$.

Point B:

When, $V_{DS} = 0$, again from the same equation we get,

$$I_D = \frac{V_{DD}}{R_D} \quad (8-3)$$

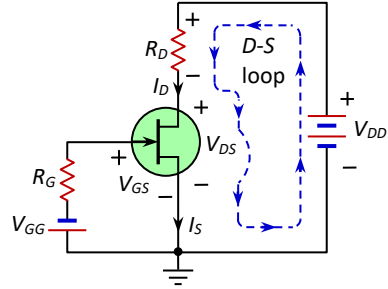


Fig.8-2: FET biasing circuit: output loop

Now, if we put these points on the output characteristics of the FET and connect them, we get the load line as shown in Fig.8-3. As the load line is determined only considering the DC condition this load line is called **DC load line**. The position and slope of the load line depend on the values of V_{DD} and R_D . As we have described for BJT, the DC load line represents all possible values of V_{DS} and I_D that a FET can achieve in a particular biasing circuit.

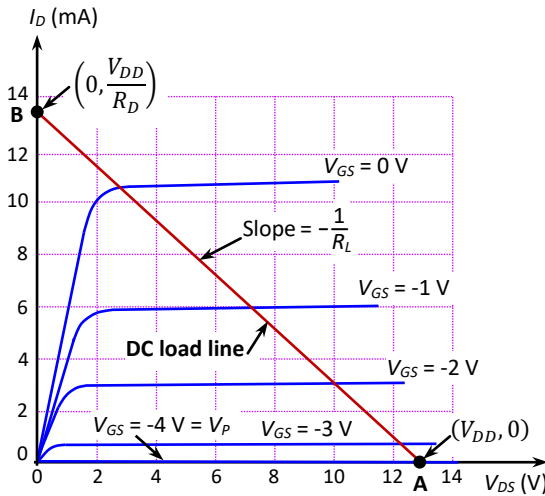


Fig.8-3: Load line on the output characteristics

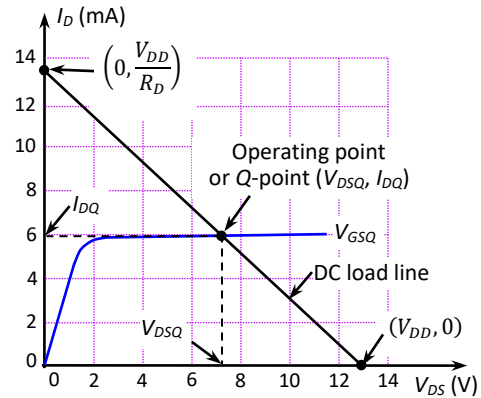


Fig.8-4: Load line and operating point (Q-point)

8.2.2 Q-Point or Operating Point

Although we find a set of curves on the output characteristics, but the device will give only one characteristic curve for a fixed value of V_{GS} which is shown in Fig.8-4. The intersecting point of the DC load line and the output characteristic curve is called the **operating point**. It is also called the **bias point**, or **quiescent point** or simply **Q-point** same as BJT. Therefore, a Q-point gives a fixed value of output voltage and output current which are represented as V_{DSQ} and I_{DQ} [Q for quiescent]. If we draw a vertical line through the Q-point it will intersect the x-axis and this is the value of V_{DSQ} . Similarly, a horizontal line through the Q-point will intersect the y-axis and will give the value of I_{DQ} . The value of the gate voltage for which the operating point is set, is called quiescent gate voltage and is denoted as V_{GSQ} .

Similar to the BJT, the load line is not the property of the device. In fact, it is the circuit property, because it depends on the values of V_{DD} and R_D . On the other hand, Q-point depends both on the properties of device and circuit. Without knowing the properties of FET or BJT we can draw the load line, but to draw the Q-point we need the output characteristics or the value of g_m .

8.2.3 Variation of the Operating Point

Actually, the bias point or operating point identifies the device current levels and terminal voltages when there is no AC input signal. When an AC signal is applied to the gate (input), the drain current I_D varies according to the instantaneous values of the input signal. This variation in I_D also causes corresponding variation in V_{DS} . The situation is illustrated in Fig.8-5. As we know, to be used as an amplifier, the FET must work in the active region. That is, the value of V_{DS} must not less than $V_{DS(sat)}$. If V_{DS} falls below $V_{DS(sat)}$, the device will operate in the voltage-controlled resistance

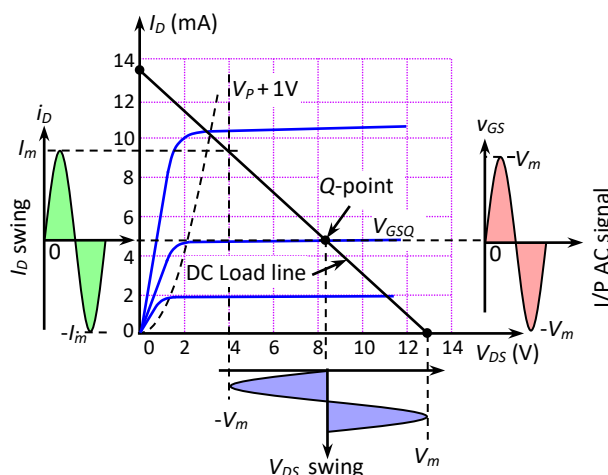


Fig.8-5: Variation of Q-point due to the variation of input voltage V_{GS}

region. To ensure the operation of a FET within the active region, we can keep V_{DS} always greater than $|V_P|$. One good practice is, to maintain the minimum value of V_{DS} , 1 V higher than the pinch-off value $|V_P|$. That is,

$$V_{DS(min)} = |V_P| + 1 \text{ V} \quad (8-4)$$

Actually, the above condition is necessary only when the gate voltage is zero when $V_{DS(sat)} = |V_P|$. If we apply a reverse bias to the gate (V_{GS}), the value of saturation voltage ($V_{DS(sat)}$) decreases by the same amount. That is, $V_{DS(sat)} = |V_P| - V_{GS}$. In that case, the minimum value of V_{DS} will be,

$$V_{DS(min)} = |V_P| - V_{GS} + 1 \text{ V} \quad (8-5)$$

Therefore, the minimum swing span of the Q -point will be within the range of $V_{DS} = |V_P| + 1 \text{ V}$ to $V_{DS} = 0 \text{ V}$ as shown in Fig.8-5.

Example 8-1

Draw the load line and determine the operating point for the FET circuit of Fig.8-6. Assume $V_P = -4 \text{ V}$, and $I_{DSS} = 10 \text{ mA}$.

Solution:

To draw the load line, we need two points. For this circuit, the point on the x -axis is $(V_{DD}, 0) = (12 \text{ V}, 0)$ and the point on the y -axis is $(0, \frac{V_{DD}}{R_D}) = (0, \frac{12 \text{ V}}{1 \text{ k}\Omega}) = (0, 12 \text{ mA})$. Now putting these two points on a $I_D - V_{DS}$ coordinate system and connecting the points we will get the load line as shown in Fig.8-7.

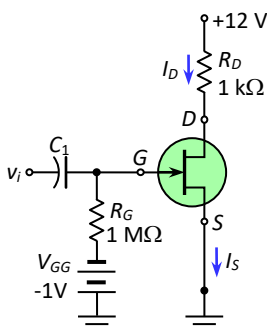


Fig.8-6: FET circuit for Example 7-1

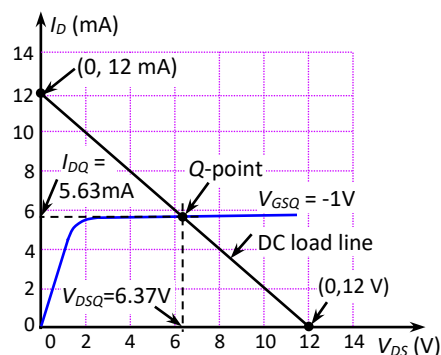


Fig.8-7: Load line and operating point for Example 7-1

From the circuit, we find $V_{GS} = -1 \text{ V}$ and given $V_P = -4 \text{ V}$. Putting these values in Shokley's equation the value of quiescent drain current can be calculated as,

$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = 10 \text{ mA} \left(1 - \frac{-1}{-4} \right)^2$$

$$\text{or,} \quad I_{DQ} = 10 \text{ mA} \left(\frac{3}{4} \right)^2 = 5.63 \text{ mA}$$

Therefore, the value of V_{DSQ} can be calculated as,

$$V_{DSQ} = V_{DD} - I_{DQ} R_D = 12 \text{ V} - 5.63 \text{ mA} \times 1 \text{ k}\Omega = 6.37 \text{ V}$$

The operating point is plotted on the same load line. Here a tentative output characteristic curve is shown through the Q-point for $V_{GSQ} = -1\text{V}$.

Comments: V_{DD} and R_D are the circuit parameters and V_P and I_{DSS} are the FET's parameters. The load line can be drawn using the circuit parameters but to determine the Q-point, we need the device parameters or device characteristics.

8.3 JFET Biasing Circuits

We have already discussed that, to use a BJT as an amplifier the device must be properly biased. Biasing is necessary for faithful amplification of the signal, i.e., amplification with no (or minimum) distortion of the input signal. **Biasing circuits** set suitable levels of currents and voltages for the device, so that the **Q-point** oscillates in the active region when AC input signal is applied to the circuit.

In Chapter 4, we found that the biasing levels for a silicon transistor configuration can be obtained using the approximate characteristic equations: $V_{BE} = 0.7\text{ V}$, $I_C = \beta I_B$, and $I_C \approx I_E$. The relation between input current and output current is provided by β , which is assumed to be fixed (constant) in magnitude for the analysis to be performed. In fact, the constant value of β establishes a linear relationship between I_C and I_B . For this reason, when a transistor works as an amplifier, it is called the **linear operation** and when works as a switch it is called **non-linear operation**. For FET, the relation between the output parameter I_D and input parameter V_{GS} is non-linear which is represented by the transfer characteristics. But when used as an amplifier, a small linear portion of the transfer characteristics is used. Unlike the BJT, FETs are voltage-controlled devices. For biasing JFET, the input current (I_G) is assumed to be zero. That is,

$$I_G \approx 0\text{ A} \quad (8-6)$$

As there is no gate-current, the drain current and the source current will be equal. That is,

$$I_D = I_S \quad (8-7)$$

For JFETs and depletion MOSFETs the relation between I_D and V_{GS} is given by the Shockley's equation as,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 \quad (8-8)$$

But for enhancement MOSFETs the relation between I_D and V_{GS} is given by the following equation, which is described in Chapter 6.

$$I_D = k(V_{GS} - V_T)^2 \quad (8-9)$$

Unlike the BJT, the biasing circuits for FETs are simple and only a few biasing circuits are used for FETs. These are:

1. Fixed Biasing Circuit
2. Self-biasing Circuit, and
3. Voltage Divider Biasing Circuit

8.3.1 Fixed Biasing Circuit

This is the simplest biasing circuit for FET and is similar to the base-bias or fixed-bias circuit for BJT. This biasing circuit is also called the **gate bias**. The fixed bias circuit of an N-channel JFET in common-source configuration is shown in Fig.8-8. The output circuit (drain-source circuit) is biased in the same way as BJT using voltage source V_{DD} . Drain is connected to the positive terminal of the supply voltage through a load resistor R_D and the source is connected to the ground. The input circuit (gate-source circuit) is reverse biased using another voltage source V_{GG} . The gate is connected to the negative terminal of the supply voltage through another resistor R_G . The input voltage (v_{in}) will be applied to the gate terminal via the coupling capacitor C_1 and the output will be taken from the drain terminal via another coupling capacitor C_2 . The function of the coupling capacitor is same as described for BJT biasing circuits. The negative gate voltage, V_{GS} , sets a DC drain current, I_D , depending on the properties of the JFET (transconductance). For this drain current the output circuit sets a voltage level across drain and source (V_{DS}).

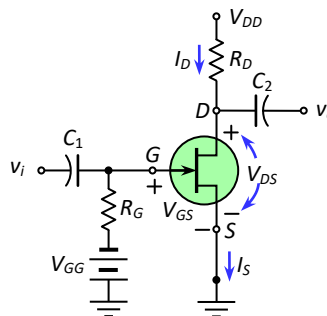


Fig.8-8: Fixed biasing or gate bias circuit

Calculation of Operating Point (Analytical Method):

Applying KVL in the input circuit (gate-source) we get,

$$-V_{GG} - I_G R_G - V_{GS} = 0 \quad (8-10)$$

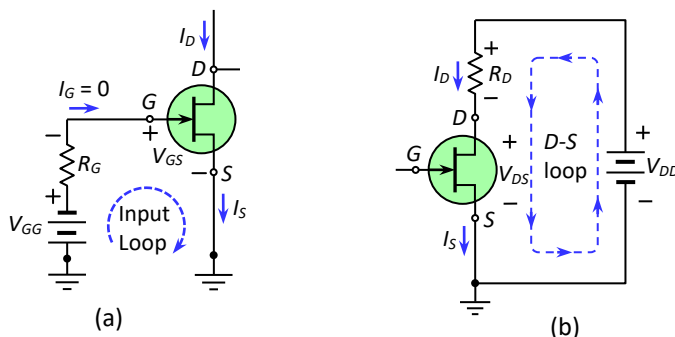


Fig.8-9: KVL to FET biasing circuit: (a) Input loop, and (b) Output loop

But for any FET, the value of gate current is approximately zero. That is $I_G = 0$. Therefore,

$$\boxed{V_{GS} = -V_{GG}} \quad (8-11)$$

Now for a particular FET (given value of V_P and I_{DSS}), the value of drain current can be calculated from Shockley's equation as,

$$I_D = I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$\therefore \quad I_D = I_{DQ} = I_{DSS} \left(1 - \frac{|V_{GG}|}{|V_P|} \right)^2 \quad (8-12)$$

Again applying KVL in the output (drain-source) circuit, we get,

$$V_{DD} - I_D R_D - V_{DS} = 0$$

$$\text{or,} \quad V_{DS} = V_{DSQ} = V_{DD} - I_D R_D \quad (8-13)$$

Using Equ.(8-12) and (8-13), the DC operating point of a fixed bias circuit can be determined. The values of drain current and drain-source voltage found from these equations are the Quiescent current and voltage. So, they are sometimes represented as I_{DQ} and V_{DSQ} .

Calculation of Operating Point (Graphical Method):

In addition to the analytical method, the operating point of FET biasing circuit can be determined using graphical method. For this purpose, we need the transfer characteristics of the FET. We can use the transfer characteristic curve provided by the manufacturer, or using V_P and I_{DSS} we can

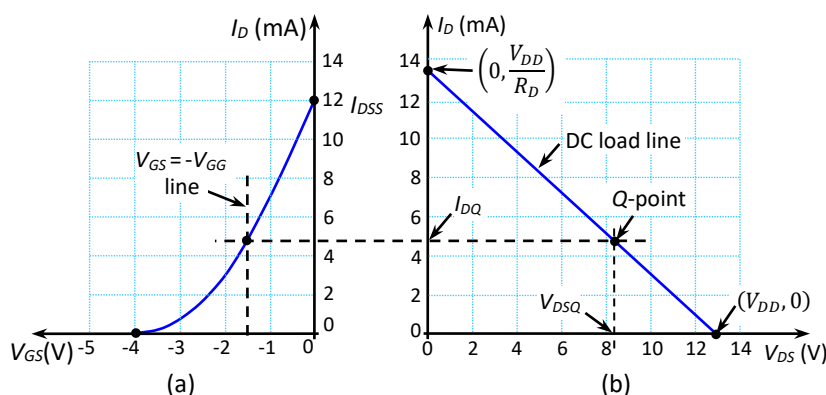


Fig.8-10: Determination of operating point: (a) bias line and the intersecting point on the transfer characteristics, and (b) Load line and operating point on the output characteristics

draw this curve as described in Chapter 6. Once we got the transfer characteristics, a vertical line (called bias line) is drawn at the point $V_{GSQ} = -V_{GG}$. The intersecting point of this straight line with the transfer characteristic curve will give the value of quiescent drain current I_{DQ} . After determining the value of I_{DQ} , the value of V_{DSQ} can be calculated using Equ.(8-13). We can determine the value of V_{DSQ} graphically as well. In that case, the transfer characteristics and the output characteristics should have the same leveling in the y-axis as shown in the Fig.8-10. Then we have to draw the load line on the output characteristics, and a horizontal line through the intersecting point on the transfer characteristics. This horizontal line will intersect the load line and the abscissa of this point is the value of V_{DSQ} . In fact this point is the DC biasing point of the FET circuit. Determination process of operating point using this graphical method is illustrated in Fig.8-10.

Example 8-2

Determine the operating point of common-source JFET circuit of Fig.8-11 both analytically and graphically. Assume $V_P = -6\text{ V}$, and $I_{DSS} = 15\text{ mA}$.

Solution:**Analytical method:**

Using Equ.(8-12),

$$I_{DQ} = 15\text{ mA} \left(1 - \frac{1.5\text{ V}}{6\text{ V}} \right)^2$$

or, $I_{DQ} \approx 8.44\text{ mA}$

Using Equ.(8-13), $V_{DSQ} = V_{DD} - I_D R_D = 16\text{ V} - 1\text{ k}\Omega \times 8.44\text{ mA} = 7.56\text{ V}$

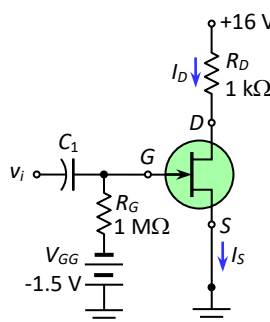
Graphical method:

Fig.8-11: Fixed biasing bias circuit for Example 8-2

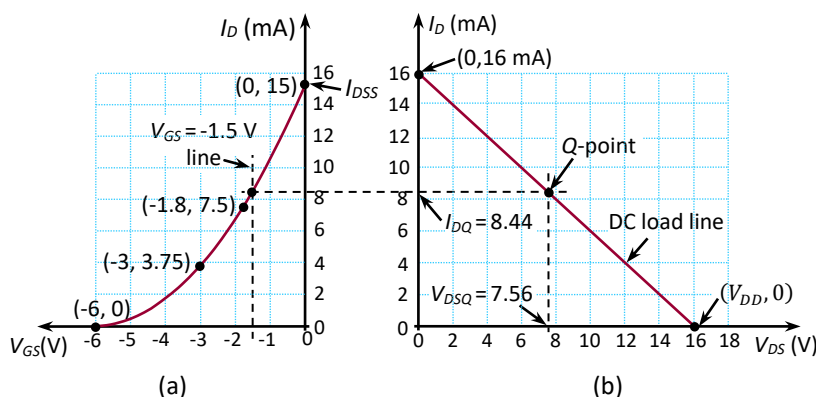


Fig.8-12: Determination of operating point using graphical method

First, we have to draw the transfer characteristic curve using the process described in Chapter 6. The values of the points to draw the transfer characteristics will be: $(0, I_{DSS}) = (0, 15\text{ mA})$, $(V_P, 0) = (-6\text{ V}, 0)$, $(0.3V_P, I_{DSS}/2) = (-1.8\text{ V}, 7.5\text{ mA})$, and $(0.5V_P, I_{DSS}/4) = (-3\text{ V}, 3.75\text{ mA})$. Putting these values on a I_D vs. V_{GS} coordinate system and connecting them using curved lines, we get the transfer characteristic curve of Fig.8-12. Now, the vertical line $V_{GS} = -1.5\text{ V}$ is drawn on the same transfer characteristic curve. The intersecting point of this line with the transfer characteristic gives the value of quiescent current I_{DQ} .

To determine the position of the Q-point the load line is drawn on the I_D vs. V_{DS} coordinate system which is shown in the same figure. Now, we have to draw a horizontal line through the previous intersecting point of vertical line and transfer characteristic curve. This horizontal line intersects the load line at point $(7.56, 8.44)$. This point is the operating point of the circuit of Fig.8-11. From this point the value of $V_{DSQ} = 7.56\text{ V}$, and the value of $I_{DQ} = 8.44\text{ mA}$.

Answer: $V_{DSQ} = 7.56\text{ V}$, and $I_{DQ} = 8.44\text{ mA}$.

Comments: Analytical method always gives the exact value of the operating point, whereas the graphical method gives approximate value. Graphical method is time consuming also. But for some biasing circuit graphical method is easier than analytical method.

8.3.2 Performance of Fixed-Bias Circuit

So far, we have considered fixed values of V_P and I_{DSS} for a particular FET. But practically the datasheets mention the minimum and maximum values of both V_P and I_{DSS} . It means that any FET of a particular model number will have the values of V_P and I_{DSS} within the minimum and maximum values same as the value of β of a BJT. Due to these variations, the DC bias condition of a circuit varies for different FETs (even though FETs of same number). Therefore, any circuit designed for a particular FET must be analyzed for the minimum and maximum values of V_P and I_{DSS} . A poor-quality circuit cannot adjust the variation of bias condition, whereas a good circuit has the ability to adjust the variation of the operating point though the values of V_P and I_{DSS} are changed. Actually, this property of the circuit is referred to as the **stability** of the biasing circuit as describe in BJT biasing chapter.

Although fixed-bias circuit is simple and easy to analyze, it has two disadvantages. One disadvantage is that the circuit needs two power supplies (V_{GG} and V_{DD}). The second disadvantage is the poor stability of the operating point. In fact, fixed-bias circuit has no capability to adjust the variation of operating point due to the change of device parameters (V_P and I_{DSS}). Among the three biasing circuits, fixed-bias produces maximum variation of biasing point for similar variation in V_P and I_{DSS} . The situation is explained in Example 8-3.

Example 8-3

Graphically determine the position of the operating points of an N-channel JFET fixed bias circuit like Fig.8-11 (circuit of Example 8-2). The JFET used in the circuit has a range of V_P and I_{DSS} . Assume, $V_{DD} = 15\text{ V}$, $V_{GG} = -1.5\text{ V}$, $R_D = 1.5\text{ k}\Omega$, $V_{P(min)} = -3\text{ V}$ and $V_{P(max)} = -6\text{ V}$, and $I_{DSS(min)} = 8\text{ mA}$ and $I_{DSS(max)} = 16\text{ mA}$.

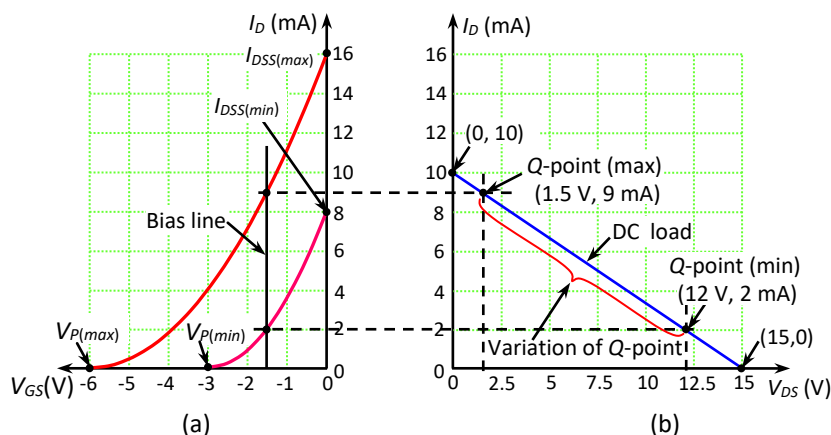


Fig.8-13: Determination of operating points for Example 7-3

Solution:

Using $V_{P(min)}$ and $I_{DSS(min)}$ the lower transfer characteristic curve is drawn and using $V_{P(max)}$ and $I_{DSS(max)}$ the upper transfer characteristic curve is drawn exactly in the same procedure of Example 8-2. The bias line is drawn along $V_{GS} = -1.5$ V. This bias line intersects the transfer characteristic curves in two points as shown in Fig.8-13. The load line of the circuit is drawn using points $(0, V_{DD}/R_D) = (0, 10 \text{ mA})$ and $(V_{DD}, 0) = (15 \text{ V}, 0)$. Now let us draw two horizontal lines along the two intersecting points on the transfer characteristic curves. These horizontal lines intersect the load line in two points. These intersecting points are the operating points of the circuit. From the graph, we find that, $Q\text{-point (min)} = (12 \text{ V}, 2 \text{ mA})$ and $Q\text{-point (max)} = (1.5 \text{ V}, 9 \text{ mA})$.

Results: The position of the lower and upper Q -points are shown in Fig.8-13 and their values are: $Q\text{-point (min)} = (12 \text{ V}, 2 \text{ mA})$ and $Q\text{-point (max)} = (1.5 \text{ V}, 9 \text{ mA})$.

Comments: The fixed bias circuit produces large variation of Q -point, due to the variation of V_P , and I_{DSS} .

8.4 Fixed-Bias Circuit for P-channel JFET

Fig.8-14 shows the fixed-bias circuit for a P-channel JFET. The circuit arrangement is same as the N-channel JFET. Here the polarity of V_{DD} and V_{GS} are opposite than that of N-channel. The drain is connected to negative terminal of supply voltage V_{DD} . In P-channel JFET the channel is P-type and the gates are N-type. Hence, to reverse bias the gate, it is connected to the positive terminal of the supply voltage V_{GG} . Due to the opposite polarity the direction of drain current I_D will be reverse. Now the drain current will flow from source (S) to drain (D) due to the majority carrier holes. The calculation methods (both analytical and graphical) are exactly same as for N-channel JFET. The process is illustrated in Fig.8-15.

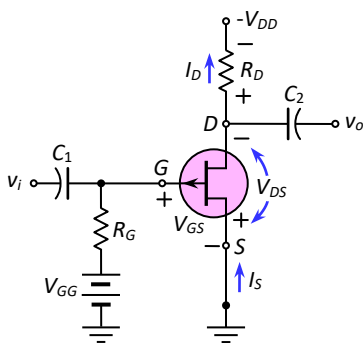


Fig.8-14: Fixed bias or gate bias circuit for P-channel JFET

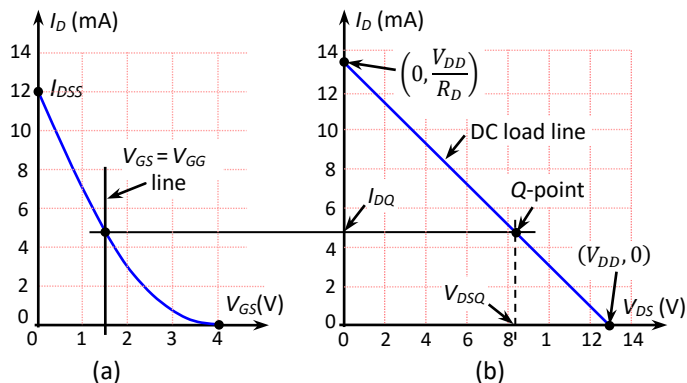


Fig.8-15: Determination of operating point for P-channel JFET: (a) Transfer characteristics, and (b) Output characteristics

8.5 Self-Bias Circuit

This biasing circuit is comparatively better than fixed-bias circuit. The **self-bias** circuit for common-source configuration of N-channel JFET is shown in Fig.8-16. This biasing circuit needs only one power supply, V_{DD} . As before, the drain is connected to the positive terminal via the load resistor R_D , but the source is not directly connected to ground. Source is connected to ground through another resistor R_S . The gate is connected to ground through a resistor R_G . The gate resistor R_G has no effect on the biasing condition, but this resistor helps the input signal to be applied at the gate terminal. If the gate were directly connected to ground, the biasing condition of the circuit would have not been changed, but input signal would have been shorted. In this circuit the reverse bias of the gate terminal is established by the voltage drop across the source resistor, R_S .

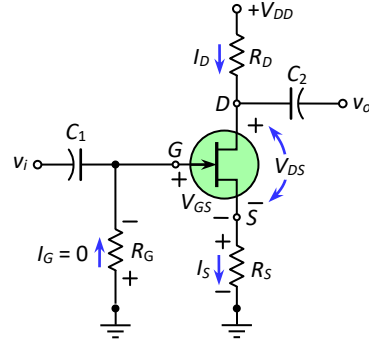


Fig.8-16: Self bias circuit for N-channel JFET

Calculation of Operating Point (Analytical Method):

Applying KVL in the input circuit (gate-source) we get,

$$\begin{aligned} -I_G R_G - V_{GS} - I_S R_S &= 0 \\ -I_G R_G - V_{GS} - I_D R_S &= 0 \quad [\because I_D = I_S] \end{aligned} \quad (8-14)$$

But for any FET the value of gate current is approximately zero. That is $I_G = 0$. Therefore,

$$V_{GS} = -I_D R_S \quad (8-15)$$

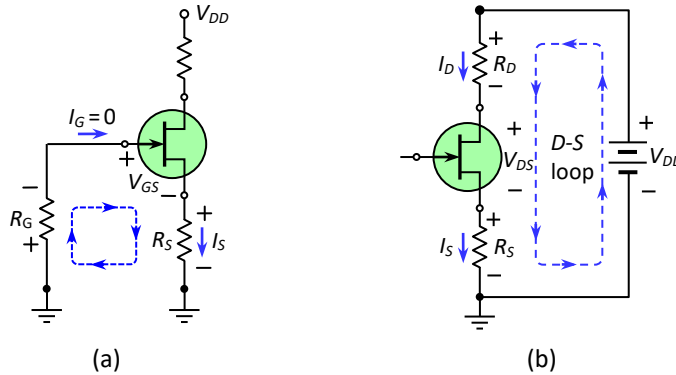


Fig.8-17: Applying KVL to FET biasing circuit:
(a) input loop, and (b) output loop

Now putting this value into Shockley's equation we get,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = I_{DSS} \left(1 - \frac{|I_D R_S|}{|V_P|} \right)^2 \quad (8-16)$$

Equ.(8-16) contains only one variable I_D and some constants. So, solving this equation we can find the value of I_D . Being a quadratic equation, it will be a little difficult to solve this equation. In that case we can use the graphical method.

Again, applying KVL in the output (drain-source) circuit, we get,

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$\therefore V_{DS} = V_{DSQ} = V_{DD} - I_D (R_D + R_S) \quad [\because I_D = I_S] \quad (8-17)$$

Using Equ.(8-16) and Equ.(8-17) the DC operating point of a self bias circuit can be determined. The Equ.(8-17) can also be used to draw the load line of the circuit.

Determination of Operating Point using Graphical Method

Sometimes it is difficult to solve Equ.(8-16). Without solving the equations, we can also determine the position of operating point graphically as described for fixed-bias circuit. For this purpose, we have to draw the transfer characteristics using V_P and I_{DSS} . On the transfer characteristic curve, we have to draw the bias line given by Eq.(8-15) as shown in Fig.8-18(a). The intersecting point of this bias line with the transfer characteristic curve will give the value of quiescent drain current I_{DQ} . After determining the value of I_{DQ} , the value of V_{DSQ} can be calculated using Equ.(817), or by graph as shown in Fig.8-18(b).

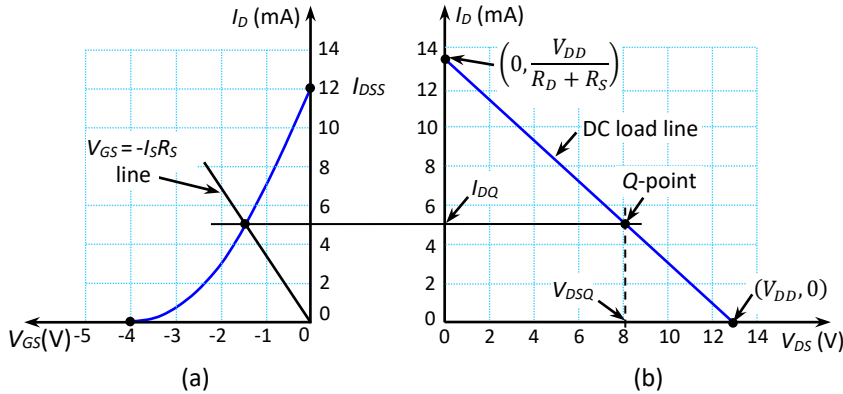


Fig.8-18: Determination of operating point using graphical method for self-bias circuit

described for self-bias. We have to draw the load line and a horizontal line through the intersecting point on the transfer characteristics. This horizontal line will intersect the load line and the abscissa of this point is the value of V_{DSQ} . In fact, this point is the DC biasing point of the FET circuit. Determination process of operating point using this graphical method is illustrated in Fig.8-18(b).

8.5.1 Performance of Self-bias

Self-bias is better than fixed-bias. It needs only one power supply. It has some ability to adjust the change of operating point due to the change of V_P and I_{DSS} . This adjustment takes place due to the source resistor R_S . For example, let us consider that the FET of a biasing circuit has been replaced with a FET of lower value of V_P or higher value of I_{DSS} or both. From the Shockley's we find that, both the cases (lower V_P or higher I_{DSS}) will result in larger drain current I_D . From Equ.(8-15), we find that with the increase in I_D , the value of $|V_{GS}|$ also increase. As this voltage increase the

reverse bias of the gate, the drain current will be reduced eventually. Similarly, if the value of V_P increase or the value of I_{DSS} decreases or both, the value of I_D will decrease. This decrease in I_D will also decrease the value of V_{GS} and finally I_D will increase. In fact, the source resistance provides a negative feedback effect as does the emitter resistance in BJT biasing circuit. In this way, the self-bias circuit can adjust the variation of operating point. The higher the value of R_S the better stability the circuit will provide. But the higher value of R_S will decrease the level of I_D and allows lower value of R_D . Lower value of R_D results in lower voltage gain. On the other hand, lower drain current level sets the operating point near the non-linear region (lower portion of transfer characteristic curve). The nonlinearity introduces distortion in the amplified output signal. Though this circuit can adjust the variation of the operating point to some extent, still its operating point changes due to the change of V_P and I_{DSS} . This situation is illustrated in Example 7-4.

Example 8-4

Graphically determine the position of the operating points of an N-channel JFET self-bias circuit like Fig.8-16. The JFET used for the circuit has a range of V_P and I_{DSS} . Assume, $V_{DD} = 15\text{ V}$, $R_S = 0.5\text{ k}\Omega$, $R_D = 1\text{ k}\Omega$, $V_{P(min)} = -3\text{ V}$ and $V_{P(max)} = -6\text{ V}$, and $I_{DSS(min)} = 8\text{ mA}$ and $I_{DSS(max)} = 16\text{ mA}$.

Solution:

Using $V_{P(min)}$ and $I_{DSS(min)}$ the lower transfer characteristic curve is drawn and using $V_{P(max)}$ and $I_{DSS(max)}$ the upper transfer characteristic curve are drawn exactly in the same procedure discussed in Example 8-2. From Equ.(8-15), the bias line of this circuit will be,

$$V_{GS} = -I_D R_S = -0.5\text{ k} \times I_D$$

To draw this line, let us determine two points on it as,

When $I_D = 0$, $V_{GS} = 0$. This is the point on the origin.

When $I_D = 2\text{ mA}$, $V_{GS} = -1\text{ V}$. This point will be on the 2nd quadrant.

Now putting these points and joining them, the bias line is drawn on the transfer characteristic curves drawn before. As shown in Fi, this bias line intersects the transfer characteristic curve in two points. These points determine the values of minimum and maximum value of I_D .

The load line is drawn using the following points:

$$\left(0, \frac{V_{DD}}{R_D + R_S}\right) = \left(0, \frac{15\text{ V}}{1\text{ k} + 0.5\text{ k}}\right) = (0, 10\text{ mA})$$

and $(V_{DD}, 0) = (15\text{ V}, 0)$

Now, two horizontal lines are drawn through the intersecting points on the transfer characteristic curves. These horizontal lines intersect the load line in two points. These intersecting points are the operating points of the circuit. From the graph, we find that, Q-point (min) = (11.1 V, 2.6 mA) and Q-point (max) = (7.3 V, 5.2 mA).

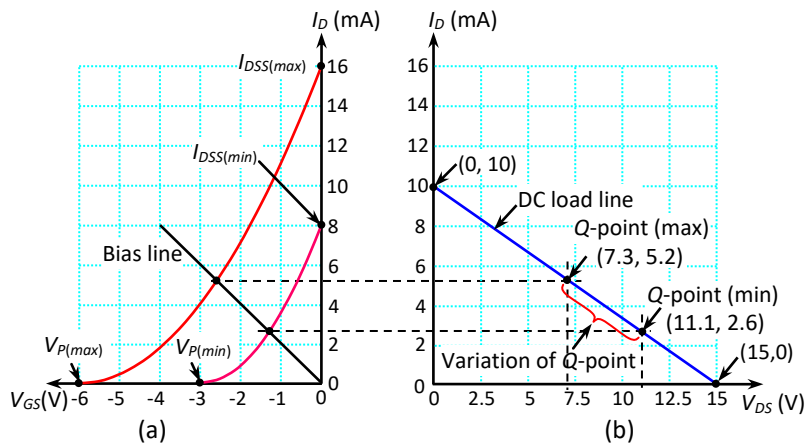


Fig.8-19: Variation of operating point of self bias circuit

Results: The position of the lower and upper Q-points are shown in Fig.8-19 and their values are: Q-point (min) = (11.1 V, 2.6 mA) and Q-point (max) = (7.3 V, 5.2 mA).

Comments: The self-bias circuit produces less variation of Q-point than fixed-bias circuit.

8.6 Self-Bias Circuit for P-channel JFET

Fig.8-20 shows the self-bias circuit for a P-channel JFET. The self-bias circuit for P-channel FET needs same components as for the N-channel JFET. Here the polarity of V_{DD} and V_{GS} are opposite than that of N-channel. The drain

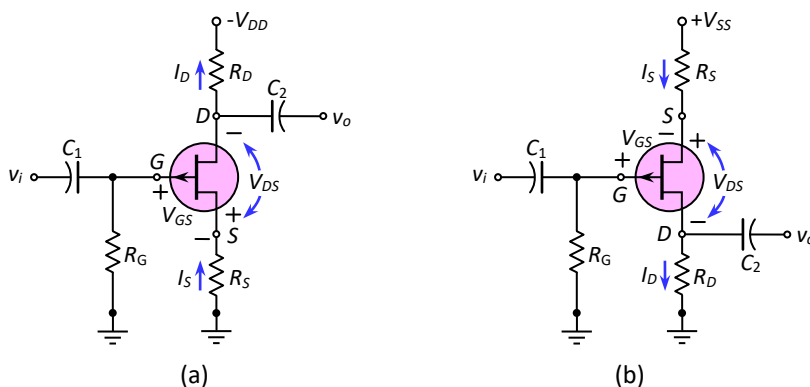


Fig.8-20: Self-bias circuit for P-channel JFET: (a) Using negative voltage, and (b) Using positive voltage

is connected to negative terminal of supply voltage V_{DD} . In P-channel JFET the channel is P-type and the gates are N-type. Hence, to reverse bias the gate, it is connected to the positive terminal of

the supply voltage V_{GG} . Due to the opposite polarity the direction of drain current I_D will be reverse. Now the drain current will flow from source to drain due to the majority carrier holes. The calculation methods (both analytical and graphical) are exactly same as for N-channel JFET.

For the second circuit [Fig.8-20(b)], the equation of bias-line will be, $V_{GS} = V_{DD} - I_D R_S$.

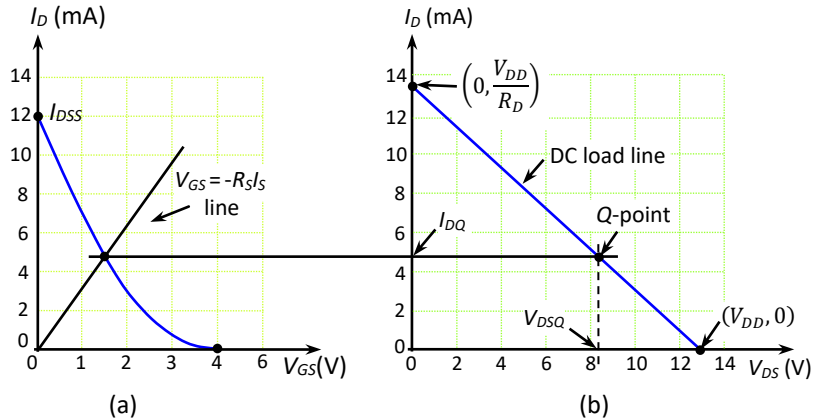


Fig.8-21: Graphical method for determining the operating point for P-channel JFET

8.7 Voltage Divider Bias

In self-bias circuit we have seen that, though the circuit can provide some stability of the operating point, but it has some limitations. The voltage divider bias can solve all these

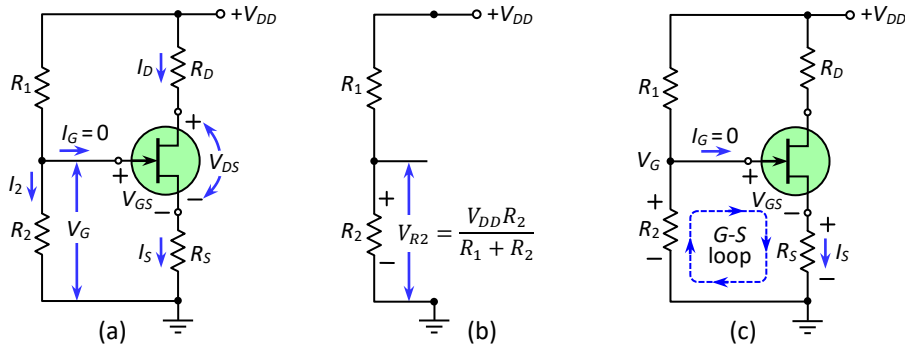


Fig.8-22: Voltage divider biasing circuit: (a) Circuit's schematic diagram, (b) Voltage division in the biasing network, and (c) KVL in the input circuit

limitations and gives the best stability of operating point. It gives better stability of operating point, and allows to set the operating point near the linear portion of the transfer characteristics. Moreover, it provides better voltage gain. The voltage divider bias circuit for a common-source amplifier with N-channel JFET is shown in Fig.8-22(a). The network is similar to that of BJT, but here the analysis of this network is different. In BJT we have to use Thevenin's equivalent circuit to get exact values of currents and voltages. But in JFET, as the input current is almost zero (for MOSFET

exactly zero) we do not need to use Thevenin's equivalent circuit. The gate voltage can be calculated only using the voltage divider rule as shown in Fig.8-22(b).

Determination of Operating Point (Analytical Method)

To analyze the circuit let us separate the voltage divider network from the JFET [Fig.8-22(b)]. The voltage across resistor R_2 can be found using voltage divider rule. Its value will be,

$$V_{R2} = \frac{V_{DD}R_2}{R_1 + R_2}$$

Now, if we reconnect the voltage divider network to the JFET, the voltage across R_2 will not change as the gate terminal will draw negligibly small current from the voltage divider network. So the gate voltage will be,

$$V_G = V_{R2} = \frac{V_{DD}R_2}{R_1 + R_2} \quad (8-18)$$

Now, applying KVL in the input circuit we get,

$$V_G - V_{GS} - I_S R_S = 0$$

$$\boxed{V_{GS} = V_G - I_D R_S} \quad [\because I_S = I_D] \quad (8-19)$$

To make the gate reverse biased, the value of $I_D R_S$ should be greater than V_G .

Using this value of V_{GS} , drain current can be calculated from Shockley's equation, which is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = I_{DSS} \left(1 - \frac{|I_D R_S - V_G|}{|V_P|} \right)^2 \quad (8-20)$$

This is a quadratic equation with only one variable I_D . Though difficult, but we can solve it for I_D , and this value of I_D is the quiescent drain current I_{DQ} .

Again, applying KVL in the output (drain-source as shown in Fig.8-23) circuit, we get,

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$\therefore V_{DS} = V_{DD} - I_D (R_D + R_S) \quad [\because I_D = I_S] \quad (8-21)$$

Using Equ.(8-18) to Equ.(8-21), the DC operating point of a voltage divider bias circuit can be determined.

Determination of Operating Point (Graphical Method)

Sometimes it may be difficult to calculate the value of I_D solve Equ.(8-20). Without solving this equation, we can also determine the position of operating point graphically, as described for the previous two biasing circuits. For this purpose, we have to draw the transfer characteristics using V_P and I_{DSS} . On the transfer characteristic curve, we have to draw the bias line given by Equ.(8-19).

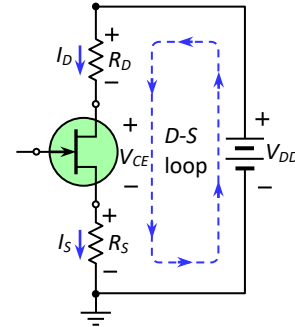


Fig.8-23: Applying KVL in the D-S of voltage divider bias

$$V_{GS} = V_G - I_D R_S \quad (8-22)$$

To draw this bias line, we need two points on it. We can easily determine them as,

Point on y-axis:

When $V_{GS} = 0$, from Equ.(8-19), we get, $I_D = V_G/R_S$. This point $(0, V_G/R_S)$ will be on y-axis.

Point on x-axis:

When $I_D = 0$, Equ.(8-19) gives, $V_{GS} = V_G$. This point $(V_G, 0)$ will be on the x-axis.

Now putting these points on the transfer characteristic curve of the JFET, and joining them the biasing line will be found. The intersecting point of this bias line with the transfer characteristic curve will give the value of quiescent drain current I_{DQ} . After determining the value of I_{DQ} , the value of V_{DSQ} can be calculated using Equ.(8-24), or by graph as described for self-bias and fixed bias. To determine the position of operating point we have to draw the load. Then a horizontal line has to be drawn through the intersecting point on the transfer characteristics. This horizontal line will intersect the load line and the x-value of this point is V_{DSQ} . Determination process of operating point using this graphical method is illustrated in Fig.8-24.

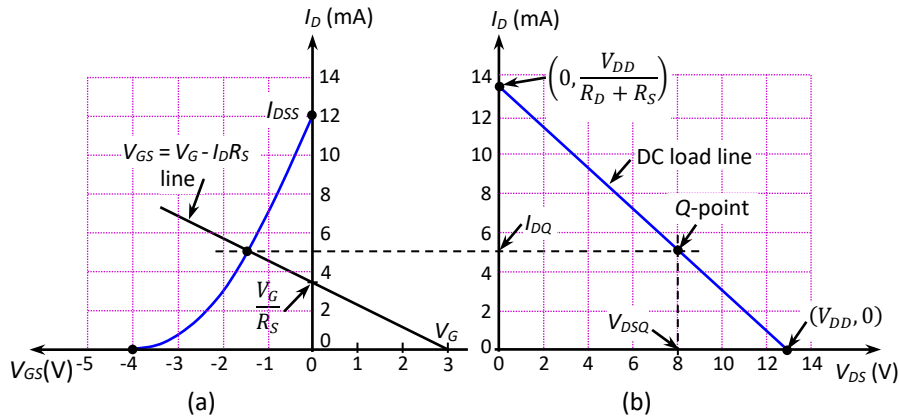


Fig.8-24: Graphical method for voltage divider biasing circuit: (a) bias line on the transfer characteristic curve, and (b) position of operating point on the output characteristic curve

8.7.1 Performance of Voltage Divider Bias

Among the three biasing circuits, discussed so far, voltage divider bias is the best biasing circuit. It provides most stable DC biasing condition for the FET. In self-bias the stability depends only on R_S . We have already discussed that R_S cannot be made very large. Large value of R_S reduces the drain current and sets the operating point near the non-linear portion of the transfer characteristics. In voltage divider bias circuit, the stability of the operating point also depends on R_S . But here the value of R_S can be made very large. The value of drain current depends on V_G as well. The higher the value of V_G the higher the value of I_D . Therefore, in spite of a larger value of R_S , the operating point can be set near the upper region (linear portion) of the transfer characteristics by increasing V_G . The higher the values of V_G the larger will be the value of I_D and the operating point will move to the upper portion of the transfer characteristics. Therefore, in voltage divider bias increasing R_S we can increase stability and increasing V_G we can increase I_D . For a fixed value of V_G and V_{DD} , a

very large value of R_S will result in lower value of R_D and lower voltage gain. So, we have to trade-off between the stability and voltage gain. The better stability of voltage divider bias is further explained in Example 8-5.

Example 8-5

Determine the position of the operating points for the voltage divider bias circuit of Fig.8-25. The JFET used in this circuit has $V_{P(min)} = -3\text{ V}$ and $V_{P(max)} = -6\text{ V}$, and $I_{DSS(min)} = 8\text{ mA}$ and $I_{DSS(max)} = 16\text{ mA}$.

Solution:

Since, graphical method visualize the position of the operating points, here this method is used. From $V_{P(min)}$ and $I_{DSS(min)}$ the lower transfer characteristic curve is drawn, and using $V_{P(max)}$ and $I_{DSS(max)}$ the upper transfer characteristic curve is drawn (using the method described before) and shown in Fig.8-26.

The value of V_G can be calculated using Equ.(8-18),

$$V_G = \frac{V_{DD}R_2}{R_1 + R_2} = \frac{15\text{ V} \times 300\text{ k}}{300\text{ k} + 1200\text{ k}} = 3\text{ V}$$

From Equ.(8-19), the bias line of this circuit will be,

$$V_{GS} = V_G - I_D R_S = 3\text{ V} - (1\text{ k})I_D$$

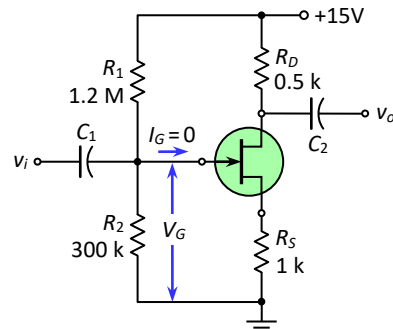


Fig.8-25: Voltage divider bias circuit for Example 8-5

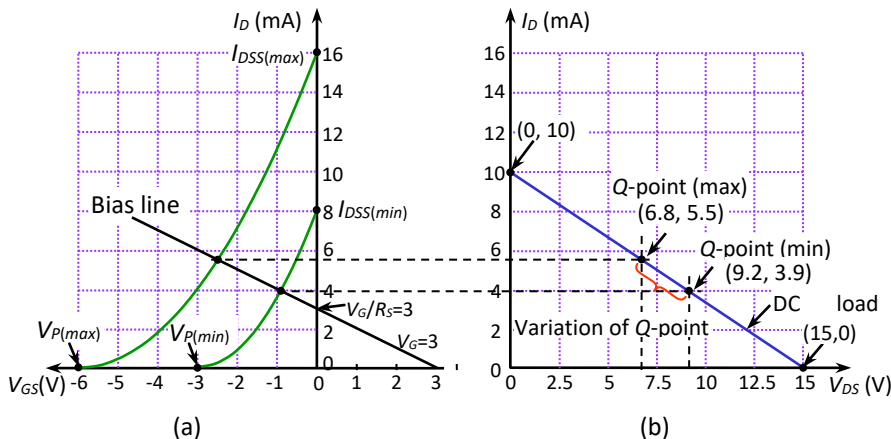


Fig.8-26: (a) Bias line on the transfer characteristic curves, and (b) Positions of the operating points on the output characteristic curve

To draw this line, let us determine two points on it:

When $I_D = 0$, $V_{GS} = 3\text{ V}$. This point will be on the x-axis.

When $V_{GS} = 0$, $I_D = 3\text{ V} / 1\text{ k} = 3\text{ mA}$. This point lies on the y-axis.

Now, putting these points and joining them, the bias line is drawn on the transfer characteristic curves. As shown in Fig.8-26(a), this bias line intersects the transfer characteristic curves in two points. These points determine the minimum and maximum values of I_D .

The load line is drawn using the following points on the Fig.8-26(b):

$$\left(0, \frac{V_{DD}}{R_D + R_S}\right) = \left(0, \frac{15 \text{ V}}{0.5 \text{ k} + 1 \text{ k}}\right) = (0, 10 \text{ mA})$$

and, $(V_{DD}, 0) = (15 \text{ V}, 0)$

Now, two horizontal lines are drawn through the intersecting points on the transfer characteristic curves. These horizontal lines also intersect the load line in two points. These intersecting points are the positions of operating points of the circuit. From the graph, we find that, Q-point (min) = (9.2 V, 3.9 mA) and Q-point (max) = (6.8 V, 5.5 mA).

Comments: Look at the positions of the output points. They are very close to each other, which ensures the better stability of voltage divider bias circuit.

8.8 Voltage Divider Bias Circuit for P-channel JFET

Fig.8-27 shows the voltage divider bias circuit for a P-channel JFET. The voltage divider bias circuit for P-channel JFET needs same components as for the N-channel JFET. Here the polarity of V_{DD} is opposite than that of N-channel as other biasing circuit for P-channel JFETs. The drain

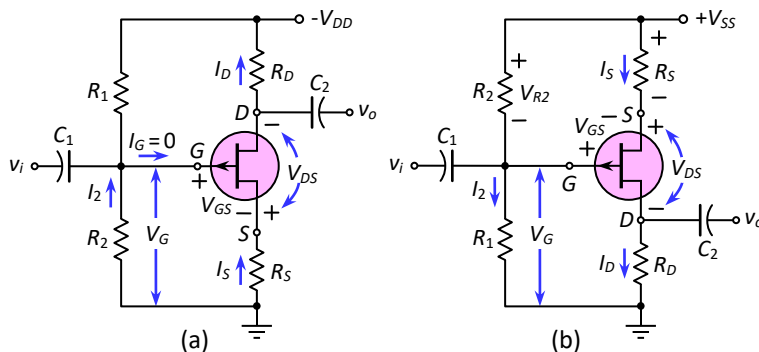


Fig.8-27: Voltage divider bias circuit for P-channel JFET: (a) Using negative voltage, and (b) Using positive voltage

is connected to the negative terminal of supply voltage V_{DD} (or lower potential). Due to the opposite polarity the direction of drain current I_D will be reverse, that is, I_D flows from source to drain and the current flow is due to the majority carrier holes. The calculation methods (both analytical and graphical) are exactly same as for N-channel JFET. In Fig.8-27(a), $-V_{DD}$ is divided by R_1 and R_2 and applied to gate. So, V_G will be negative. As the drain current flows from source to drain, the voltage drop across R_S will be negative. Like the N-channel JFET, here also the gate will be reverse biased by the difference of the gate voltage, V_G , and the source voltage $I_S R_S$. In both N-channel and P-channel circuits, the magnitude of source voltage will be greater than the magnitude of gate voltage. So, for P-channel circuit, V_{GS} will be,

$$V_{GS} = -V_G - (-I_S R_S) = I_D R_S - V_G \quad [\because I_S = I_D] \quad (8-23)$$

Using this value and the Shockley's equation, the drain current can be calculated analytically or drawing this bias line on the transfer characteristic curve, the drain current can be determined graphically. The process of analysis is same as the N-channel JFET.

For the second circuit [Fig.8-27(b)], carefully observe the connection of the JFET and R_1 and R_2 . For this circuit arrangement the voltage drop across R_2 resistor will be,

$$V_{R2} = \frac{V_{DD} R_2}{R_1 + R_2} = V_G \quad (8-24)$$

Now, apply KVL in the gate-source loop we will find,

$$V_{R2} - I_D R_S + V_{GS} = 0$$

or,

$$\boxed{V_{GS} = I_D R_S - V_G} \quad (8-25)$$

As $I_D R_S > V_G$, the gate will remain at higher potential than the source, just opposite to that of N-channel JFET. Now using this value of V_{GS} into Shockley's equation, we find the value of I_{DQ} , as

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 = I_{DSS} \left(1 - \frac{|I_D R_S - V_G|}{|V_P|} \right)^2 \quad (8-26)$$

$$\text{Again, } V_{DS} \text{ will be,} \quad V_{DS} = V_{DD} - I_D (R_D + R_S) \quad [\because I_D = I_S] \quad (8-27)$$

We can also determine the operating point by plotting the bias line on the transfer characteristics as shown in Fig.8-28.

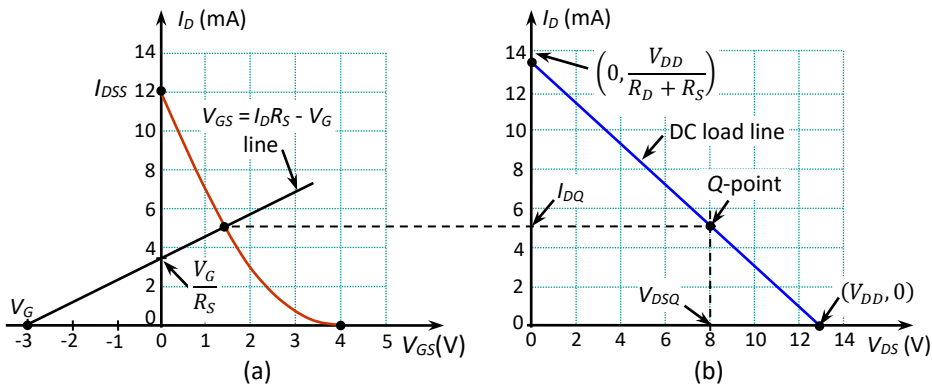


Fig.8-28: Graphical method for voltage divider biasing circuit with P-channel JFET: (a) bias line on the transfer characteristic curve, and (b) position of operating point on the output characteristic curve

Example 8-6

Using analytical method, determine the operating point of the voltage divider bias circuit of Fig.8-29. Also determine the values of V_G , V_S , V_D , and the voltage drop

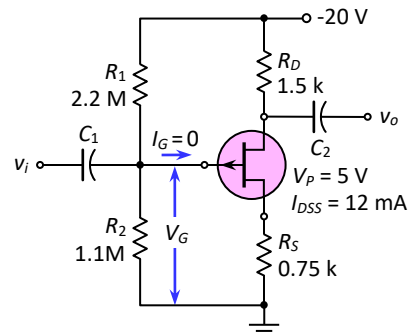


Fig.8-29: Voltage divider bias circuit for Example 8-6

across R_D , and R_S . Assume, $V_P = -5$ V, and $I_{DSS} = 12$ mA.

Solution:

Using Equ.(8-18),
$$V_G = V_{R2} = \frac{V_{DD}R_2}{R_1 + R_2} = \frac{-15 \text{ V} \times 1.1 \text{ M}}{1.1 \text{ M} + 2.2 \text{ M}} = -3 \text{ V}$$

Using Equ.(8-19),
$$V_{GS} = I_D R_S - V_G = 0.75 I_D - 3 \text{ V}$$

Using Equ.(8-20),
$$I_D = I_{DSS} \left(1 - \frac{I_D R_S - V_G}{V_P} \right)^2 = 12 \text{ mA} \left(1 - \frac{0.75 \text{ k} \times I_D - 3 \text{ V}}{5 \text{ V}} \right)^2$$

or,
$$I_D = 12 \left(\frac{5 - 0.75 \times I_D + 3}{5 \text{ V}} \right)^2$$

or,
$$I_D = 12 \times \frac{(8 - 0.75 \times I_D)^2}{25 \text{ V}}$$

or,
$$25 I_D = 12 \times (64 - 16 \times 0.75 \times I_D + 0.5625 I_D^2)$$

or,
$$25 I_D = 768 - 144 \times I_D + 6.75 I_D^2$$

or,
$$6.75 I_D^2 - 169 I_D + 768 = 0$$

or,
$$I_D = \frac{-(-169) \pm \sqrt{(-169)^2 - 4 \times 6.75 \times 768}}{2 \times 6.75}$$

or,
$$I_D = \frac{169 \pm \sqrt{28561 - 20736}}{2 \times 6.75} = \frac{169 \pm \sqrt{7825}}{2 \times 6.75}$$

or,
$$I_D = \frac{169 \pm 88.46}{2 \times 6.75}$$

or,
$$I_D = \frac{257.46}{2 \times 6.75} \text{ or } \frac{80.54}{2 \times 6.75}$$

$\therefore I_D = 19.07 \text{ or } 5.97$

Here, 19.07 mA is not acceptable, because it is greater than I_{DSS} . Therefore,

$$I_D = 5.97 \text{ mA}$$

Again, applying KVL in the output (drain-source) circuit, we get,

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

Using Equ.(8-21),
$$V_{DS} = 20 \text{ V} - 5.97 \text{ mA}(0.75 + 1.5)\text{k} \approx 6.57 \text{ V}$$

The voltage-drop across R_S will be, $I_S R_S = I_D R_S = 5.97 \text{ mA} \times 0.75 \text{ k} = 4.48 \text{ V}$ [Ans.]

The source voltage will be, $V_S = I_S R_S = 4.48 \text{ V}$ [Ans.]

The voltage-drop across R_D will be, $I_D R_D = 5.97 \text{ mA} \times 1.5 \text{ k} \approx 9 \text{ V}$ [Ans.]

Voltage at the drain terminal will be, $V_D = V_S + V_{DS} = 4.48 \text{ V} + 6.57 \text{ V} = 10.05 \text{ V}$ [Ans.]

Comments: Analysis process of voltage divider bias circuit for P-channel JFET is same as the N-channel circuit.

8.9 Saturation of Biasing Circuits

In Chapter 4 (BJT biasing), we have discussed the saturation of each biasing circuit. In different biasing circuits, different resistors were responsible to cause saturation of the transistor. In FET biasing circuit, less components are responsible for saturation of the circuits. As there is no effect of gate resistor (R_G) on the biasing circuits, R_G cannot cause the saturation of FET biasing circuit. Only the large value of R_D , or large value of drain current I_D can cause saturation of the biasing circuit. On the contrary, small value for R_S can produce saturation. For poor stability biasing circuits (like fixed bias), variation of V_P and I_{DSS} from device to device, may produce large I_D and consequently saturation of the biasing circuits. The situation is further illustrated in Example 8-7.

Example 8-7

Repeat Example 8-6 considering each of the following changes individually: i) Considering $R_D = 2.7 \text{ k}$, and ii) $R_S = 240 \Omega$, iii) $V_P = -10 \text{ V}$, and iv) $I_{DSS} = 50 \text{ mA}$.

Solution:

i) Calculation up to I_D remains unchanged. Therefore, $I_D = 5.97 \text{ mA}$ [found from Example 8-6]

$$\therefore V_{DS} = 20 \text{ V} - 5.97 \text{ mA}(0.75 + 2.7)\text{k} \approx -0.6 \text{ V} \quad \text{Using Equ. (8-21)}$$

V_{DS} cannot be negative. Actually, the FET will be saturated and the drain-source voltage will be $V_{DS} = V_{DS(sat)} \approx 0 \text{ V}$.

ii) In this case R_S has been changed, so the drain current will change. The new drain current can be calculated by solving the following equation.

$$\text{Using Equ. (8-20),} \quad I_D = I_{DSS} \left(1 - \frac{|I_D R_S - V_G|}{|V_P|} \right)^2 = 12 \text{ mA} \left(1 - \frac{0.24 \text{ k} \times I_D - 3 \text{ V}}{5 \text{ V}} \right)^2$$

Solving this equation, we get $I_D = 12.3 \text{ mA}$. But the maximum current of a JFET is I_{DSS} . So, in this case the actual value of I_D will be 12 mA .

$$\therefore V_{DS} = 20 \text{ V} - 12 \text{ mA}(0.24 + 1.5)\text{k} \approx -0.88 \text{ V} \text{ [saturated]}$$

iii) As the value of V_P has been changed, so the drain current will also change.

The new drain current can be calculated by solving the following equation.

$$\text{Using Equ. (8-20),} \quad I_D = I_{DSS} \left(1 - \frac{I_D R_S - V_G}{V_P} \right)^2 = 12 \text{ mA} \left(1 - \frac{0.75 \text{ k} \times I_D - 3 \text{ V}}{10 \text{ V}} \right)^2$$

Solving this equation, we get $I_D = 7.1 \text{ mA}$.

$$\therefore V_{DS} = 20 \text{ V} - 7.1 \text{ mA}(0.75 + 1.5)\text{k} \approx 4 \text{ V} \text{ [not saturated]}$$

iv) As the value of I_{DSS} has been changed, so the drain current will also change.

The new drain current can be calculated by solving the following equation.

Using Equ.(8-20),
$$I_D = I_{DSS} \left(1 - \frac{I_D R_S - V_G}{V_P} \right)^2 = 50 \text{ mA} \left(1 - \frac{0.75 \text{ k} \times I_D - 3 \text{ V}}{10 \text{ V}} \right)^2$$

Solving this equation, we get $I_D = 8 \text{ mA}$.

$\therefore V_{DS} = 20 \text{ V} - 8 \text{ mA}(0.75 + 1.5)\text{k} \approx 2 \text{ V}$ [not saturated]

Comments: As voltage divider biasing circuits provide the most stability of operating point, a large change in V_P and I_{DSS} cannot produce saturation of the circuit.

8.10 Biasing Circuits with Split Power Supply

So far, we have discussed three types of biasing circuits for JFETs in common source using **single-ended** power supply. Sometimes, we need to use FET circuit in a system that uses **split power supply** (power supply with three terminals- positive, negative and ground). All of the three biasing circuits can be used with split power supplies. The circuits are shown in Fig.8-30.

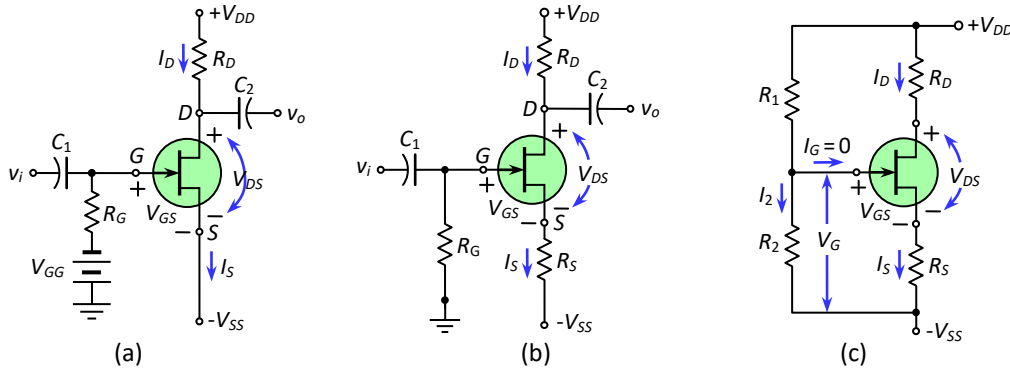


Fig.8-30: JFET biasing circuit using split power supply: (a) Fixed bias, (b) Self bias, and (c) Voltage divider bias

8.10.1 Fixed Bias with Split Power Supply

Fig.8-30(a) shows the fixed-bias circuit for N-channel JFET in common source configuration with split power supply. Here the circuit is same as before but only difference is, the source is connected to negative voltage $-V_{SS}$. So, the equation of bias line of this circuit will be,

$$V_{GS} = |V_{GG}| - |V_{SS}| \quad (8-28)$$

Here, absolute sign is used to avoid confusion, in applying this equation. To ensure reverse bias of the gate, value of V_{GG} should be greater than the value of V_{SS} . Using this value of V_{GS} in Shockley's equation drain current can be determined analytically or drawing this bias line on the transfer characteristic curve I_D can be determined graphically.

The value of V_{DS} can be calculated using KVL in the output circuit as,

$$V_{DS} = V_{DD} + |V_{SS}| - I_D(R_D + R_S) \quad (8-29)$$

V_{DS} can be calculated from the above equation, or can be determine graphically from the load line described as before.

8.10.2 Self-Bias with Split Supply

The self-bias circuit with split power supply is shown in Fig.8-30(b) From the input circuit the equation of bias line will be,

$$V_{GS} = |V_{SS}| - I_S R_S = |V_{SS}| - I_D R_S [\because I_D = I_S] \quad (8-30)$$

This equation is comparable with the equation of bias line for voltage divider bias circuit with single-ended voltage source. So, this circuit will give the advantages of the voltage divider bias.

From the output circuit the drain-source voltage will be,

$$V_{DS} = V_{DD} + |V_{SS}| - I_D (R_S + R_D) \quad (8-31)$$

The above equation can be used to determine the value of V_{DS} . The DC load line is also drawn from this equation.

8.10.3 Voltage Divider Bias with Split Supply

Though the self-bias circuit with split supply gives the advantages of voltage divider bias, we can also apply voltage divider network to control the position of operating point as necessary. The voltage divider bias circuit with split power supply is shown in Fig.8-30(c). Using superposition theorem, the gate voltage can be calculated as,

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} - \frac{R_1 V_{SS}}{R_1 + R_2} \quad (8-32)$$

Now, from the input circuit the equation of bias line will be,

$$V_{GS} = V_G + |V_{SS}| - I_D R_S \quad (8-33)$$

From the output circuit the drain-source voltage will be,

$$V_{DS} = V_{DD} + |V_{SS}| - I_D (R_S + R_D) \quad (8-34)$$

The above equation can be used to determine the value of V_{DS} and the DC load line.

Example 8-8

Using graphical method, determine the operating point of the circuit of Fig.8-31. Assume, $V_p = -3$ V, and $I_{DSS} = 12$ mA.

Solution:

To solve the problem graphically, the transfer characteristic curve is drawn using the values of $V_p = -3$ V and $I_{DSS} = 12$ mA [Fig.8-31(b)].

Using Equ.(8-30), $V_{GS} = |V_{SS}| - I_D R_S = 6$ V $- 1$ k $\times I_D$

Now this equation is plotted on the same transfer characteristic curve as shown in Fig.8-31(b). This bias line intersects the transfer characteristic curve at a point. The ordinate of this intersecting point gives the value of I_{DQ} . From the graph this value is 6.75 mA.

V_{DSQ} can be determined from the operating point graphically, but here its value is calculated using Equ.(8-31),

$$V_{DSQ} = V_{DS} = 6\text{ V} + 6\text{ V} - 6.75\text{ mA} (1\text{ k} + 0.5\text{ k}) \approx 1.88\text{ V}$$

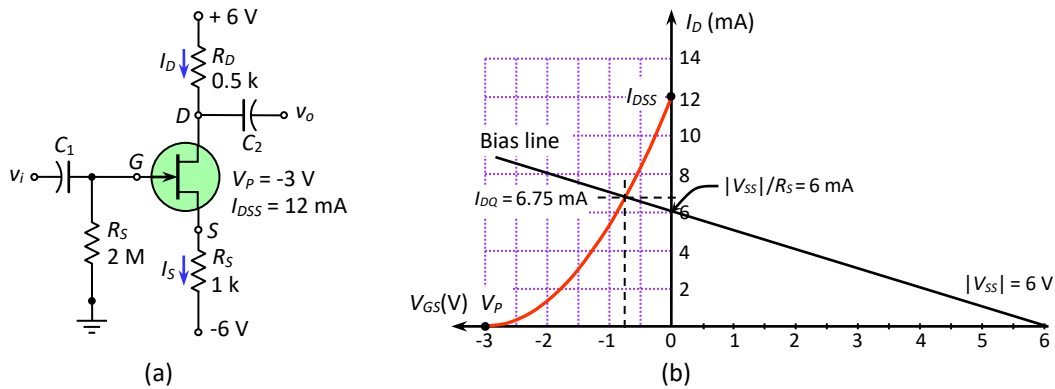


Fig.8-31: (a) Self-bias circuit with split power supply, and (b) Determination of operating point

Answer: $V_{DSQ} = 1.88\text{ V}$, and $I_{DQ} = 6.75\text{ mA}$

Comments: Large value of gate voltage ($V_G = V_{SS}$) results in a large drain current. For this large I_{DQ} , the operating point has been set near saturation region. This problem can be solved by increasing biasing voltage.

8.11 Comparison of Biasing Circuits

So far, we have discussed three main biasing circuit of JFET. These are fixed or gate bias, self-bias, and voltage divider bias. We have also discussed the performance of these biasing circuits. The performance of these FET biasing circuits is quite similar to the performance three basic BJT biasing circuits like: fixed or base bias, emitter feedback bias, and voltage divider bias. Being the bipolar device, BJT circuits have minority current components. This minority current changes with temperature. So BJT biasing circuits must have the ability to compensate both the effect of temperature variation, and the variation of device parameters (β , V_{BE}). On the other hand, FETs are unipolar device. They do not have minority carrier current. Therefore, FET biasing circuit has to compensate on the variation of device parameters. We have already seen that datasheet of any particular FET number mentions minimum and maximum values of V_P and I_{DSS} . That is, the FET has a minimum and a maximum transfer characteristic curve. A FET of that number can have any values within the range of V_P , and I_{DSS} . Therefore, as already stated, the specific minimum and maximum transfer characteristic curves must be used when analyzing (or design) a FET bias circuit. In Example 8- to 7. , we have analyzed three types of biasing circuits keeping $R_D + R_S$ almost same. In all the examples, we have considered same minimum and maximum transfer characteristic curves. From these analyses, we have seen that fixed bias or gate bias circuit produces the maximum variation of the operating point for the change in transfer characteristic curves. The self-bias circuit provided stability of operating point to some extends. But the voltage divider bias provides the most stability

of operating point. To compare the results, the circuits are redrawn here, and the variation of operating point is shown in a single graph (Fig.8-33).

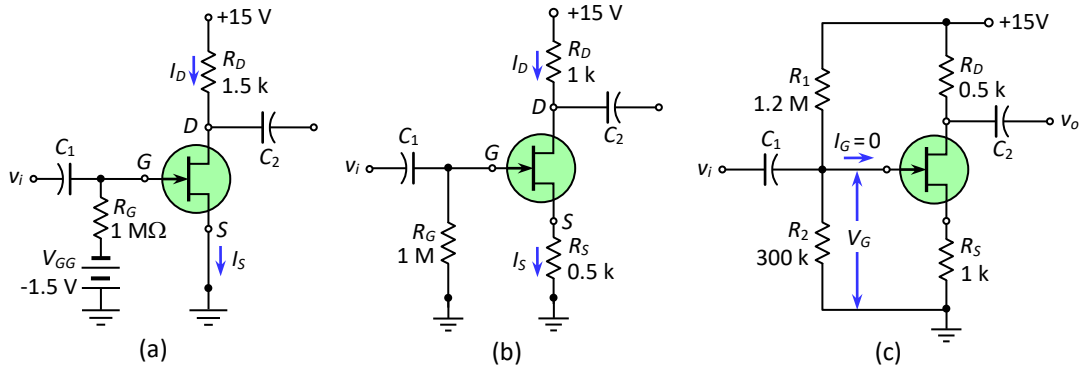


Fig.8-32: All of the three biasing circuits for N-channel JFET: (a) Fixed bias, (b) Self-bias, and (c) Voltage divider bias

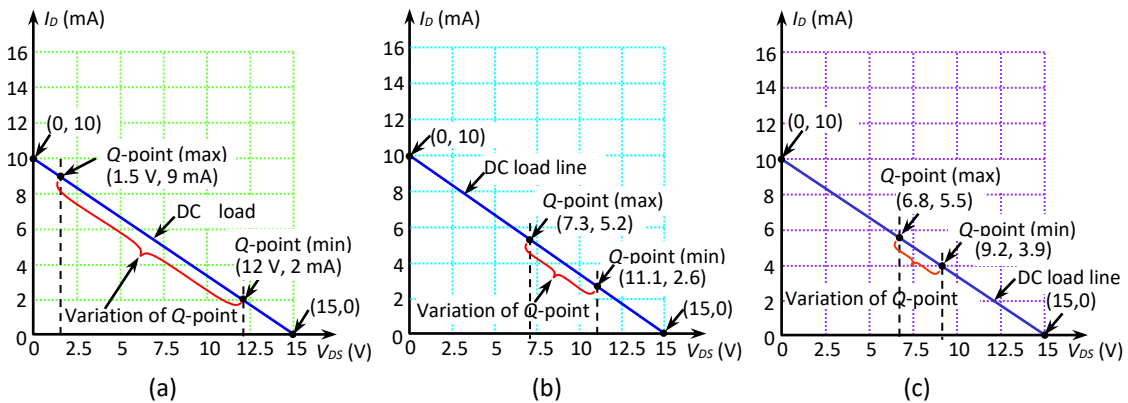


Fig.8-33: Variation of operating point due to the variations in V_P and I_{DSS} : (a) Fixed bias, (b) Self-bias, and (c) Voltage divider bias

8.12 JFET Bias Circuit Design

So far, we have discussed different types of biasing circuits for JFET in common source configuration. Now we will discuss how to design biasing circuits for JFET in CS configuration. Designing process of FET biasing circuits is almost similar to the designing process of BJT biasing circuits. One main difference is that, FET biasing circuit design generally uses a graphical approach. However, for a given drain current level, we can design FET biasing circuit using analytical method also. Graphical method involves the drawing of bias line on the device transfer characteristics, as in the case of FET circuit analysis. To design an amplifier circuit, at first, we have to select the device to be used for the amplifier. Device selection depends on the value of V_{DD} , g_m and drain current etc. Once, the FET to be used in the circuit is known, the maximum and minimum transfer characteristics should be taken from datasheet or should be drawn using maximum and minimum

values of V_P , and I_{DSS} . From these transfer characteristics, a maximum drain current ($I_{D(max)}$) is selected at a suitable position on the maximum transfer characteristics [see Fig.8-34(a)]. In selecting $I_{D(max)}$, we have to remember that higher values of I_D give higher g_m and less nonlinearity. On the other hand, higher values of I_D reduces the values of R_S and R_D . Small values of R_S and R_D result in poor stability and lower voltage gain, respectively. So, we have to make a trade of between them. Once the value of $I_{D(max)}$ is determined, a bias line is drawn through this point of maximum transfer characteristic curve. As will be explained later, this procedure applies to gate bias and self-bias circuit. For voltage divider circuit, first the value of R_S should be calculated, and then, the bias line has to be drawn using the slope of $-1/R_S$.

For the maximum drain current ($I_{D(max)}$), the drain source voltage will be minimum. The minimum value of drain source voltage should be,

$$V_{DS(min)} = V_P + 1\text{ V} - V_{GS} \quad (8-35)$$

For large V_{DD} , we can also use,
$$V_{DS(min)} = V_P + 1\text{ V} \quad (8-36)$$

The rest of the voltage will be equally divided between R_S and R_D .

Then the values of R_S and R_D are calculated by dividing these voltages by the selected $I_{D(max)}$.

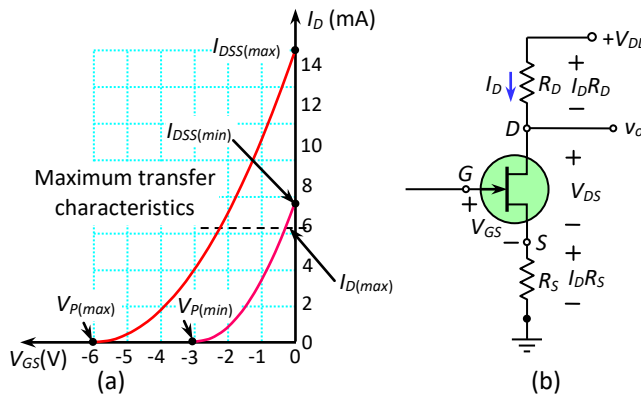


Fig.8-34: Maximum and minimum transfer characteristics, and (b) Output circuit of Self-bias circuit

It is safe to use the maximum characteristics for designing biasing circuits. If the minimum characteristics, or some characteristics between the maximum and minimum is used in the circuit design, now, if the actual FET of the implemented circuit have the maximum transfer characteristics, I_D will be larger than the design value. Consequently, the resistor voltage drops ($I_D R_D$ and $I_D R_S$) will be greater than the voltages assigned to them. In that case, V_{DS} will be smaller than its design level, as a result the FET might be forced to work in the voltage-controlled resistance region. Then the amplifier will not work. Therefore, **always use the device maximum transfer characteristic when design a FET bias circuit.**

As already explained, a FET has a very high input resistance so high-value bias resistors can be used at the gate terminal without any loading effect. However, there are some disadvantages of using extremely high resistance values. A charge accumulated at the gate can take a long time to leak off (to discharge) through a very high resistance. In that case, the gate voltage might not be a stable

quantity, and consequently, the drain current could be incorrect. High-value bias resistors also generate unwanted thermal noise (thermal noise is also known as *Johnson-Nyquist noise*, *Johnson noise*, or *Nyquist noise*) and they readily pick-up *stray radio-frequency signals*. For these reasons, 1 M Ω is normally a reasonable upper limit for bias resistors.

In all situations (designs) where resistor values are calculated, the closest standard-value components are selected, (see appendix). Normally, $\pm 10\%$ tolerance resistors are always used, and the more expensive $\pm 5\%$ and $\pm 1\%$ components are employed where greater precision is required.

8.12.1 Gate Bias Circuit Design:

Design of gate bias circuit is very simple. As explained, an appropriate maximum I_D level is marked on the maximum transfer characteristics for the FET. The bias line is drawn vertically through this point on the transfer characteristic curve. The intersecting point of this bias line, on the V_{GS} axis gives the value the required gate bias voltage ($-V_G$). Alternately, V_{GS} can be calculated analytically using the following equation.

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) \quad (8-37)$$

Because the circuit design uses $I_{D(max)}$, the voltage drop across R_D is also maximum, and consequently, the drain-source voltage (V_{DS}) is a minimum. So, the drain resistor is calculated as,

$$R_D = \frac{V_{DD} - V_{DS(min)}}{I_{D(max)}} \quad (8-38)$$

As discussed, the bias resistor (R_G) is usually selected as 1 M Ω , although a lower resistance value may be used.

Fig.8-35 shows the equations involved in designing a gate bias circuit, and Example 8-9 demonstrates the design procedure.

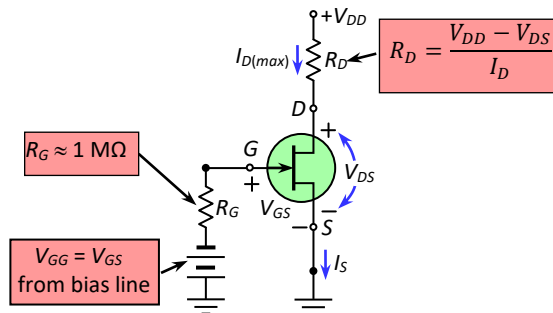


Fig.8-35: Designing process of fixed bias circuit

Example 8-9

The maximum and minimum transfer characteristics of a JFET are shown in Fig.8-36(a). Using these characteristics design a gate bias circuit for the JFET. Assume $V_{DD} = 16\text{ V}$.

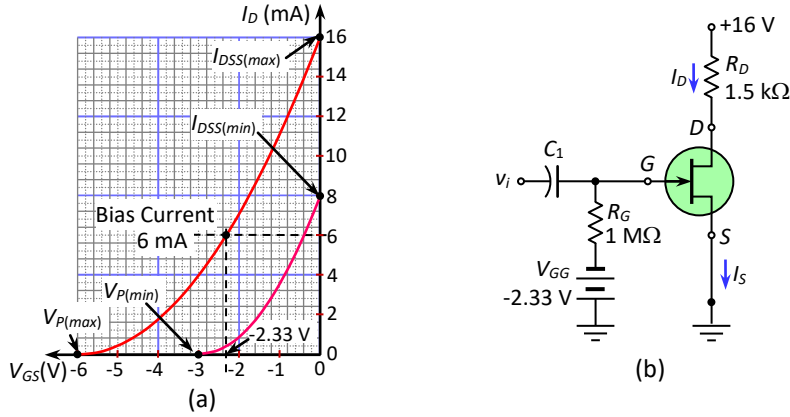


Fig.8-36: (a) Transfer characteristic curves given for Example 8-9, and (b) The designed circuit for this example

Solution:

From the transfer characteristics, we can set $I_{DQ} = 6\text{ mA}$. Now we have to draw a bias line through $I_{DQ} = 6\text{ mA}$ point of the maximum transfer characteristic curve as shown in the given figure. This line intersects the x -axis at $V_{GSQ} = -2.33\text{ V}$. Therefore, the value of V_G will be -2.33 V . We can use $R_G = 1\text{ M}\Omega$.

From the maximum transfer characteristic curve, we find that $V_{P(max)} = -6\text{ V}$. So, we can assume $V_{DS(min)} = V_{P(max)} + 1\text{ V} = 7\text{ V}$.

The voltage drop across R_D will be, $16\text{ V} - 7\text{ V} = 9\text{ V}$.

Therefore, the value of the drain resistor will be,

$$R_D = \frac{V_{DD} - V_{DS(min)}}{I_{D(max)}} = \frac{9\text{ V}}{6\text{ mA}} = 1.5\text{ k}\Omega$$

Answer: The designed biasing circuit is shown in Fig.8-36(b).

Comments: The main problem of a fixed bias circuit is that it needs two power supplies.

8.12.2 Self-Bias Circuit Design

Like gate bias, the procedure for designing a self-bias circuit (Fig.8-37) commences with $I_{D(max)}$ being marked on the FET maximum transfer characteristic as shown in Fig.8-38(a). The bias line is then drawn through this point from the origin ($V_{GS} = 0, I_D = 0$). The value of the source resistor (R_S) is determined from the slope of the bias line as,

$$R_S = \frac{\Delta V_{GS}}{\Delta I_D} \quad (8-39)$$

The total resistance in series with the FET drain-source terminals [Fig.8-37(b)] is $R_D + R_S$. So, Equ.(8-38) can be modified to calculate this resistance,

$$R_D + R_S = \frac{V_{DD} - V_{DS(min)}}{I_{D(max)}} \quad (8-40)$$

With R_S already known, R_D is easily determined. Gate resistor R_G is normally selected as $1 \text{ M}\Omega$, or lower, as already discussed.

Fig.8-37 shows the equations for designing a self-bias circuit and Example 8-10 demonstrates the design procedure.

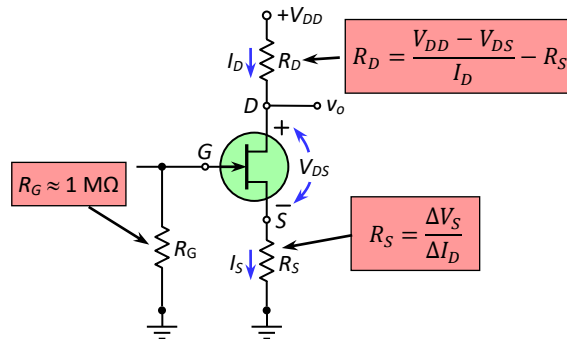


Fig.8-37: Designing self-bias circuit for JFET

Example 8-10

The maximum and minimum transfer characteristics of a JFET are shown in Fig.8-38. Using these characteristics design a self-bias circuit for the JFET. Assume $V_{DD} = 16 \text{ V}$.

Solution:

From the transfer characteristic curves, we can assume $I_{DQ} = 6 \text{ mA}$. Now let us draw a bias line through $I_D = 6 \text{ mA}$ point as shown in Fig.8-38(a). Value of R_S can be calculated from the slope of this bias line. We can consider any suitable value of ΔV_{GS} and its corresponding ΔI_D . As shown in the figure let $\Delta V_{GS} = 2 \text{ V}$, hence $\Delta I_D = 5.2 \text{ mA}$ (calculated from the graph).

$$\therefore R_S = \frac{\Delta V_{GS}}{\Delta I_D} = \frac{2 \text{ V}}{5.2 \text{ mA}} = 385 \Omega, \text{ use standard } 390 \Omega$$

Let, $R_G = 1 \text{ M}\Omega$.

From the maximum transfer characteristic curve, we find that $V_{P(max)} = -6 \text{ V}$. So, we can assume $V_{DS(min)} = V_{P(max)} + 1 \text{ V} = 7 \text{ V}$.

$$\text{Using Equ.(8-40), } R_D + R_S = \frac{V_{DD} - V_{DS(min)}}{I_{D(max)}} = \frac{16 \text{ V} - 7 \text{ V}}{6 \text{ mA}} = 1.5 \text{ k}$$

$$\therefore R_D = 1.5 \text{ k} - R_S = 1.5 \text{ k} - 0.39 \text{ k} = 1.11 \text{ k}, \text{ use standard } 1.2 \text{ k}$$

Results: The designed biasing circuit is shown in Fig.8-38(b).

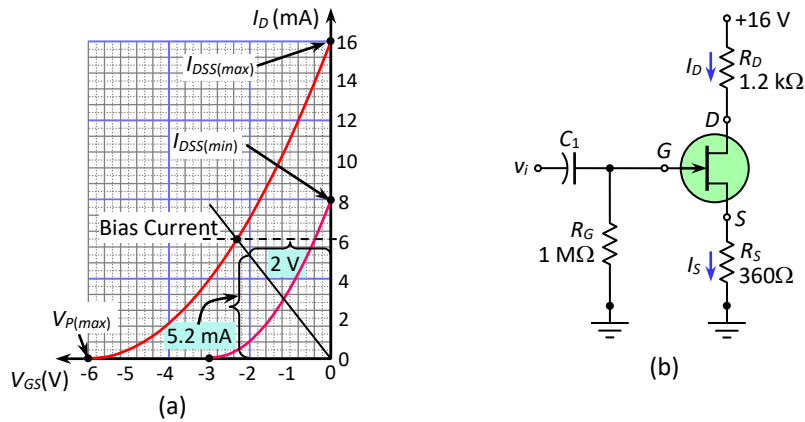


Fig.8-38: (a) Transfer characteristics for Example 8-10, and (b) The designed circuit for this Example

Comments: As we have used a higher value source resistor, I_D will be little lower than the assumed value. Again, a higher drain resistor than the calculated value will result in smaller $V_{DS(min)}$.

8.12.3 Voltage Divider Bias Circuit Design:

Here also, the JFET has to be selected first, depending on the amplifier requirements. The design of the biasing circuit will start from drawing the maximum and minimum transfer characteristic curve. From these curves a suitable $I_{D(max)}$ is decided. $V_{DS(min)}$ is calculated using Equ.(8-35) or Equ.(8-36). The maximum drain current is plotted on the device maximum transfer characteristic, then either the resistance of R_S or the level of V_G must be determined before the bias line can be drawn for a voltage divider bias circuit.

With $V_{DS(min)}$ specified, the remaining portion of the supply voltage ($V_{DD} - V_{DS(min)}$) is split between V_S and V_{RD} . The voltage gain of a FET amplifier

stage (described in Chapter 11) is directly proportional to R_D . Maximum gain is achieved by selecting R_D as large as possible, and for a given level of I_D this requires the greatest possible voltage for R_D . However, the best bias stability is achieved by selecting R_S as large as possible, and this dictates maximum V_S . A reasonable compromise between these two conflicting requirements (maximum V_{RD} and maximum V_S) is to simply make the two voltage drops (and the two resistors) equal. Equ.(8-40) is used to calculate $R_D + R_S$, and this is equally divided to give two equal resistors.

Once R_S is determined, the bias line can be drawn through $I_{D(max)}$ with the slope of the line set by R_S . This requires that the V_{GS} scale be extended positively, as shown in Fig.8-24(a).

$$\text{From Equ.(39),} \quad \Delta V_{GS} = \Delta I_D \times R_S \quad (8-41)$$

The gate bias voltage (V_G) is read from the intersection of the bias line and the (extended) V_{GS} scale [as in Fig.8-31(b)]. R_2 is normally the smaller resistor in the voltage divider. So, R_2 is usually selected as 1 MΩ, for the reasons already discussed for R_G in the gate bias and self-bias circuits. R_1 is then calculated from R_2 and the resistor voltage drops as,

$$R_1 = \frac{(V_{DD} - V_G) \times R_2}{V_G} \quad (8-42)$$

Another approach that can be used to determine R_S is to specify a minimum drain current, and mark $I_{D(min)}$ on the minimum drain characteristic. The bias line is then drawn through the $I_{D(max)}$ point on the maximum transfer characteristic curve, and $I_{D(max)}$ point on the minimum transfer characteristic curve. The slope of the bias line and its interaction with the V_{GS} scale dictates R_S and V_G , respectively.

The design equations for a voltage divider bias current are shown on Fig.8-39 and Example 8-11 demonstrates the design procedure.

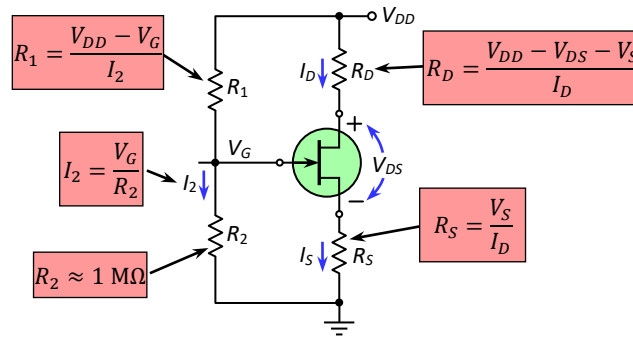


Fig.8-39: Designing process of voltage divider bias circuit

Example 8-11

A P-channel JFET has $V_P = 5 \text{ V}$, and $I_{DSS} = 10 \text{ mA}$. Using this JFET design a voltage divider biasing circuit for a common source amplifier. Assume, $V_{DD} = 20 \text{ V}$, and $I_{D(max)} = 5 \text{ mA}$. Use analytical method.

Solution:

As $V_P = 5 \text{ V}$, therefore $V_{DS(min)} = 5 \text{ V} + 1 \text{ V} = 6 \text{ V}$

Let,

$$V_{RS} = V_{RD} = \frac{1}{2}(V_{DD} - V_{DS(min)}) = \frac{1}{2}(20 \text{ V} - 6 \text{ V}) = 7 \text{ V}$$

$$\therefore R_S = R_D = \frac{7 \text{ V}}{5 \text{ mA}} = 1.4 \text{ k}$$

Use R_D = standard 1.5 k, and R_S = standard 1.3 k

$$\therefore V_{RD} = 1.5 \text{ k} \times 5 \text{ mA} = 7.5 \text{ V}, \text{ and } V_{RD} = 1.3 \text{ k} \times 5 \text{ mA} = 6.5 \text{ V}$$

Using Equ.(8-37),

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) = 5 \text{ V} \left(1 - \sqrt{\frac{5 \text{ mA}}{10 \text{ mA}}} \right) = 1.47 \text{ V}$$

∴

$$V_G = V_{RS} - V_{GS} = 6.5 \text{ V} - 1.47 \text{ V} = 5.03 \text{ V}$$

Let, $R_1 = 1 \text{ M}\Omega$.

$$\therefore R_2 = \frac{(V_{DD} - V_G) \times R_1}{V_G} = \frac{(20 \text{ V} - 5.03 \text{ V}) \times 1 \text{ M}\Omega}{5.03 \text{ V}} = 2.98 \text{ M}\Omega$$

use $R_2 = \text{standard } 3 \text{ M}\Omega$

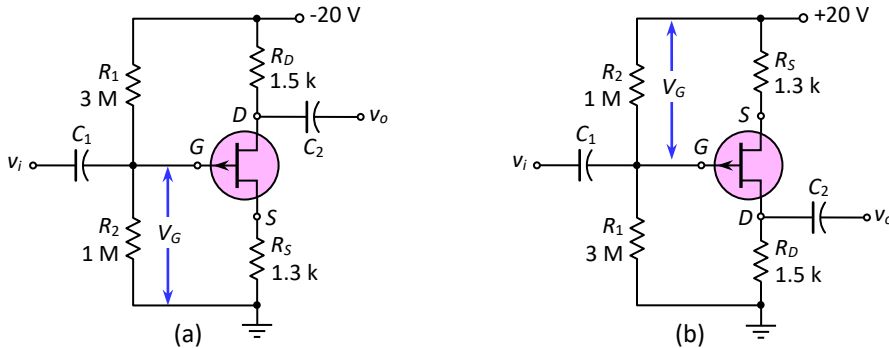


Fig.8-40: P-channel JFET circuits for Example 8-11: (a) Negative supply, and (b) Positive supply

Results: $R_D = 1.5 \text{ k}$, $R_S = 1.3 \text{ k}$, $R_1 = 1 \text{ M}\Omega$, and $R_2 = 3 \text{ M}\Omega$. The designed circuits are shown in Fig.8-40(a).

Comments: If the transfer characteristics of the FET is fixed, and $I_{D(max)}$ is given, we can use analytical method also. Design procedure for P-channel FETs is same as that of N-channel FETs.

8.13 FET Common Drain Configuration

So far, we have discussed different biasing circuits of N-channel and P-channel JFET in common source configuration. Now we will discuss the biasing circuit for common drain configuration. Like the common collector amplifier, the common drain amplifier gives unity voltage gain and the highest current gain. **This configuration gives the highest input impedance and lowest output impedance.** In this configuration, the output signal is in phase with the input. The common drain amplifier is also called *source follower* or *voltage follower*. For its very high input impedance, it is used as a buffer amplifier.

The biasing circuits for common drain configuration are similar to those of common source configuration. Only difference is the output is taken from the source terminal and no resistance is used in the drain terminal. Generally, the *self-bias* and *voltage divider bias* circuits are used for common drain configurations. These biasing circuits are shown in F.

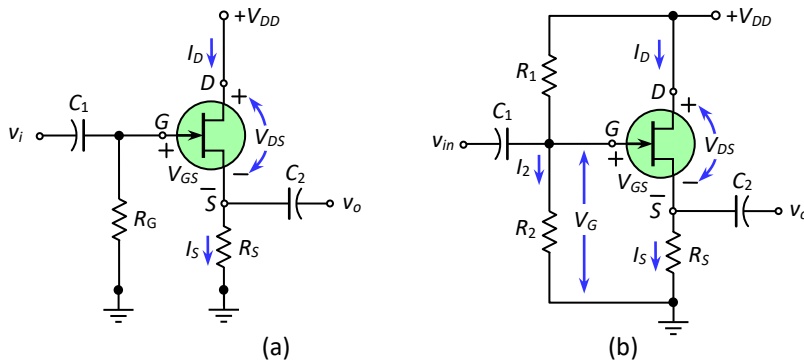


Fig.8-41: Common drain circuits: (a) Self bias, and (b) Voltage divider bias

Here also, the input gate current will be almost zero, and the equation of biasline of the self-bias circuit will be

$$V_{GS} = -I_D R_S \quad (8-43)$$

Now putting this value into Shockley's equation we get,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = I_{DSS} \left(1 - \frac{|I_D R_S|}{|V_P|}\right)^2 \quad (8-44)$$

As there is no drain resistance, the equation of drain source voltage will be,

$$\therefore I_D = I_S, \quad \therefore V_{DS} = V_{DD} - I_D R_S \quad (8-45)$$

Using Equ.(8-43) to Equ.(8-45), the DC operating point of a fixed bias circuit for common drain configuration can be determined. The operating point can also be determined graphically as described for the common source configuration.

Example 8-12

Assuming $V_P = -5\text{ V}$, and $I_{DSS} = 10\text{ mA}$, determine the Q-point of the following common-drain amplifier using analytical method.

Solution:

Using Equ.(8-43), $V_{GS} = -1.2 \times I_D$

Now using this value into Shockley's equation we get,

$$I_D = 10\text{ mA} \left(1 - \frac{|1.2 \times I_D|}{|5|}\right)^2$$

Solving the above equation for I_D , we get

$$I_D = 2.21\text{ mA [Ans.]}$$

Using Equ.(8-45), $V_{DS} = 6\text{ V} - 2.21\text{ mA} \times 1.2\text{ k} = 3.35\text{ V [Ans.]}$

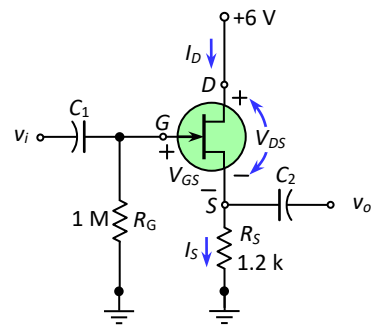


Fig.8-42: Common-drain amplifier for Example 8-12

Comments: The DC analysis of common drain circuits is same as the common source circuits.

For the voltage divider bias, the equation of bias line will be same as for common source amplifier. This is,

$$V_{GS} = V_G - I_D R_S \quad (8-46)$$

Where,

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2} \quad (8-47)$$

Using this value of V_{GS} , drain current can be calculated from Shockley's equation, which is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = I_{DSS} \left(1 - \frac{|I_D R_S - V_G|}{|V_P|}\right)^2 \quad (8-48)$$

From the output (drain-source) circuit, we get,

$$V_{DS} = V_{DD} - I_D R_S \quad (8-49)$$

The DC operating of the common drain voltage divider bias circuit can be determined analytically using the above equations or it can be determined graphically, exactly in the same procedure of common source configuration.

Example 8-13

Design a voltage divider bias circuit for common drain JFET amplifier. Assume $V_P = -5$ V, and $I_{DSS} = 10$ mA, $V_{DD} = 12$ V.

Solution:

Let $I_{DQ} = 6$ mA, $V_{DSQ} = V_S = \frac{1}{2} V_{DD} = 6$ V.

$$\therefore R_S = \frac{V_S}{I_{DQ}} = \frac{6 \text{ V}}{6 \text{ mA}} = 1 \text{ k}$$

Using Equ.(8-37),

$$V_{GS} = -5 \text{ V} \left(1 - \sqrt{\frac{6}{10}}\right) = -1.13 \text{ V}$$

Value of V_G will be,

$$V_G = V_S - V_{GS} = 6 \text{ V} - 1.13 \text{ V} = 4.87 \text{ V}$$

Let $R_1 = 1$ M.

$$\therefore R_2 = \frac{V_G \times 1 \text{ M}}{(V_{DD} - V_G)} = \frac{4.871 \times 1 \text{ M}}{(12 - 4.87)} \approx 140 \text{ k}\Omega$$

Answer: The designed circuit is shown in Fig.8-43.

Comments: Here, we have consider $I_{DQ} = 6$ mA, and $V_S = 6$ V. By decreasing I_{DQ} and increasing V_S , we can design this circuit. In that case, the value of R_S will be high, that will give high voltage gain.

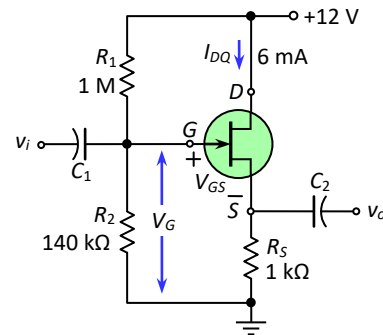


Fig.8-43: Designed voltage divider bias circuit for common drain JFET

8.14 Biasing of Common Gate JFET Circuit

The FET common gate amplifier circuit is the least widely used, but it does possess some characteristics that has important application in some practical circuits. Common gate FET configuration provides a low input impedance while offering a high output impedance. As the gate is grounded, this acts as a barrier between input and output providing high levels of isolation, preventing feedback, especially at very high frequencies. Although the voltage gain is high, the current gain is low and the overall power gain is also low when compared to other FET circuit configurations. Like the common drain configuration, this circuit has no phase shift between the input and output signals.

The common gate circuit configuration does find uses with amplifiers that require low input impedance levels. One such application is VHF and UHF RF amplifiers where the low input impedance of the amplifier allows accurate matching to the feeder impedance which is typically 50Ω or 75Ω . Another application is moving-coil microphones preamplifiers - these microphones have very low impedance levels.

The biasing circuits of JFET in common gate configuration are shown in F. The first one is the self-bias circuit and the second one is the voltage divider bias circuit. The equations of bias lines for these circuits will same as before. For the self-bias, the equation of bias line is,

$$V_{GS} = -I_D R_S \quad (8-50)$$

and for the voltage divider bias the equation of bias line is,

$$V_{GS} = V_G - I_D R_S \quad (8-51)$$

The value of V_G is calculated as before.

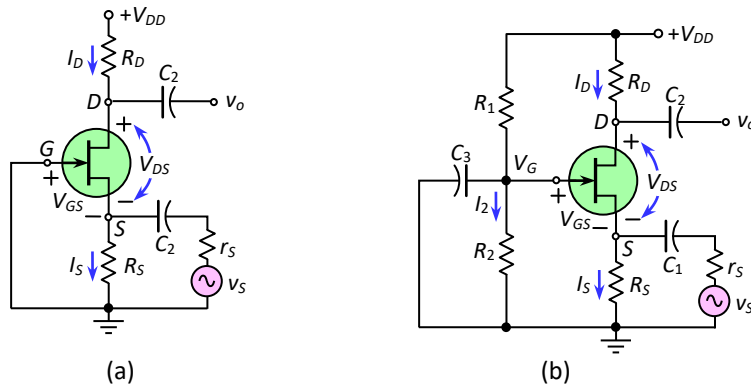


Fig.8-44: Common gate circuits: (a) Self bias, and (b) Voltage divider bias

The equation of quiescent drain-source voltage will also be same as common source amplifier. For both the circuits,

$$\therefore I_D = I_S, \quad \therefore V_{DSQ} = V_{DD} - I_D(R_D + R_S) \quad (8-52)$$

Example 8-14

Assume that the JFET of Fig.8-44(a) has $V_P = -6\text{ V}$, and $I_{DSS} = 12\text{ mA}$. Using graphical approach determine the quiescent point of the circuit. Also determine voltage levels at every terminal of the JFET for DC condition.

Solution:

For the given values of V_P , and I_{DSS} the transfer characteristic curve is drawn using the same procedure described as before. The bias line of this circuit is,

$$V_{GS} = -I_D R_S = -0.51 \times I_D$$

This bias-line intersects the transfer characteristic curve at point A.

The load-line of the circuit is drawn using the following two points.

$$(V_{DD}, 0) = (12\text{ V}, 0) \text{ and } (0, \frac{V_{DD}}{R_D + R_S}) = (0, 7.02\text{ mA})$$

Now, a horizontal line is drawn through point A on the transfer characteristic curve. This horizontal line intersects the load-line at the Q-point. From this Q-point, we get,

$$I_{DQ} = 4.53\text{ mA}, \text{ and } V_{DSQ} = 4.25\text{ V}$$

Voltage at source terminal will be,

$$V_S = V_{RS} = I_D R_S = 4.53\text{ mA} \times 0.51\text{ k} \approx 2.31\text{ V}$$

Voltage at gate terminal will be zero.

Voltage at drain terminal will be,

$$V_D = V_{DD} - I_D R_D = 12\text{ V} - 4.53\text{ mA} \times 1.2\text{ k} \approx 6.56\text{ V}$$

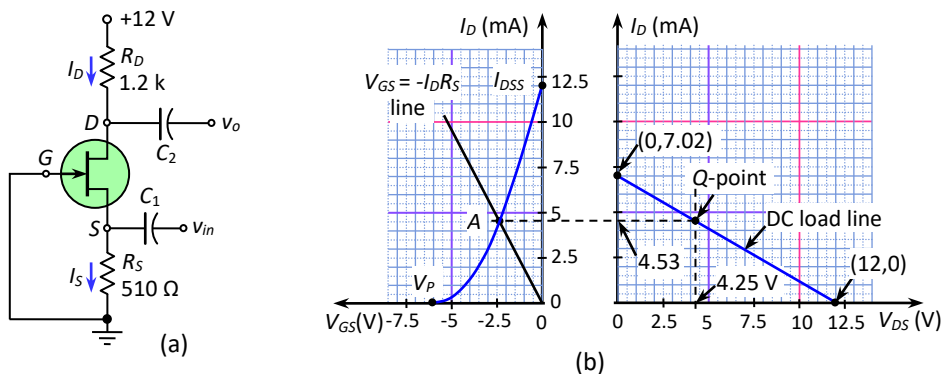


Fig.8-45: (a) CGC JFET circuit for Example 8-14, and (b) Graphs for Q-point determination

Answer: $I_{DQ} = 4.53\text{ mA}$, and $V_{DSQ} = 4.25\text{ V}$, $V_G = 0$, $V_S = 2.31\text{ V}$, and $V_D = 5.65\text{ V}$.

Comments: Analysis of common gate amplifiers is same as common source amplifiers.

8.15 Biasing Circuit of Depletion MOSFET

All of the three biasing circuits described so far, can be used for depletion MOSFET. The biasing circuits of the DMOSFET are shown in Fig.8-47. From the operation of enhancement MOSFET, we have seen that it can work in depletion mode and enhancement mode. That is, drain current can be controlled by applying both positive and negative voltage at gate terminal. Therefore, to bias a DMOSFET we can use both positive and negative gate voltage. For N-channel DMOSFET, the device will be biased in the depletion mode region for negative gate voltage and the device will be biased in the enhancement mode region for positive gate voltage. For P-channel DMOSFET, the device will operate in depletion mode for positive gate voltage and in enhancement mode for negative gate voltage.

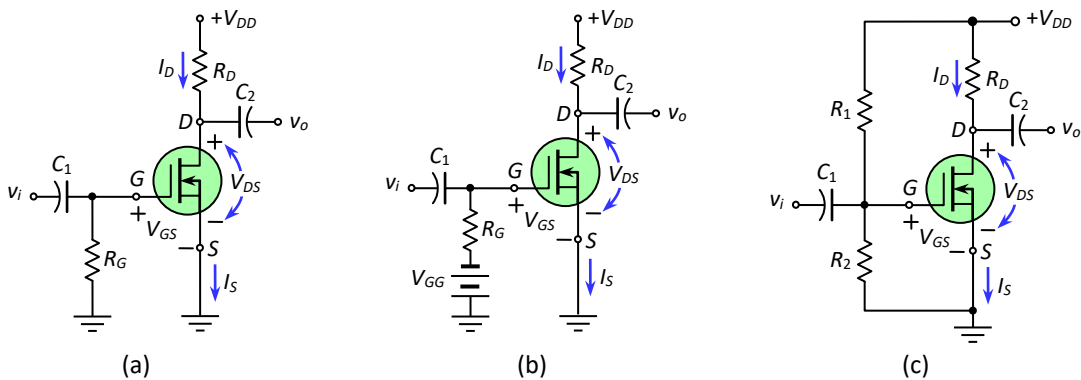


Fig.8-47: Different forms of fixed bias circuits for depletion MOSFET

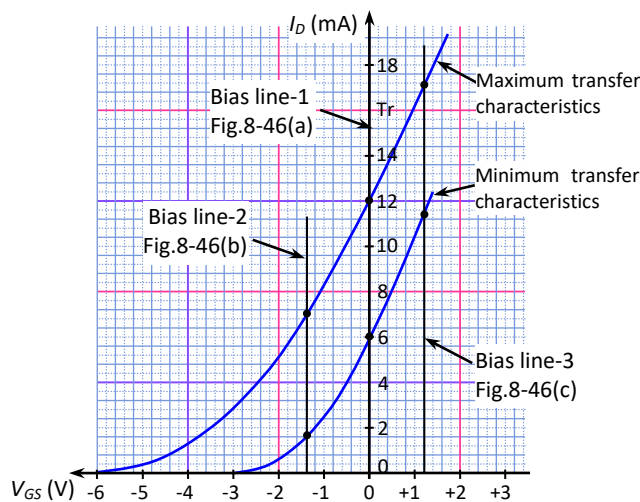


Fig.8-48: Different bias lines for fixed bias circuits intersecting the minimum and maximum transfer characteristic curves

8.15.1 Gate Bias or Fixed Bias

Fig.8-47 shows the fixed bias circuit for an N-channel DMOSFET in common source configuration. Unlike the JFET, DMOSFET can be biased at $V_G = 0$. For this reason, no extra source is connected to the gate in the circuit of Fig.8-46(a). As with zero AC signal, the gate voltage is zero, the DC operating point will be on the y-axis and the value of I_{DQ} will be I_{DSS} as shown in Fig.8-47. The drain source voltage will be same as JFET.

$$V_{DSQ} = V_{DD} - I_D R_D \quad (8-53)$$

However, by using a positive or negative voltage at the gate terminal, shown in Fig.8-46(b) and (c), fixed bias circuit can also set the operating point in depletion mode or enhancement mode region. The bias lines for these circuits are shown in Fig.8-47 as well.

Example 8-15

Fig.8-49(a) shows a fixed bias circuit for DMOSFET and Fig.8-49(b) is its transfer characteristics. Determine the operating point.

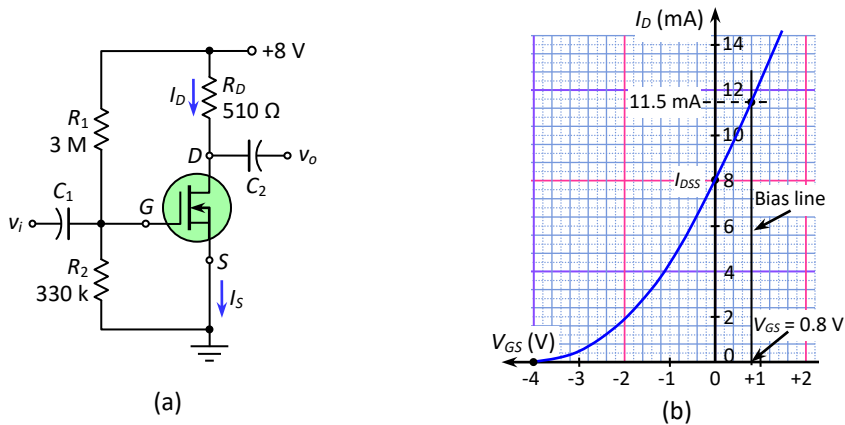


Fig.8-49: (a) Fixed bias circuit for DMOSFET, (b) Its transfer characteristics

Solution:

Using Equ.(8-24),

$$V_G = V_{GS} = \frac{V_{DD} R_1}{R_1 + R_2} = \frac{8 \text{ V} \times 0.33 \text{ M}}{0.33 \text{ M} + 3 \text{ M}} \approx 0.8 \text{ V}$$

On the transfer characteristic curve, a bias line is drawn along $V_{GS} = +0.8 \text{ V}$. This bias line interests the transfer characteristic curve at $I_D = 11.5 \text{ mA}$.

Using Equ.(8-53),

$$V_{DS} = V_{DD} - I_D R_D = 8 \text{ V} - 0.51 \text{ k} \times 11.5 \text{ mA} = 2.14 \text{ V}$$

Comments: This type of biasing circuit always set the operating point on the enhancement region. The drain current is very large. Therefore, V_{DS} is very small, even for a small value of R_D .

8.15.2 Self-Bias

The fixed bias circuit set the operating point on the y -axis and the quiescent drain current becomes I_{DSS} which may be sometimes very large. To reduce the drain current, we have to bias the EMOSFET in the depletion mode region. The self-bias circuit can set the operating point in the depletion mode. This circuit is exactly same as the JFET and all graphical and analytical analysis process will be same. The equation of bias line is

$$V_{GS} = -I_D R_S \quad (8-54)$$

The value of I_{DQ} will be,

$$I_{DQ} = I_{DSS} \left(1 - \frac{|I_D R_S|}{|V_P|} \right)^2 \quad (8-55)$$

$$\text{The value of } V_{DSQ} \text{ will be,} \quad V_{DSQ} = V_{DD} - I_D (R_S + R_D) \quad (8-56)$$

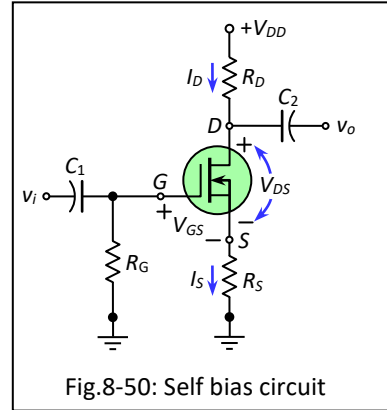


Fig.8-50: Self bias circuit

Example 8-16

The self-bias circuit of Fig.8-51 uses a P-channel depletion MOSFET of $V_P = +5$ V, and $I_{DSS} = 10$ mA. Determine the operating point, and the levels of voltage and currents at three terminals of the circuit using analytical method.

Solution:

The equation of bias line of this self-bias circuit is,

$$V_{GS} = -I_D R_S = -1.2 I_D$$

$$\text{Using Equ.(8-55),} \quad I_D = 12 \text{ mA} \left(1 - \frac{|1.2 I_D|}{|5|} \right)^2$$

Solving the above equation, we get

$$I_D = 2.21 \text{ mA [Ans.]}$$

As the drain current and source current are equal, $I_S = 2.21 \text{ mA [Ans.]}$.

Voltage at drain terminal will be, $V_D = I_D R_D = 2.21 \text{ mA} \times 4.7 \text{ k} = 10.39 \text{ V [Ans.]}$.

Voltage across R_S is, $V_{RS} = I_D R_S = 2.21 \text{ mA} \times 1.2 \text{ k} = 2.65 \text{ V}$

Therefore, $V_{DS} = V_{DD} - I_D R_D - I_D R_S = 20 \text{ V} - 10.39 \text{ V} - 2.56 \text{ V} = 7.05 \text{ V [Ans.]}$

Voltage at source terminal will be, $V_S = V_{DD} - I_D R_S = 20 \text{ V} - 2.65 \text{ V} = 17.44 \text{ V [Ans.]}$

The gate current will be almost zero. Therefore, $V_{RG} = 0 \text{ V}$. As the gate is connected to ground through R_G , and there is no voltage drop across R_G , the gate voltage will be zero. That is, $V_G = 0 \text{ [Ans.]}$

Comments: The analysis of biasing circuit with P-channel FET is similar to N-channel circuits.

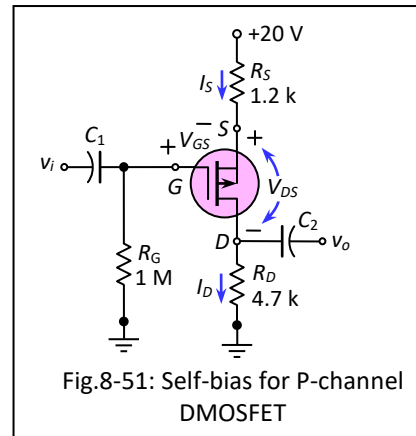


Fig.8-51: Self-bias for P-channel DMOSFET

8.16 Voltage divider Bias

The voltage divider bias circuit for DMOSFET is given in Fig.8-52. Voltage divider bias can set the operating point in depletion mode or in enhancement mode. Larger value of R_S gives better stability of operating point and shifts the operating point to the depletion mode region, while smaller value of R_S provides less stability and shifts the operating point to the enhancement region, as illustrated in Fig.8-51. The analytical and graphical analysis of this circuit will be exactly same as JFET.

Gate voltage will be,

$$V_G = V_{R2} = \frac{V_{DD} R_2}{R_1 + R_2} \quad (8-57)$$

The equation of bias line will be,

$$V_{GS} = V_G - I_D R_S \quad [\because I_S = I_D] \quad (8-58)$$

Using this value of V_{GS} , quiescent drain current can be calculated from the following equation.

$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = I_{DSS} \left(1 - \frac{|I_D R_S - V_G|}{|V_P|}\right)^2 \quad (8-59)$$

The quiescent drain source voltage will be,

$$V_{DSQ} = V_{DD} - I_D (R_D + R_S) \quad (8-60)$$

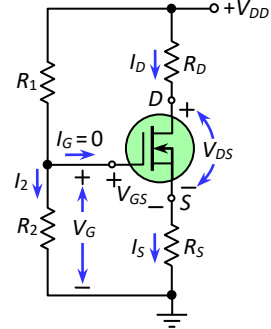


Fig.8-52: Voltage divider bias for DMOSFET

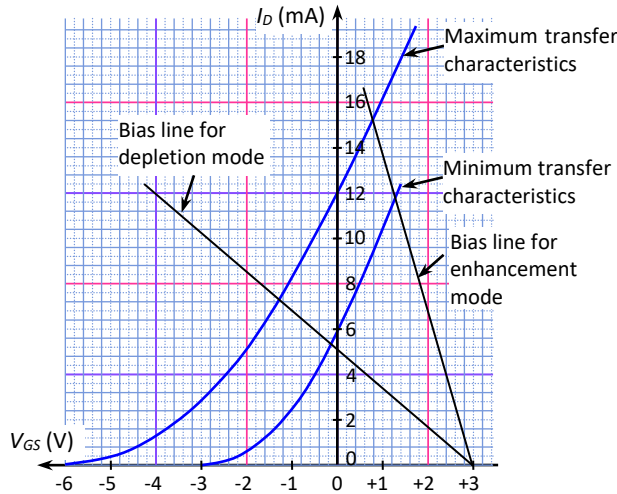


Fig.8-53: Bias line of voltage divider biasing circuit

Example 8-17

Fig.8-52(a) shows a fixed bias circuit for DMOSFET and Fig.8-52(b) is its transfer characteristics. Determine the operating point.

Solution:

Using Equ.(8-57),
$$V_G = \frac{V_{DD}R_2}{R_1 + R_2} = \frac{20\text{ V} \times 1\text{ M}}{1\text{ M} + 4\text{ M}} = 4\text{ V}$$

Bias line of this circuit,
$$V_{GS} = V_G - I_D R_S = 4\text{ V} - 1\text{ k} \times I_D \quad \text{Using Equ.(8-58)}$$

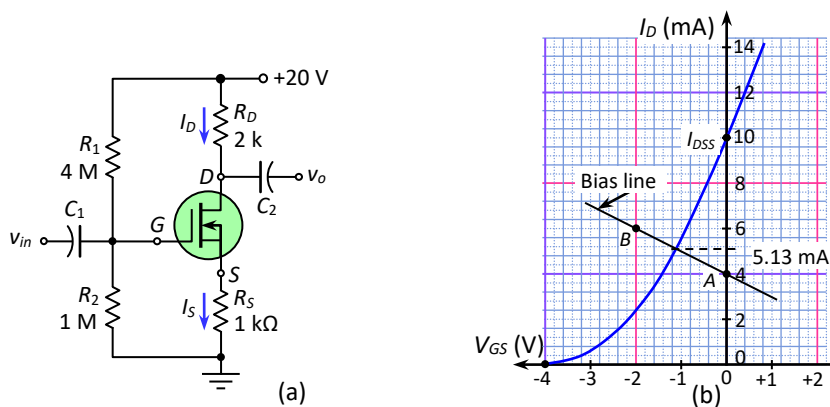


Fig.8-54: (a) Voltage divider bias circuit for DMOSFET, (b) Its transfer characteristics

Let us determine two points on this bias line.

Point A: When $V_{GS} = 0$, $I_D = 4\text{ mA}$

Point B: When $V_{GS} = -2\text{ V}$, $I_D = 6\text{ mA}$

Using these points, the bias line is drawn on the given transfer characteristic curve [Fig.8-54(a)]. The bias line intersects the transfer characteristic curve at $I_D = 5.13\text{ mA}$ [Ans.]

Using Equ.(8-60),
$$V_{DSQ} = 20\text{ V} - 5.13\text{ mA} (1\text{ k} + 2\text{ k}) = 4.61\text{ V} \text{ [Ans.]}$$

Comments: The bias line of a voltage divider biasing circuit can be drawn using any two points, not necessarily from V_G .

Example 8-18

The P-channel DMOSFET of the voltage divider biasing circuit of Fig.8-55(a) has the transfer characteristic curve as given in Fig.8-55(b). Determine the operating point of the circuit.

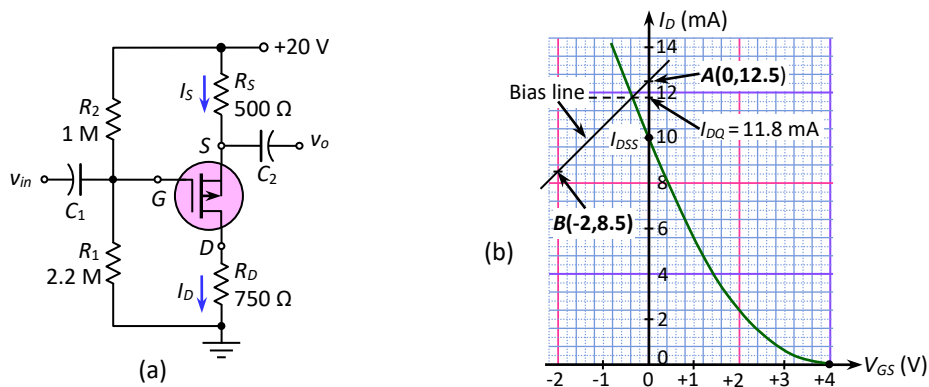


Fig.8-55: (a) Voltage divider bias circuit for DMOSFET, (b) Its transfer characteristics

Solution:

Using Equ.(8-57),

$$V_{R2} = \frac{V_{DD}R_2}{R_1 + R_2} = \frac{20\text{ V} \times 1\text{ M}}{1\text{ M} + 2.2\text{ M}} = 6.25\text{ V}$$

Bias line of the circuit (using Equ.(8-58), $V_{GS} = 0.5\text{ k} \times I_D - 6.25\text{ V}$

Let us determine two points on this bias line.

Point A: When $V_{GS} = 0$, $I_D = 12.5\text{ mA}$

Point B: When $V_{GS} = -2\text{ V}$, $I_D = 8.5\text{ mA}$

Using these points, the bias line is drawn on the given transfer characteristic curve. The bias line intersects the transfer characteristic curve at $I_D = 11.8\text{ mA}$ [Ans.]

Using Equ.(8-60), $V_{DS} = 20\text{ V} - 11.8\text{ mA} (0.5\text{ k} + 0.75\text{ k}) = 5.25\text{ V}$ [Ans.]

Comments: Here the DMOSFET works in the enhancement mode and the quiescent drain current is very high. For this very high level of drain current values of R_D and R_S are very small.

8.17 Biasing Circuits for Enhancement MOSFET

We have already known that the enhancement MOSFET works only when the threshold voltage (V_T) is applied to the gate. For N-channel EMOSFET, gate has to be positively biased and for P-channel EMOSFET, gate has to be negatively biased. Actually, the operation of enhancement MOSFET is similar to BJT. Therefore, all biasing circuits for BJT can be used for EMOSFET. However, as there is no gate current, it is not possible to bias EMOSFETs using a single resistor connected from gate to V_{DD} . In that case the full V_{DD} will be applied to the gate. However, for EMOSFETs we can use fixed bias by applying a voltage less than V_{DD} to the gate. This can be done by applying a voltage divider network as shown in Fig.8-54(a). The voltage divider bias and drain-feedback bias circuits are shown in Fig.8-54(b) and Fig.8-54(c) can also be used of this type of MOSFETs. Moreover, the self-bias circuit cannot be used for EMOSFETs. For the first two circuits, the value of gate voltage will be,

$$V_G = V_{R2} = \frac{V_{DD} R_2}{R_1 + R_2} \quad (8-61)$$

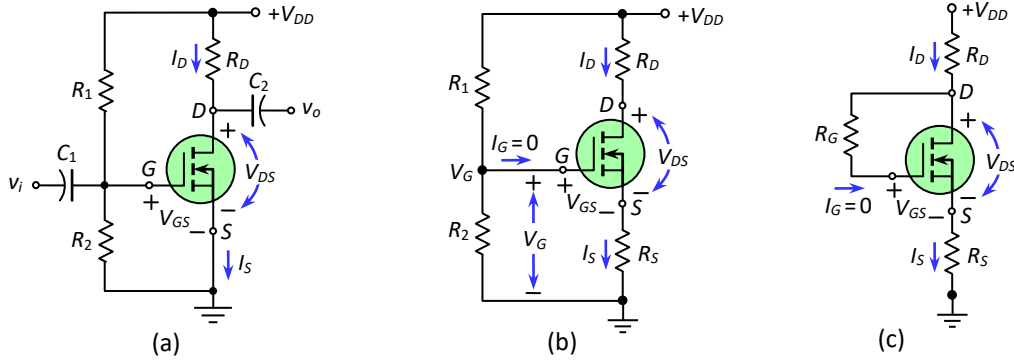


Fig.8-56: Biasing circuits for EMOSFET: (a) Fixed bias, (b) Voltage divider bias, and (c) Drain feedback bias

For the fixed bias circuit the equation of bias line will be,

$$V_{GS} = V_G \quad (8-62)$$

For the voltage divider bias, and the drain feedback bias, the bias lines will be,

For circuit (b):
$$V_{GS} = V_G - I_D R_S \quad (8-63)$$

For circuit (a):
$$V_{GS} = V_{DD} - I_D (R_D + R_S) \quad (8-64)$$

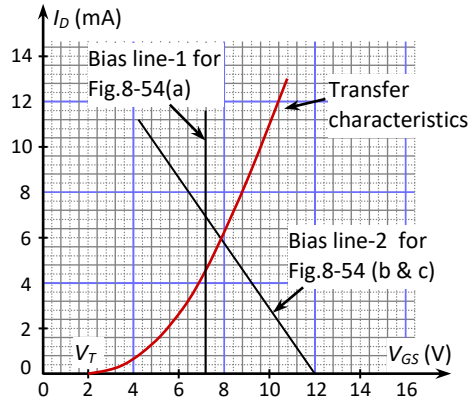


Fig.8-57: Bias lines on the transfer characteristics of EMOSFET

Using these values of V_{GS} in the following equation [equation of transfer characteristic curve], the quiescent drain current can be calculated analytically.

Correct table
$$I_D = k(V_{GS} - V_T)^2 \quad (8-65)$$

As other biasing circuit, the drain-source voltage will be,

$$V_{DSQ} = V_{DD} - I_D (R_D + R_S) \quad (8-66)$$

On the other hand, to determine the operating point graphically, the bias lines have to be plotted on the transfer characteristic curve as shown in Fig.8-55.

Example 8-19

The transfer characteristic curve of the EMOSFET used in the circuit of Fig.8-56(a), is given in Fig.8-56(b). Determine the operating point of the circuit.

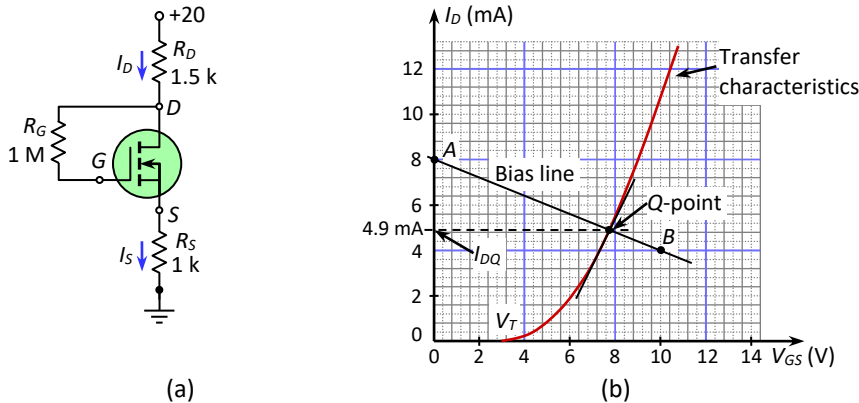


Fig.8-58: (a) Drain feedback bias, and (b) Transfer characteristics curve of the EMOSFET

Solution:

The bias line of this circuit is,

$$V_{GS} = V_{DD} - I_D(R_D + R_S)$$

To draw this bias line, let us determine two point on it.

Point A: When $V_{GS} = 0$,

$$I_D = \frac{V_{DD}}{R_D + R_S} = \frac{20\text{ V}}{1.5\text{ k} + 1\text{ k}} = 8\text{ mA}$$

Point B: When $V_{GS} = 10\text{ V}$,

$$I_D = \frac{V_{DD} - V_{GS}}{R_D + R_S} = \frac{20\text{ V} - 10\text{ V}}{1.5\text{ k} + 1\text{ k}} = 4\text{ mA}$$

Using these two points, the bias line is drawn on the given transfer characteristic curve. This bias line intersects the transfer characteristics at $I_{DQ} = 4.9\text{ mA}$ [Ans.]

Using Equ.(8-66),

$$V_{DSQ} = V_{DD} - I_D(R_D + R_S)$$

or,

$$V_{DSQ} = 20\text{ V} - 4.9\text{ mA}(1.5\text{ k} + 1\text{ k}) = 7.75\text{ V [Ans.]}$$

Comments: The equation of load line (equation of V_{DSQ}) is same as the equation of bias line. So, the load line and bias line will be exactly same. Therefore, the intersecting point of the bias line with the transfer curve will be the operating point, and the value of V_{DSQ} can be determined from this point. **Gain of this circuit should be 1**

8.18 Universal Transfer Characteristics for FET

Due to the time consuming and tedious calculation of analytical method (see Example quadratic solution problem.), FET circuits are generally analyzed using graphical method. To analyze a FET circuits using graphical method, every time we need to draw the transfer characteristic curves of the devices for given values of V_P and I_{DSS} . It takes time to draw the transfer characteristic curve

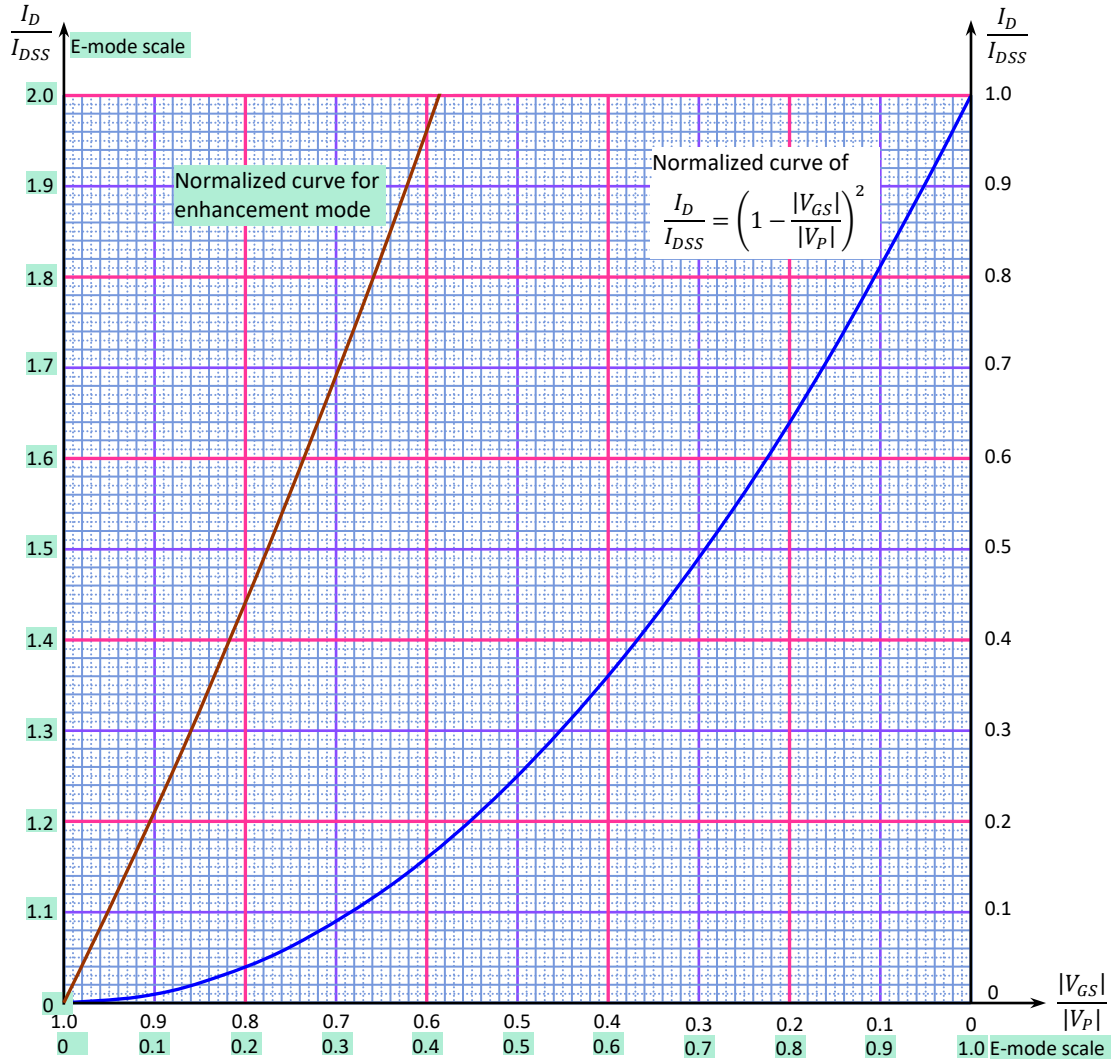


Fig.8-59: Universal transfer characteristic curve

for each circuit. To solve this problem a universal transfer characteristic curve has been developed, that can be used for all circuits of JFETs and DMOSFETs. The universal transfer characteristics for FET is simply a transfer characteristic plotted with normalized drain current, that is I_D/I_{DSS} , and normalized gate-source voltage that V_{GS}/V_P . So, the maximum value of I_D/I_{DSS} , and V_{GS}/V_P will be 1. But in case of enhancement mode operation of DMOSFET the value of I_D/I_{DSS} will be greater

than 1. We know the Shockley's equation is used to draw the transfer characteristics. The normalized form of Shockley's equation will be,

$$\frac{I_D}{I_{DSS}} = \left(1 - \frac{|V_{GS}|}{|V_P|}\right)^2 \quad (8-67)$$

The transfer characteristic curve is drawn considering $\frac{V_{GS}}{V_P}$ as the independent parameter and $\frac{I_D}{I_{DSS}}$ as the dependent parameter. The graph is shown in Fig.8-57. The graph on the right can be used to solve any circuits with JFETs and DMOSFET when works in depletion mode. But the graph on the left side can be used for DMOSFET when works in enhancement mode. The universal transfer characteristic curve cannot be used for EMOSFET circuits.

The y-axis scale on the right, and the upper x-axis scale are used for JFETs and DMOSFET when works in depletion mode. On the other hand, the left y-axis scale, and the lower x-axis scale has to be used for DMOSFET when works in enhancement mode.

The enhancement portion of the graph is drawn in the same method using the following equation,

$$\frac{I_D}{I_{DSS}} = \left(1 + \frac{|V_{GS}|}{|V_P|}\right)^2 \quad (8-68)$$

The application of universal transfer characteristic curve to solve FET circuits, are demonstrated in the following examples.

Example 8-20

Using universal characteristic curve, solve the problem of Example 8-4. For better understanding, the problem is mentioned here again.

"Graphically determine the position of the operating points of an N-channel JFET self-bias circuit like Fig.8-. The JFET used for the circuit has a range of V_P and I_{DSS} . Assume, $V_{DD} = 15\text{ V}$, $R_S = 0.5\text{ k}\Omega$, $R_D = 1\text{ k}\Omega$, $V_{P(min)} = -3\text{ V}$ and $V_{P(max)} = -6\text{ V}$, and $I_{DSS(min)} = 8\text{ mA}$ and $I_{DSS(max)} = 16\text{ mA}$."

Solution:

The equation of bias line of the self-bias circuit is,

$$\text{Using Equ.(8-15),} \quad V_{GS} = -I_D R_S = 0.5\text{ k} \times I_D$$

This bias line passes through the origin (0,0). To determine another point, let $V_{GS} = 2\text{ V}$. So, $I_D = 4\text{ mA}$. Now, normalizing this point using the maximum values, $V_P = 6\text{ V}$ and $I_{DSS} = 12\text{ mA}$, the second point will be,

$$\frac{V_{GS}}{V_P} = \frac{2\text{ V}}{6\text{ V}} = 0.33, \text{ and } \frac{I_D}{I_{DSS}} = \frac{4\text{ mA}}{16\text{ mA}} = 0.25$$

Now, let us put this point on the universal transfer characteristic curve and draw the bias line passing through the origin. As shown in Fig, this bias line intersects the characteristic curve at $I_D/I_{DSS} = 0.325$. Therefore,

$$I_D = 0.325 \times 16\text{ mA} = 5.2\text{ mA [Ans.]}$$

Drain-source voltage will be,

$$V_{DS} = 15\text{ V} - 5.2\text{ mA}(1\text{ k} + 0.5\text{ k}) = 7.2\text{ V [Ans.]}$$

Again, normalizing the points of bias line using $V_{P(\min)} = -3\text{ V}$, and $I_{DSS(\min)} = 8\text{ mA}$ and repeating the above process the minimum Q-point can be determined.

Comments: The maximum Q-point determined using the actual transfer characteristic curve was (7.3 V, 5.2 mA). Now, using the universal transfer characteristic curve we got the maximum Q-point = (7.2 V, 5.2 mA). Difference is very small.

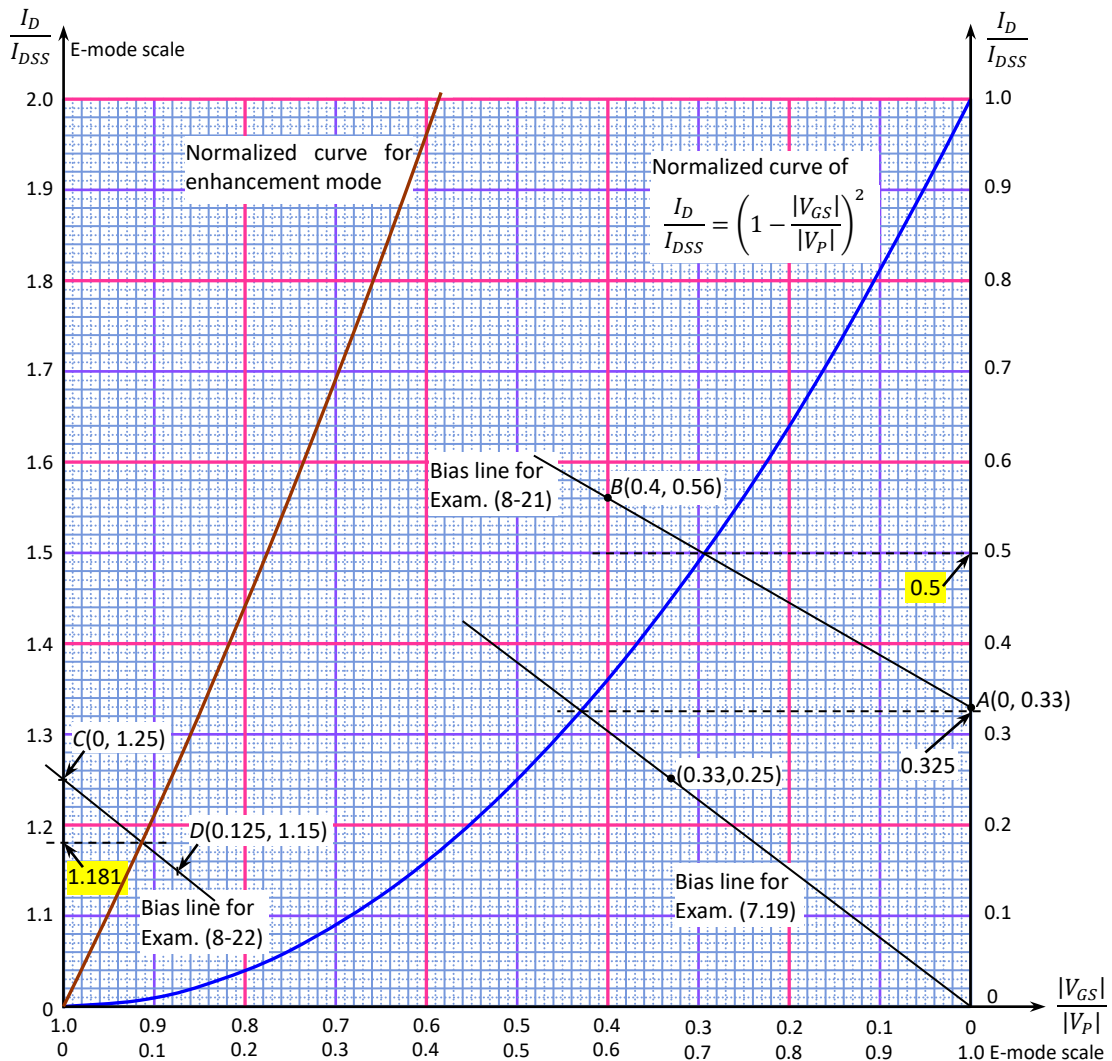


Fig.8-60: Bias line and operating point of different biasing circuits on the Universal Transfer characteristic curve

Example 8-21

Using the universal graph, determine the Q -point of the circuit of Example 8-6. The circuit is redrawn in Fig.8-61.

Solution:

Using Equ.(8-18), $V_G = V_{R2} = \frac{V_{DD} \times R_2}{R_1 + R_2} = \frac{-15 \text{ V} \times 1.1 \text{ M}}{1.1 \text{ M} + 2.2 \text{ M}} = -3 \text{ V}$

Bias line of this circuit is, $V_{GS} = I_D R_S - V_G = 0.75 \text{ k} \times I_D - 3 \text{ V}$ Using Equ.(8-19)

Let us determine two points on the line (values will be normalized using V_P and I_{DSS}).

Point A:

When $V_{GS} = 0$, $I_D = \frac{3 \text{ V}}{0.75 \text{ k}} = 4 \text{ mA}$

The normalized point will be, $(0, 4 \text{ mA}/12 \text{ m}) = (0, 0.33)$

Point B:

When $V_{GS} = 2 \text{ V}$, $0.75 \text{ k} I_D = 2 \text{ V} + 3 \text{ V}$ or, $I_D = \frac{5 \text{ V}}{0.75 \text{ k}} = 6.67 \text{ mA}$

The normalized point will be, $(2 \text{ V}/5 \text{ V}, 6.67 \text{ mA}/12 \text{ m}) = (0.4, 0.56)$

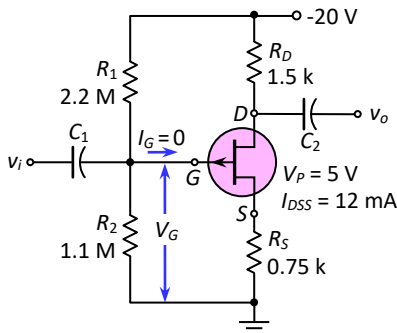


Fig.8-61: Voltage divider biasing circuit for P-channel JFET

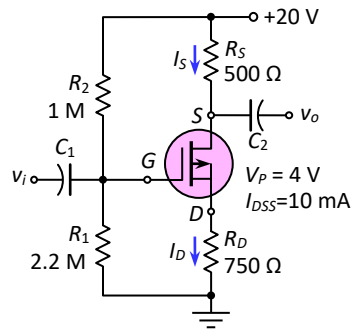


Fig.8-62: Voltage divider biasing circuit for P-channel DMOSFET

Now let us put these two points on the universal characteristic curve and joining them draw the bias line. As shown in the Fig.8-60, the bias line intersects the universal transfer characteristic curve at $I_D/I_{DSS} = 0.5$. Therefore, the value of I_{DQ} will be,

$$I_{DQ} = 0.5 \times 12 \text{ mA} = 6 \text{ mA [Ans.]}$$

Now, V_{DSQ} can be calculated as,

$$V_{DSQ} = V_{DD} - I_{DQ}(R_D + R_S)$$

or,

$$V_{DSQ} = 20 \text{ V} - 6 \text{ mA}(1.5 \text{ k} + 0.75 \text{ k}) \approx 6.5 \text{ V [Ans.]}$$

Comments: The actual result of Example was, $Q\text{-point} = (6.57 \text{ V}, 5.97 \text{ mA})$, now using the universal curve gives, $Q\text{-point} = (6.5 \text{ V}, 6 \text{ mA})$. Therefore, difference is very small.

Example 8-22

Using the universal graph, determine the $Q\text{-point}$ of the depletion MOSFET of the circuit of Example 8-18. The circuit is redrawn in Fig.8-62.

Solution:

Using Equ.(8-18),
$$V_{R2} = \frac{V_{DD}R_2}{R_1 + R_2} = \frac{20 \text{ V} \times 1 \text{ M}}{1 \text{ M} + 2.2 \text{ M}} = 6.25 \text{ V}$$

Using Equ.(8-19),
$$V_{GS} = 0.5 \text{ k} \times I_D - 6.25 \text{ V}$$

Let us determine two points on this bias line.

Point C: When $V_{GS} = 0$, $I_D = 12.5 \text{ mA}$

Point D: When $V_{GS} = -0.5 \text{ V}$, $I_D = 11.5 \text{ mA}$

Normalized values of these points will be,

Point C:
$$\left(0, \frac{12.5 \text{ mA}}{10 \text{ mA}}\right) = (0, 1.25)$$

Point D:
$$\left(\frac{0.5 \text{ V}}{4 \text{ V}}, \frac{11.5 \text{ mA}}{10 \text{ mA}}\right) = (0.125, 1.15)$$

Now using these two points the bias line is drawn on the universal characteristic curve ($C\text{-}D$ line). As the device works in enhancement mode, the left curve and left y -axis scale is used. As shown in the Fig.8-60, the bias line intersects the universal transfer characteristic curve at $I_D/I_{DSS} = 1.181$. Therefore, the value of I_{DQ} will be,

$$I_{DQ} = 1.181 \times 10 \text{ mA} = 11.81 \text{ mA [Ans.]}$$

Now, V_{DSQ} can be calculated as,

$$V_{DSQ} = 20 \text{ V} - 11.81 \text{ mA} (0.5 \text{ k} + 0.75 \text{ k}) = 5.24 \text{ V [Ans.]}$$

Comments: The result using the actual transfer characteristic curve was $(5.25 \text{ V}, 11.8 \text{ mA})$, and using universal transfer characteristic curve the result is $(5.24 \text{ V}, 11.8 \text{ mA})$, almost same. Therefore, we can say that the universal transfer characteristic curve can also be used to analyze DMOSFETs when they work in enhancement mode. Remember, to draw the bias line for enhancement mode, both points must have current levels greater than I_{DSS} . Otherwise this graph cannot be used.

8.19 Trouble Shooting FET Biasing Circuits

Already, we have discussed about different biasing circuits of FETs. Biasing process of different FETs in three configurations (common source, common drain, and common gate) have been covered. Before implementing these circuits, we must test the FET that will be used in the circuit. The testing

procedure of FETs has been discussed in Chapter 6. In laboratory we implement a circuit in bread board. But after implementing, a circuit may not work properly or may not work at all. In this section, we will discuss how to find out the reasons of malfunctioning (trouble shooting) of three biasing circuit (fixed or gate bias, self-bias, and voltage divider bias) with JFET. But the trouble shooting procedure of circuits with MOSFET will be almost similar. Here on-line trouble shooting process (voltage measurement at different terminals) will be discussed.

8.19.1 Fixed Bias or Gate Bias

First of all, we have to check the voltage at the positive and negative terminals of the circuit using a multimeter (in appropriate voltage range). If we do not get any voltage reading, it means that power is not coming from the power supply (PS). We have to check the PS switch, connecting wires, connections to bread board. Thus, we must ensure the voltage at the terminals of the circuit. Then, we have to measure the voltage levels at three terminals of the fixed bias circuit. Normally, the gate should be negative with respect to ground, and the drain voltage should be anywhere in the range of perhaps 3 V to $(V_{DD} - 3)$ V. If meter reading shows that $V_D \approx V_{DD}$ as shown in Fig.8-61(a), the FET terminals might be open-circuited, or drain resistor R_D might be short. However, when drain voltage is approximately equal to zero, as indicated in Fig.8-61(b), drain and source terminals might be shorted or R_G and R_D might be interchanged. Another possibility is that R_D is open-circuited.

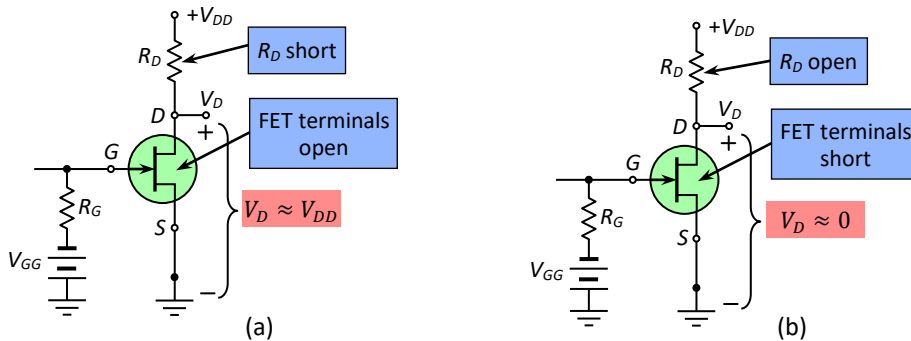


Fig.8-63: Trouble shooting fixed bias circuit for JFET: (a) JFET open, and (b) JFET short

8.19.2 Self-Bias Circuit

Trouble shooting process of self-bias is almost similar to that of fixed bias circuit. First of all, we have to ensure the required amount of voltage at the positive and negative terminals of the biasing circuit as described for fixed bias circuit. Then, we have to measure different voltage levels at three terminals of the FET. Possible reasons for incorrect voltage readings in a self-bias circuit is illustrated in Fig.8-62. If meter reading shows, $V_D \approx V_{DD}$ [Fig.8-62(a)], R_S or drain and source terminals might be open. Other reasons are, R_D might be short, or R_G and R_S might be interchanged. On the other hand, if meter reading shows $V_S = V_D$ [Fig.8-62(b)], the drain and source terminal could be shorted or R_G and R_S might be interchanged.

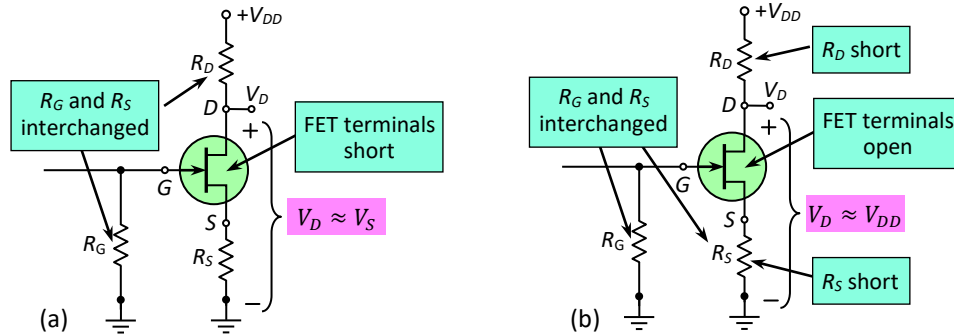


Fig.8-64: Trouble shooting self bias circuit for JFET: (a) JFET open, and (b) JFET short

8.19.3 Voltage Divider Bias

For this circuit, trouble shooting process is similar to the previous two circuits. At first, we have to ensure the required amount of voltage at the positive and negative terminals of the biasing circuit. Then, we have to measure different voltage levels at three terminals of the FET. Possible reasons for incorrect voltage readings in a voltage divider bias circuit is illustrated in Fig.8-63. If voltage measurement shows that V_D is almost equal to V_{DD} , then the drain and source terminal, or R_S or R_1 might be open-circuited. Alternately, R_D or R_2 might be short circuited. On the other hand, if meter reading shown that $V_S \approx V_D$ [as given in Fig.8-63(b)], drain and source terminals are shorted. Some other reasons are, R_2 might be open circuited, or R_1 , and R_2 might be interchanged.

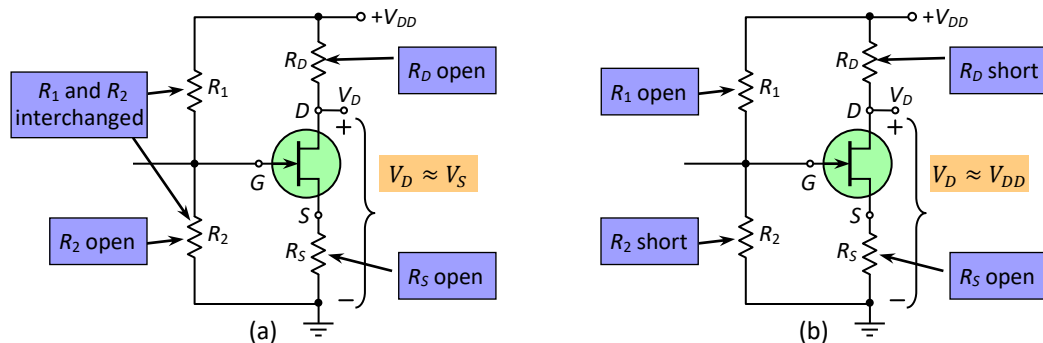


Fig.8-65: Trouble shooting self bias circuit for JFET: (a) JFET open, and (b) JFET short

Miscellaneous FET Biasing Circuits

So far, we have discussed mainly four types of biasing circuits for FETs. These are: gate bias or fixed bias, self-bias, voltage divider bias, and drain feedback bias. Here some other biasing circuits are described. These biasing circuits are rarely used.

8.19.4 Voltage Divider Bias with Drain Feedback

The voltage divider bias with drain feedback is shown in Fig.8-64. In normal voltage divider bias V_{DD} is divided by two resistors R_1 and R_2 . But here, V_D is divided by the voltage divider network (R_1 and

R_2). The main advantage of this circuit is: in addition to R_S , R_D also provides stability of DC operating point. If (due to device parameters change or any other reasons) I_D increases, V_D will decrease ($\because V_D = V_{DD} - I_D R_D$), consequently V_G will also decrease. But we know, $V_{GS} = -(V_S - V_G)$. This negative voltage controls the drain current. Now, as V_G has decreased, V_{GS} will be more negative and the drain current will be reduced back. Similarly, if I_D decrease, V_D will increase, consequently V_G will also increase. So, V_{GS} will be less negative and the drain current will be increased back. In this way, R_D provides negative feedback. Here, R_S also produces negative feedback in same process as described earlier. The two feedback effects (V_{RD} , and V_{RS} changes) tend to keep $I_{D(min)}$, and $I_{D(max)}$ (hence, the operating points) closer than in ordinary voltage divider bias circuits.

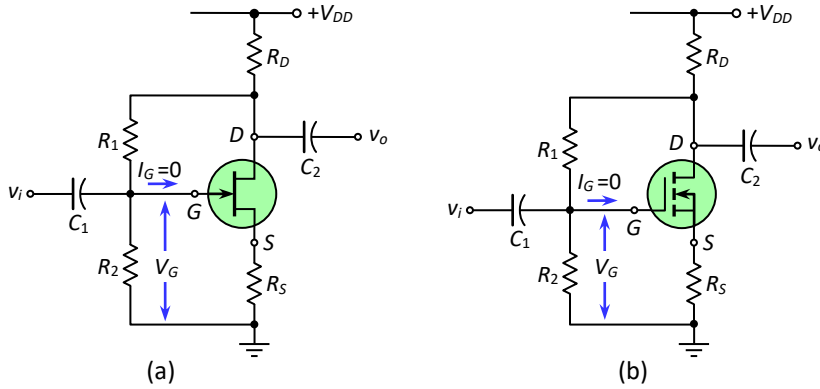


Fig.8-66: Voltage divider bias circuits with drain feedback for (a) JFET, and (b) EMOSFET

One major disadvantage of this current is reduced AC voltage gain. As, R_D is not bypassed using any capacitor, it produces negative feedback for AC signal as well. For the negative feedback AC gain will be lower than normal voltage divider bias circuit.

The gate voltage of this circuit will be,

$$V_G = \frac{V_D R_2}{R_1 + R_2} = \frac{(V_{DD} - I_D R_D) R_2}{R_1 + R_2} \quad (8-69)$$

The equation of bias line will be,

$$V_{GS} = V_G - I_D R_S = \frac{(V_{DD} - I_D R_D) R_2}{R_1 + R_2} - I_D R_S$$

or,

$$V_{GS} = \frac{V_{DD} R_2 - I_D (R_D R_2 + R_S R_1 + R_S R_2)}{R_1 + R_2} \quad (8-70)$$

Now, using this value in Shockley's equation, the value of I_D can be calculated. The equation is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = I_{DSS} \left[1 - \frac{V_{DD} R_2 - I_D (R_D R_2 + R_S R_1 + R_S R_2)}{V_P (R_1 + R_2)}\right]^2 \quad (8-71)$$

This equation contains only one variable I_D . Therefore, it can be solved for I_D .

As the gate current is zero, the drain-source voltage can be calculated as before,

$$V_{DS} = V_{DD} - I_D(R_D + R_S) \quad (8-72)$$

Alternately, drawing the bias line of Equ., on the transfer characteristic curve, the value of I_D can be determined graphically.

Example 8-23

The JFET used in the following circuit (Fig.8-67), has $V_P = 7\text{ V}$, and $I_{DSS} = 9\text{ mA}$. Graphically determine the operating point of the biasing circuit.

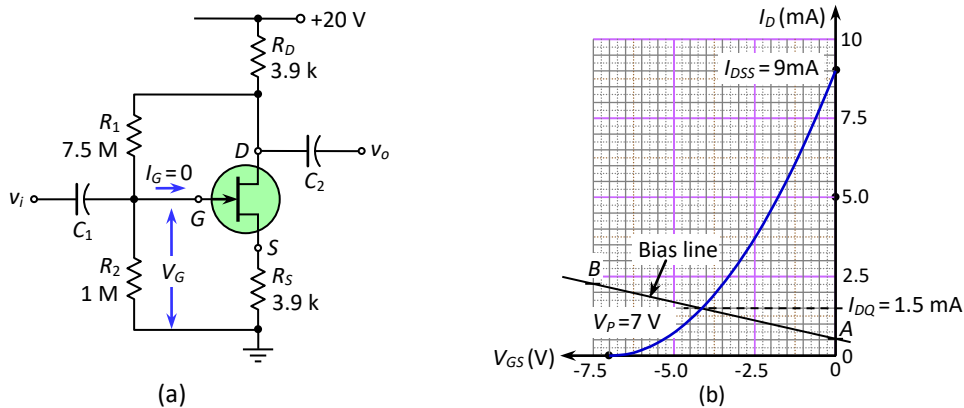


Fig.8-67: Voltage divider bias circuits with drain feedback for (a) JFET, and (b) EMOSFET

Solution:

Using $V_P = 7\text{ V}$, and $I_{DSS} = 9\text{ mA}$, the transfer characteristic curve of the device is drawn as shown in Fig.8-56(a).

$$\text{or,} \quad V_{GS} = \frac{V_{DD}R_2 - I_D(R_DR_2 + R_SR_1 + R_SR_2)}{R_1 + R_2}$$

The equation of bias line will be,

$$V_{GS} = \frac{20\text{V} \times 1\text{ M} - I_D(3.9\text{ k} \times 1\text{ M} + 3.9\text{ k} \times 1\text{ M} + 3.9\text{ k} \times 7.5\text{ M})}{1\text{ M} + 7.5\text{ M}}$$

$$\text{or,} \quad V_{GS} = \frac{20 - I_D(3.9 + 3.9 + 3.9 \times 7.5)}{1 + 7.5} = \frac{20 - 37.05I_D}{8.5}$$

Let us determine two points on this bias line.

Point A: When $V_{GS} = 0$,
$$I_D = \frac{20}{37.05} \approx 0.54\text{ mA}$$

Point B: When $V_{GS} = -7.5\text{ V}$,
$$-7.5 = \frac{20 - 37.05I_D}{8.5}$$

Using these points, the bias line is drawn on the transfer characteristic curve. The bias line intersects the transfer characteristic curve at $I_D = 1.5\text{ mA}$ [Ans.]

Using Equ.(8-82), $V_{DS} = 20 \text{ V} - 1.5 \text{ mA} (3.9 \text{ k} + 3.9 \text{ k}) = 8.3 \text{ V}$ [Ans.]

Comments: Due to the large values of R_D , and R_S the quiescent drain current is very small.

8.20 Biasing Circuit with Constant Current Source

In different biasing circuits, we have found that the stability of operating point depends on the slope of the bias line. The lower the slope of bias line the higher the stability of the operating point which produces closest levels of $I_{D(min)}$, and $I_{D(max)}$. To reduce the slope of the bias line, we have to use larger source resistance. Large source resistance results in lower value of drain current, lower value of drain resistance and hence lower voltage gain. But, using a constant current source, the bias line can be made horizontal to x -axis, that is, zero slope. This is equivalent to use an infinite resistance (r_o of BJT) at the source terminal. This infinite source resistance will not reduce the drain current. Moreover, drain current can be set at any level, by adjusting the constant current source.

Among different arrangements, Fig.8-66 shows two biasing circuits with constant current source. Here the BJT (Q_1) is used as the constant current source. The circuit in Fig.8-66(a) uses a voltage divider network to divide V_{DD} and produce the base voltage V_B . On the other hand, the circuit in Fig.8-66(b), uses $-V_{EE}$ to bias the base of the transistor. In both cases, the gate is connected to base via R_G , and the source is directly connected to the collector of BJT. Therefore, in both cases, the gate-source voltage will be,

$$V_{GS} = -V_{CB} = V_B - V_S \quad (8-73)$$

The levels of drain currents will be determined by this value of V_{GS} . Once V_{GS} is known, analysis of the FET current will be same as described before.

As described in Chapter 4, for the first circuit the value of base current will be (Equ.),

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \quad (8-74)$$

And the collector current which is also the source current will be,

$$I_C = I_E = I_S = \beta I_B + (\beta + 1) I_{CBO} \approx \beta I_B \quad (8-75)$$

For the second circuit, the base current will be,

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E} \quad (8-76)$$

Similarly, the collector current and the source current will be,

$$I_C = I_E = I_S = \beta I_B + (\beta + 1) I_{CBO} \approx \beta I_B \quad (8-77)$$

Now look at the circuits. I_S , and I_C flows in series and will be equal. But I_S is determined by V_{GS} , while I_C is determined by V_{BE} . The values of these two currents must be same. Otherwise, the minimum one will be the actual current in the circuit.

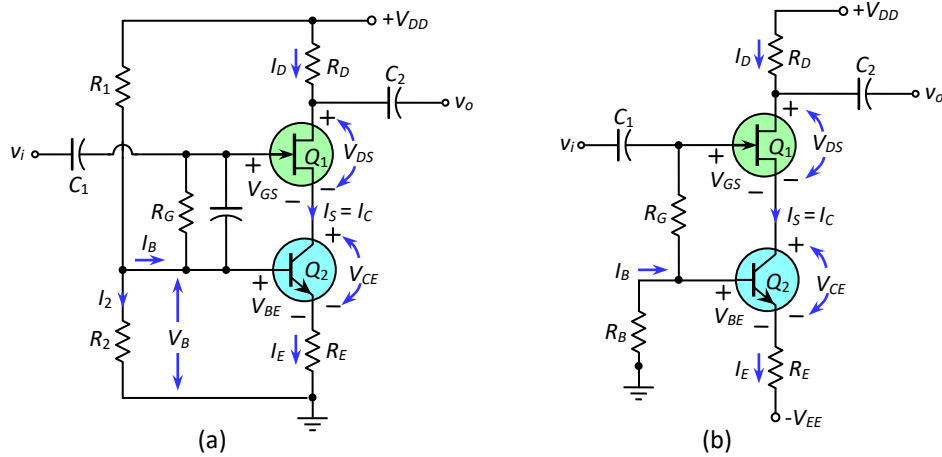


Fig.8-68: Biasing circuit with constant current source: (a) With single polarity supply, and (b) Dual polarity supply

Regarding the design of these circuits, the same methods as described for FET and BJT can be used. The bypass capacitor across R_G will ensure the same AC signal at the gate and base. So that I_D , and I_C can increase or decrease simultaneously. To get better AC voltage gain, a bypass capacitor has to be used at the source terminals of the FETs.

Example 8-24

Design a voltage divider bias circuit with constant current source as shown in Fig.8-67(a). Assume, $V_{DD} = 24 \text{ V}$, $I_{D(max)} = 3 \text{ mA}$, and $V_{DS(min)} = 7 \text{ V}$. The FET has $V_P = -6 \text{ V}$, and $I_{DSS} = 10 \text{ mA}$. The transistor has $\beta = 100$.

Solution:

Let $I_2 = I_D/10 = 3 \text{ mA} / 10 = 0.3 \text{ mA}$, and $V_E = 5 \text{ V}$. So, $V_B = V_E + 0.7 \text{ V} = 5.7 \text{ V}$.

$$\therefore R_E = \frac{V_E}{I_E} = \frac{5 \text{ V}}{3 \text{ mA}} = 1.67 \text{ k}, \text{ use standard } 1.6 \text{ k}$$

$$\therefore R_2 = \frac{V_B}{I_2} = \frac{5.7 \text{ V}}{0.3 \text{ mA}} = 19 \text{ k}, \text{ use standard } 20 \text{ k}$$

$$\text{Base current will be} \quad I_B = \frac{I_C}{\beta} = \frac{3 \text{ mA}}{100} = 0.03 \text{ mA}$$

Current through R_1 will be,

$$\text{Base current will be,} \quad I_1 = I_2 + I_B = 0.3 \text{ mA} + 0.03 \text{ mA} = 0.33 \text{ mA}$$

$$\therefore R_1 = \frac{(V_{DD} - V_B)}{I_1} = \frac{24 \text{ V} - 5.7 \text{ V}}{0.33 \text{ mA}} = 55.5 \text{ k}, \text{ use standard } 56 \text{ k}$$

Using Equ.(8-37),
$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) = -6 \text{ V} \left(1 - \sqrt{\frac{3 \text{ mA}}{10 \text{ mA}}} \right) = -2.71 \text{ V}$$

Using Equ.(8-73),

$$V_S = V_B - V_{GS} = 5.7 \text{ V} - (-2.71 \text{ V}) = 8.41 \text{ V}$$

$$\therefore V_{RD} = V_{DD} - V_S - V_{DS(min)} = 24 \text{ V} - 8.41 \text{ V} - 7 \text{ V}$$

$$\text{or, } V_{RD} = 8.59 \text{ V}$$

$$\therefore R_D = \frac{V_{RD}}{I_D} = \frac{8.59 \text{ V}}{3 \text{ mA}} = 2.86 \text{ k}, \text{ use a standard } 2.7 \text{ k}$$

$$\therefore V_{RD} = 2.7 \text{ k} \times 3 \text{ mA} = 8.1 \text{ V, Let } V_{DS(min)} \approx 7.5 \text{ V}$$

The value of R_G will usually be $1 \text{ M}\Omega$.

Answer: The designed circuit is shown in Fig.8-69.

Comments: Instead of using the calculated values, nearest standard values have been used for different resistors. So, actual voltage levels and current levels will be slide different.

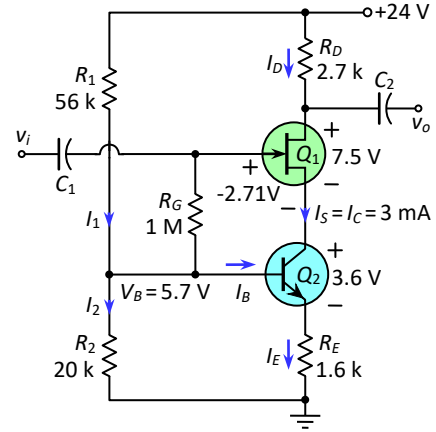


Fig.8-69: Designed circuit for Example 8-24

8.21 FET as Constant Current Source

In Chapter 4, we have discussed BJT as a constant current source. Similarly, FET can be used as a constant current source. A JFET can also be used to supply constant current to any type of load (constant or variable resistance). The circuit arrangements are shown in Fig.8-70. Here, the gate is directly

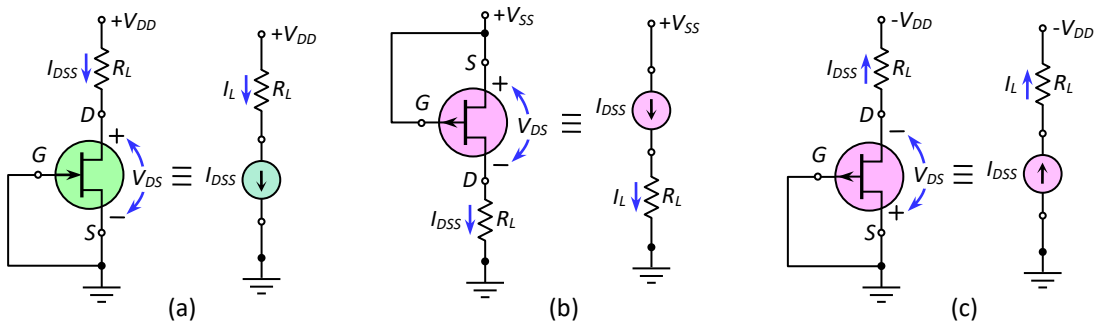


Fig.8-70: Constant source using (a) N-channel JFET, (b) P-channel JFET (Positive voltage), and (c) P-channel JFET (Positive voltage)

connected to ground and no resistance is used in the source terminal. The load resistance $R_L (= R_D)$ has to be connected in the drain terminal. The drain current (I_D) will be the load current and its value will be I_{DSS} . The P-channel current source can be constructed using both negative voltage [Fig.8-70(b)] or negative voltage [Fig.8-68(c)]. To keep the current constant irrespective of the

values of R_L , the FET must operate in its pinch-off region. The condition for pinch-off is, $V_{DS} > V_P - V_{GS}$. Since, $V_{GS} = 0$ in this case, the condition reduces to,

$$V_{DS} > V_P \quad (8-78)$$

The constant current produced by the JFET is, then, $I_D = I_{DSS}$, because it is the drain current in the pinch-off region when $V_{GS} = 0$.

So long as the JFET works in its pinch-off region (active region), the line corresponding $V_{GS} = 0$ is almost horizontal, meaning that the same current flows through the drain (load) regardless the value of V_{DS} as shown in Fig.8-71. However, in reality the line rises slightly to the right, which means that the current source is not ideal. Of course, it is not possible to develop an ideal current source. The internal resistance of an ideal current source is infinite. But here the internal resistance of the JFET current source will be $r_d = \Delta V_{DS} / \Delta I_D$. Obviously, to be an ideal constant current source $\Delta I_D = 0$, and hence, $r_d = \infty$.

In this circuits (Fig.8-70), the value of constant current is I_{DSS} which is a device property (solely depends on JFET). There is no way to change this device property. However, we can decrease the current level below I_{DSS} by applying appropriate reverse bias in the gate terminal as shown in Fig.8-72.

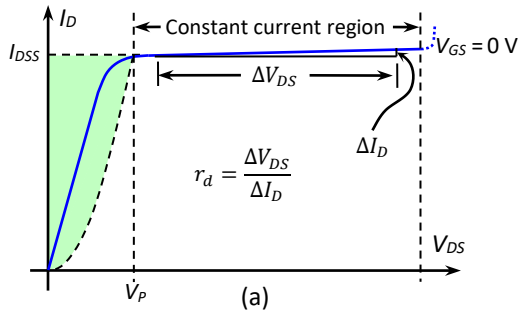


Fig.8-71: I - V characteristics of the current source

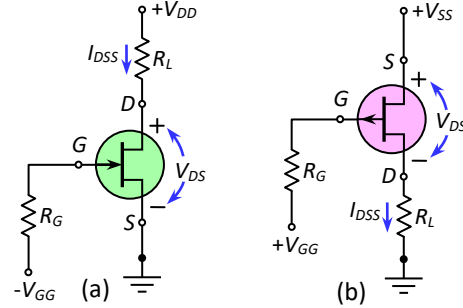


Fig.8-72: Constant current sources with $I_D < I_{DSS}$: (a) N-channel JFET, and (b) P-channel JFET

8.21.1 Constant Current Sink or Current Mirror Circuit

Fig.8-73 shows a current mirror circuit. The circuit is made using two identical enhancement MOSFET. The current in the second MOSFET will be exactly same as in the first MOSFET. The value of current can be changed by changing the variable resistance (R_D).

According to Ohm's law,

$$I_{D1} = \frac{V_{DD} - V_{DS1}}{R_D} = \frac{V_{DD} - V_{GS1}}{R_D} \quad (8-79)$$

But, according to the function of EMOSFET,

$$I_{D1} = k(V_{GS1} - V_T)^2 \quad (8-80)$$

Combining these two equations,

$$\frac{V_{DD} - V_{GS1}}{R_D} = k(V_{GS1} - V_T)^2 \quad (8-81)$$

In this equation, the only variable is V_{GS1} . So, solving this equation we will get the value of V_{GS1} , and putting this value into Equ(*), the drain current of Q_1 can be calculated.

The gate voltage of the second EMOSFET will be same as V_{GS1} . As the two FETs are identical, so the drain current of Q_2 will be same as I_{D1} . That is $I_L = I_{D1}$.

Now, if we change the value of R_D of the first MOSFET, according to Equ, V_{GS1} will be change. Consequently, V_{GS2} will also change and I_L will change. In fact, I_{D1} and I_L will be same.

Thus, by changing the current in first circuit the current in the second circuit can be changed. For this reason, the second circuit is called a current mirror circuit. As this circuit will sink a constant current through the load resistance, it is also called a **constant current sink** circuit.

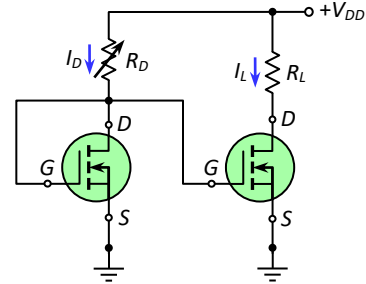


Fig.8-73: Constant current sink or current mirror circuit

Example 8-25

Assume the EMOSFETs used in Fig.8-73 have $V_T = 2$ V, and $k = 0.15$ mA/V². Considering $V_{DD} = 12$ V, and $R_D = 1$ k, calculate the value of I_L .

Solution:

Using Equ.(8-81),
$$\frac{V_{DD} - V_{GS1}}{R_D} = k(V_{GS1} - V_T)^2$$

Putting the values
$$\frac{12 - V_{GS1}}{1 \times 10^3} = 1.5 \times 10^{-3} (V_{GS1} - 2)^2$$

or
$$12 - V_{GS1} = 1.5(V_{GS1} - 2)^2$$

or
$$12 - V_{GS1} = 1.5(V_{GS1}^2 - 4V_{GS1} + 4)$$

or
$$12 - V_{GS1} = 1.5V_{GS1}^2 - 6V_{GS1} + 6$$

or
$$1.5V_{GS1}^2 - 5V_{GS1} - 6 = 0$$

or
$$V_{GS1} = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1.5)(-6)}}{2 \times 1.5} = \frac{5 \pm \sqrt{25 + 36}}{3}$$

∴
$$V_{GS1} = +4.27 \text{ or } -0.94$$

For N-channel EMOSFET, V_{GS} must be positive.

∴
$$V_{GS1} = +4.27 \text{ V}$$

Putting this value in Equ.(8-79),
$$I_{D1} = \frac{12 \text{ V} - 4.27 \text{ V}}{1 \text{ k}} = 7.73 \text{ mA}$$

Being the current mirror, $I_L = 7.73 \text{ mA}$ [Ans.]

Comments: Current mirror circuits are used in Integrated Circuits (ICs). They are further discussed in Chapter 14.

8.22 FET as a Voltage Controlled Resistor

As we have seen that the output characteristics of JFET and DMOSFET, have a region called voltage-controlled-resistance region or Ohmic region. The output characteristics of a JFET is redrawn in Fig. From this characteristic curve, we find that, the device works in the Ohmic region, when $V_{DS} > V_P - V_{GS}$. In this region, the output characteristics are almost linear, which means that current I_D increases with the increase of V_{DS} . It is the property of a resistance. Actually here, the channel resistance becomes effective and is given by,

$$R_D = \frac{V_{DS}}{I_D} \quad (8-82)$$

For a fixed value of V_{DS} (below $V_P - V_{GS}$), the resistance R_D decreases with the increase in V_{GS} . To understand this, let us draw a vertical line along $V_{DS} = 0.75 \text{ V}$. This vertical line intersects the output characteristic curves as shown in Fig. The intersecting points determine the values of I_D for the same value of $V_{DS} = 0.75 \text{ V}$. For $V_{GS} = -2 \text{ V}$, current is $\approx 2 \text{ mA}$, for $V_{GS} = -1 \text{ V}$ the current is $\approx 3 \text{ mA}$ and for $V_{GS} = 0 \text{ V}$ the value of current is $\approx 4.2 \text{ mA}$. Therefore, for $V_{GS} = -2 \text{ V}$, $R_D = V_{DS}/I_D = 0.75 \text{ V}/2 \text{ mA} = 375 \Omega$, for $V_{GS} = -1 \text{ V}$, $R_D = 0.75 \text{ V}/3 \text{ mA} = 250 \Omega$, and for $V_{GS} = 0 \text{ V}$, $R_D = 0.75 \text{ V}/4.2 \text{ mA} = 179 \Omega$. Therefore, we can say that when JFET works in Ohmic region, the drain resistance can be decreased by increasing gate voltage (V_{GS}) and vice-versa. For this reason, this region of output characteristics is called the voltage-controlled-resistance region.

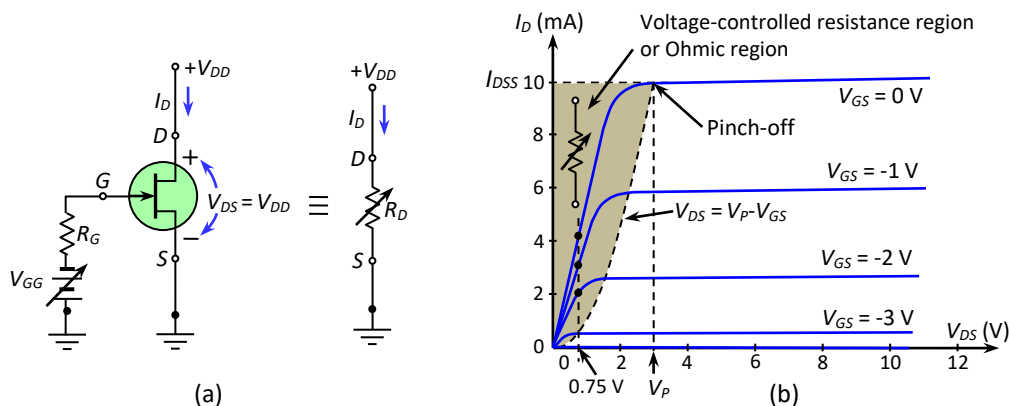


Fig.8-74: JFET used as voltage controlled resistance: (a) Circuit diagrams, and (b) Characteristics

8.23 Switching Circuits using FET

In Chapter 4, we have discussed how to use BJT as a switch. For low power controlling, BJT is a very good switch, because of its very small ON state voltage drop and moderate switching speed. But

the main problem of BJT switch is that they produce wastage of power in two ways. First the collector-emitter saturation voltage ($V_{CE(sat)}$) and the second one is the base current. Being a current-controlled device, it draws base current in ON state which results in wastage of power. As power-BJT has very low value of β , in case of high-power switching, the loss is very high. ON the other hand, being a voltage-controlled device, FET does not draw any input current. **Moreover, the ON-state voltage drop ($V_{DS(ON)}$) of FET is comparatively lower than that of BJT.**

Though, theoretically, all FETs can be used as switch, JFETs and EMOSFETs are generally used. For low power switching JFETs are used and for high power switch EMOSFETs (VMOS) are used.

8.23.1 JFET as Switch

Like the BJT, when used as a switch, the load is connected to the drain terminal, the source is directly connected to ground for N-channel or to V_{DD} for P-channel device as shown in Fig.8-76 and Fig.8-77. The controlling voltage is applied to the gate terminal. In N-channel JFET, zero voltage at the gate terminal drives the switch ON, and a negative voltage greater than $|V_P|$ will turn the switch OFF. On the other hand, in case of P-channel JFET, zero voltage at the gate terminal drives the switch OFF, and a positive voltage greater than $|V_P|$ will turn the switch ON. In ON state, the voltage-drop across drain and source ($V_{DS(ON)} = I_D r_{DS(ON)}$) is almost zero. So, the whole V_{DD} will be dropped across the load. The ON state drain current will be controlled by the load, but it will never be greater than I_{DSS} . Therefore, to design a switch, a suitable the JFET has to be selected with $I_{DSS} > I_L$, and $V_{DS(max)} > V_L$. The supply voltage, V_{DD} of the switching circuit should be equal to load voltage V_L .

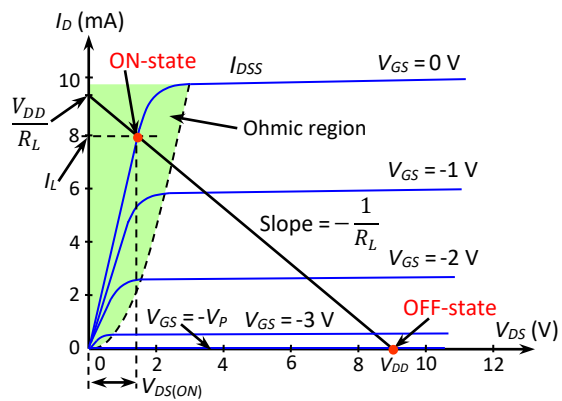


Fig.8-75: Operating point of a JFET switch

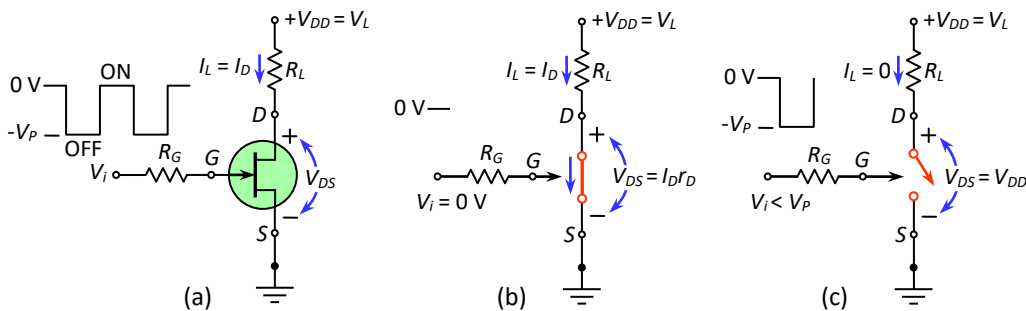


Fig.8-76: N-channel JFET used as switch (a) Switching circuit, (b) ON state equivalent, and (c) OFF state equivalent

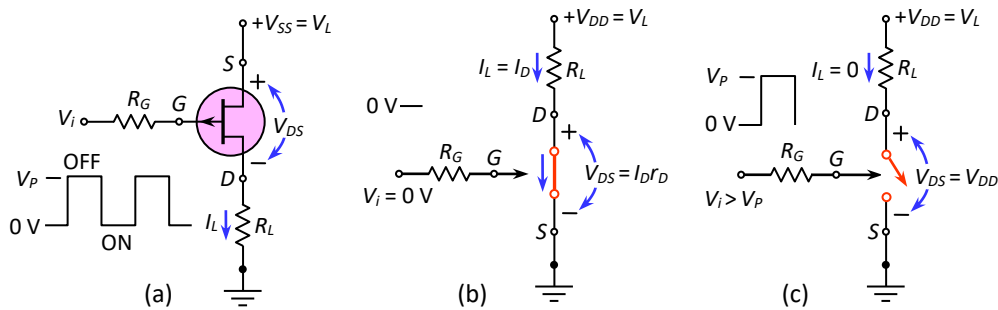


Fig.8-77: P-channel JFET used as switch (a) Switching circuit, (b) ON state equivalent, and (c) OFF state equivalent

However, if V_{DD} of a switching circuit is greater than load voltage (V_L), resistance has to be used in series with the load to drop the additional voltage. This situation is further illustrated in Example 8- As the input voltage is directly connected to the gate, this type of switch is called direct-coupled switch. Here, R_G has no effect on the current operation. It is used for safety purpose. For example, in any case, if the FET becomes short, R_G protect the driving circuit from being shorted. As the gate current is very low, generally large values for R_G is selected. Typically, 10 k to 100 k can be used.

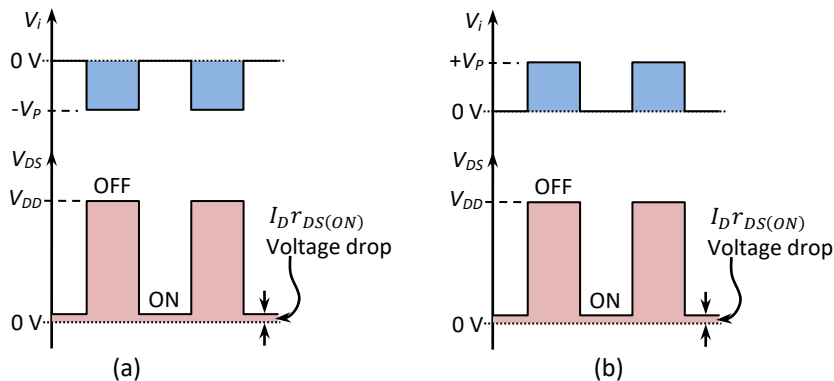


Fig.8-28: (a) Input and output voltage (V_{DS}) of N-channel JFET and (b) P-channel JFET

Example 8-26

Generally, a normal LED requires 10 mA to 20 mA to glow. The forward voltage drop of LED depends on color. For example, a red LED needs 1.2 V, while a blue LED needs 3.6 V. Assuming $V_{DD} = 12$ V, design a switching circuit using JFET to drive a red LED.

Solution:

To design the switching circuit, we need an N-channel JFET with $I_{DSS} \geq 20$ mA. Let us use J111, an N-channel JFET with $I_{DSS(min)} = 20$ mA, and $V_{P(max)} = -10$ V.

As the supply voltage is greater than the load voltage, we have to use a resistor in series with the LED as shown in Fig.8-79. Considering 10 mA driving current in ON state, the value of the resistor will be,

$$R_D = \frac{V_{DD} - V_L}{I_L} = \frac{12 \text{ V} - 1.2 \text{ V}}{10 \text{ mA}} = 1.08 \text{ k}, \text{ use standard } 1 \text{ k}$$

As the gate current is very small, we can use a very large value for R_G . Let $R_G = 100 \text{ k}\Omega$. For $V_i = 0 \text{ V}$, the LED will be ON, and for $V_i = -10 \text{ V}$, the LED will be OFF.

Answer: The designed circuit is shown in Fig.8-79.

Comments: To ensure the OFF state, input voltage should be less than or equal to -10 V .

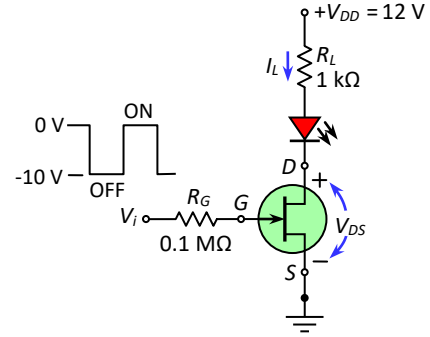


Fig.8-79: Switching circuit designed for Example 8-26

8.24 MOSFET as a Switch

To control (ON/OFF) a very large power load (large V_L and/or large I_L), EMOSFETs are used. N-channel EMOSFETs are suitable for easily controlling the non-grounded load, while P-channel EMOSFETs are used to control ground-connected loads. Both the circuits are shown in Fig.8-80. The circuit configurations are same as JFETs. In N-channel EMOSFET, positive input voltage less than V_T will turn the switch OFF, while a positive input voltage greater than V_T will turn the switch ON. In P-channel EMOSFET the source is connected to supply voltage (V_{EE}), and the drain is connected to ground through the load. To turn the P-channel EMOSFET ON, V_{GS} must be equal or less than $-V_T$. Therefore, any gate voltage ($V_G = V_i$), less than or equal to $V_{SS} - V_T$, will turn it ON. It will be clear if we apply KVL as,

$$\begin{aligned} V_{SS} - V_{GS} - V_G &= 0 \\ V_{SS} - V_{GS} - (V_{SS} - V_T) &= 0 \\ V_{SS} - V_{GS} - V_{SS} + V_T &= 0 \end{aligned}$$

or,

$$V_{GS} = -V_T \quad (8-83)$$

On the other hand, the P-channel EMOSFET will turn OFF for any input voltage greater than $V_{SS} - V_P$. However, in case of EMOSFET barely turning it ON will not do. We have to saturate the transistor to minimize power loss.

To design a particular switching circuit, we must know the specifications of the load. That is load voltage (V_L) and load current (I_L) or load power. Then a MOSFET has to be selected with $I_{D(max)} > I_L$ and $V_{DS(max)} > V_L$. The supply voltage (V_{DD} or V_{SS}) should be equal to load voltage (V_L). In case of high-power switch, resistance is not used in series with the load, as series resistor will produce power loss.

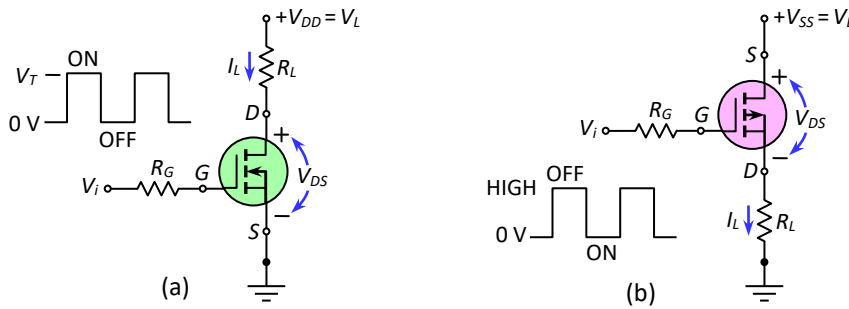


Fig.8-80: Switching circuits using EMOSFET: (a) N-channel MOSFET, and (b) P-channel MOSFET

From the datasheet of the MOSFET, we must produce the minimum transfer characteristic curve or take the minimum values of k , and V_T . On the transfer characteristic curve, the level of load current has to be marked and its corresponding value of V_{GS} has to be calculated. Instead of the transfer characteristic curve, if we know the value of k , using the following equation, the value of V_{GS} has to be calculated.

$$I_D = I_L = k(V_{GS} - V_T) \quad (8-84)$$

The value of V_{GS} determined in this way sets the level of controlling voltage. The controlling voltage V_i must be equal to or greater than V_{GS} . But, it should not be greater than $V_{GS(max)}$.

If it is not possible to determine the level of V_i , we can set its value as high as possible. A thumb rule is V_i should be near about $V_{GS(max)}$. In this case, the drain current of the MOSFET will be set to very large value, but actual current will be determined by the load.

Example 8-27

Design a switch to control (ON/OFF) a 120 V, 1.2 kW DC load. The switch will be driven by a controlling circuit of 5 V output.

Solution:

The load current is,

$$I_L = \frac{P_L}{V_L} = \frac{1200 \text{ W}}{120 \text{ V}} = 10 \text{ A}$$

To design the switching circuit, we need a power MOSFET of $I_{D(max)} \geq 10 \text{ A}$, and $V_{DS(max)} \geq 120 \text{ V}$. IRF244 is an enhancement MOSFET with $I_{D(max)} = 14 \text{ A}$, and $V_{DS(max)} = 250 \text{ V}$. Let us use this MOSFET. The threshold voltage (V_T) of this transistor is 2 V to 4 V, but to operate the device in the Ohmic region, the controlling voltage should be larger than this V_T . The maximum V_{GS} of this device is 20 V. So, we will apply about 16 V to control this device.

As the controlling voltage is only 5 V, a Zener diode (1N4745A) with $V_Z = 16 \text{ V}$ is used as shown in Fig. The breakdown current of the Zener diode is 5 mA. Here, let $I_Z = 12 \text{ mA}$. So, the value of R_Z will be,

$$R_Z = \frac{V_{DD} - V_Z}{I_Z} = \frac{120 \text{ V} - 16 \text{ V}}{12 \text{ mA}} = 8.67 \text{ k}, \text{ use standard } 9.1 \text{ k}$$

As in ON state the full I_Z will flow through the transistor, its maximum current capacity of the transistor will be 12 mA. Let us use C828 transistor. Minimum value of h_{FE} of this transistor is 130. Therefore, maximum base current will be,

$$I_B = \frac{I_C}{h_{FE(min)}} = \frac{12 \text{ mA}}{130} \approx 92 \mu\text{A}$$

To ensure saturation of the transistor let, $I_B = 1.5 \times 92 \mu\text{A} = 138 \mu\text{A}$

$$\therefore R_B = \frac{V_i - V_{BE}}{I_B} = \frac{5 \text{ V} - 0.7 \text{ V}}{138 \mu\text{A}} = 3.1 \text{ k}, \text{ use standard } 3 \text{ k}$$

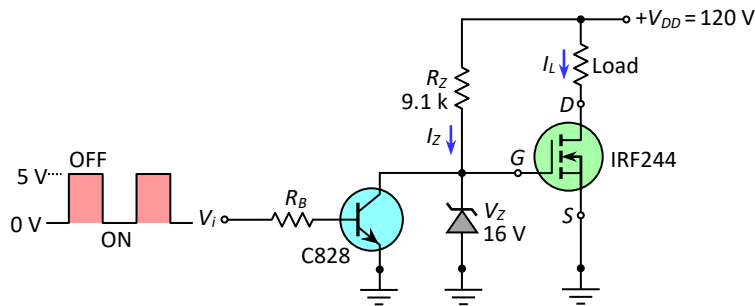


Fig.8-83: MOSFET switching circuit for Example 8-27

Comments: This type of switching is used in solar photovoltaic (PV) power generating system, where the output voltage of PV module is very large.

8.25 Sample Questions

1. What is load line and operating point or Q-point? What is DC load line?
2. Explain how to draw the DC load line of a JFET circuit.
3. With a suitable diagram explain the changing of the operating point on the load line due to an input sinusoidal signal.
4. Write the names of common biasing circuits used for FET.
5. Draw the diagram of a fixed biasing circuit for an N-channel JFET and explain how to determine the operating point.
6. Explain how to determine the position of the operating point of fixed biasing circuit using graphical method. What is the advantage of graphical method over analytical method?
7. What are the advantages and disadvantages of fixed biasing circuit?
8. Draw the circuit diagram of a fixed biasing circuit for a P-channel JFET and explain how to determine the operating point.
9. Draw the circuit diagram of a self-bias circuit for an N-channel JFET and explain how to determine the operating point.

10. Explain how to determine the position of the operating point of self-bias circuit using graphical method. What is the advantage of graphical method over analytical method?
11. What are the advantages and disadvantages of a self-bias circuit?
12. Draw the diagram of a self-bias circuit for a P-channel JFET and explain how to determine the operating point.
13. Draw the circuit diagram of a voltage divider biasing circuit for an N-channel JFET and explain how to determine the operating point.
14. Explain how to determine the position of the operating point of voltage divider biasing circuit using graphical method. What is the advantage of graphical method over analytical method?
15. What are the advantages and disadvantages of voltage divider biasing circuit?
16. Draw the figure of a voltage divider biasing circuit for a P-channel JFET and explain how to determine the operating point.
17. What do you mean by saturation and cut-off? Explain the component that can cause saturation of the FET in voltage divider bias and self-bias circuits.
18. In a voltage divider biasing circuit of JFET, is there any possibility of the FET to operate at the cut-off point. Explain your answer.
19. Draw the schematic diagram of a self-bias circuit for an N-channel JFET with split power supply and explain how to determine the operating point.
20. Draw the diagram of a fixed bias circuit for an N-channel JFET with split power supply and explain how to determine the operating point.
21. Draw the diagram of a voltage divider bias circuit for an N-channel JFET with split power supply and explain how to determine the operating point.
22. What is stability of biasing circuit? Discuss the stability factor for different biasing circuits of FET.
23. Describe how to design a fixed bias circuit for JFET.
24. Describe how to design a self bias circuit for JFET.
25. Describe how to design a voltage divider bias circuit for JFET.
26. Draw the circuit diagram of self-bias circuit for JFET in common drain configuration (CDC) and describe how to determine the position of Q -point.
27. Draw the circuit diagram of voltage divider bias circuit for JFET in CDC and describe how to determine the position of Q -point.
28. Draw the circuit diagram of self-bias circuit for JFET in common gate configuration (CGC) and describe how to determine the position of Q -point.
29. Draw the circuit diagram of a voltage divider bias for JFET in CGC and describe how to determine the position of Q -point.
30. Draw the circuit diagram of a fixed-bias circuit for depletion MOSFET and describe how to determine the position of Q -point.
31. Draw the circuit diagram of a self-bias circuit for depletion MOSFET and describe how to determine the position of Q -point.
32. Draw the circuit diagram of a voltage divider bias circuit for depletion MOSFET and describe how to determine the position of Q -point.
33. Draw the diagram of a fixed-bias circuit for enhancement MOSFET and describe how to determine the position of Q -point.

34. Draw the circuit diagram of a self-bias circuit for enhancement MOSFET and describe how to determine the position of Q -point.
35. Draw the circuit diagram of a voltage divider bias circuit for enhancement MOSFET and describe how to determine the position of Q -point.
36. What is universal transfer characteristic of the field effect transistor? What is use of the this characteristics curve?
37. Explain some methods for trouble shooting of self-bias circuit of N-channel JFET.
38. Explain some methods for trouble shooting of fixed-bias circuit of N-channel JFET.
39. Explain some methods for trouble shooting of voltage divider bias circuit of N-channel JFET.
40. Draw the circuit diagram of a voltage divider bias with drain feedback for a N-channel JFET and explain how to determine the operating point.
41. Draw the circuit diagram of a voltage divider bias with drain feedback for a N-channel EMOSFET and explain how to determine the operating point.
42. Describe how the design a voltage divider bias with drain feedback for JFET.
43. What are the advantages of voltage divider bias with drain feedback?
44. What is constant current source? Explain how to bias JFET using constant current source. How to calculate the operating point of this circuit?
45. Explain with appropriate diagram how to use FET as constant current source.
46. What is constant current sink? Draw the circuit diagram of constant current sink and explain how to determine the value of current of constant current sink.
47. What is voltage controlled resistor? Explain how FET can be used as voltage controlled resistor.
48. Explain with diagram how N-channel JFET can be used as a switch.
49. Explain with diagram how P-channel JFET can be used as a switch.
50. Explain with diagram how P-channel MOSFET can be used as a switch.
51. Explain the design procedure of a MOSFET switch.
52. What are the advantages and disadvantages of MOSFET switch over BJT switch?
53. In Chapter 7, we found that there is a parasitic diode in VMOSFET. Is there any use of this diode-explain.
54. How we will design a MOSFET switching circuit, if the driving voltage is greater than $V_{GS(max)}$ of this FET - explain.
55. How we will design a MOSFET switching circuit, if the driving voltage is less than V_T of this FET but V_L is much greater than V_T - explain with diagram.
56. In terms of distortion which amplifiers are better BJT or FET? Justify your answer.

8.26 Exercise

1. Draw the load line and determine the operating point for the FET circuit of Fig.8-82. Assume $V_P = -5\text{ V}$, and $I_{DSS} = 12\text{ mA}$.

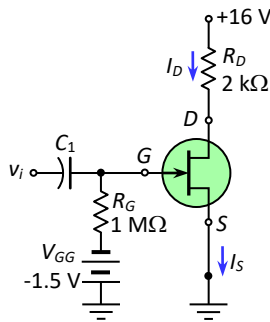


Fig. 8-82: FET circuit for Exercise no.

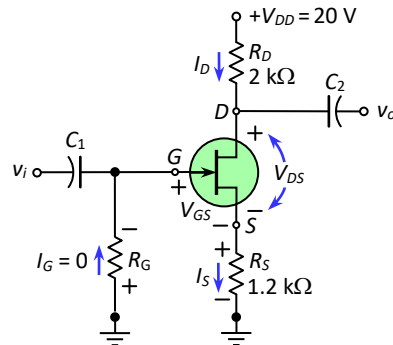


Fig. 8-83: Self bias circuit for Exercise

- Determine the operating point of common-source JFET circuit of Fig. 8-82 both analytically and graphically. Assume, $V_P = -5$ V, and $I_{DSS} = 16$ mA.
- Graphically determine the position of the operating points of an N-channel JFET self-bias circuit like Fig. 8-83.
- The JFET used for the circuit of Fig. 8-83 has a range of V_P and I_{DSS} . Assume, $V_{DD} = 15$ V, $R_S = 0.5$ kΩ, $R_D = 1$ kΩ, $V_{P(min)} = -3$ V and $V_{P(max)} = -6$ V, and $I_{DSS(min)} = 8$ mA and $I_{DSS(max)} = 16$ mA. Determine the position of the minimum and maximum operating point.
- Repeat Exercise no. 3 using analytical method.
- Graphically determine the position of the operating points of an N-channel JFET self-bias circuit like Fig. 8-83. The JFET used for the circuit has a range of V_P and I_{DSS} . Assume, $V_{DD} = 15$ V, $R_S = 0.5$ kΩ, $R_D = 1$ kΩ, $V_{P(min)} = -3$ V and $V_{P(max)} = -6$ V, and $I_{DSS(min)} = 8$ mA and $I_{DSS(max)} = 16$ mA.
- For the biasing circuits of P-channel JFET of Fig. 8-84, draw the load line, calculate the operating point and show the position of the operating point on the load line.

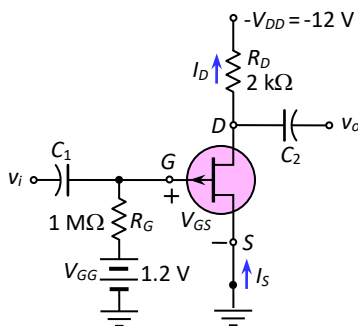


Fig. 8-84: Fixed bias or gate bias circuit for P-channel JFET

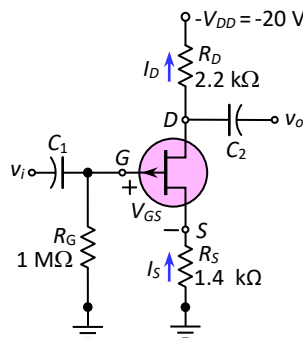


Fig. 8-85: Self-bias circuit for P-channel JFET for Exe. No. 10

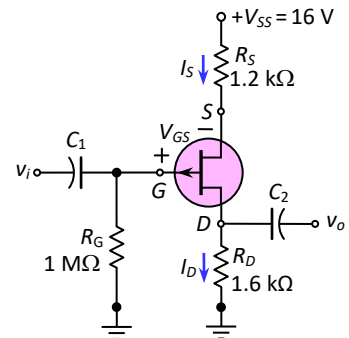


Fig. 8-86: Self-bias circuit for P-channel JFET for Exe. No. 10

- For the biasing circuits of P-channel JFET of Fig. 8-85, draw the load line, calculate the operating point and show the position of the operating point on the load line.
- For the biasing circuits of P-channel JFET of Fig. 8-86, draw the load line, calculate the operating point and show the position of the operating point on the load line.

10. Determine the position of the operating points for the voltage divider bias circuit of Fig.8-87. The JFET used in this circuit has $V_{P(min)} = -3\text{ V}$ and $V_{P(max)} = -6\text{ V}$, and $I_{DSS(min)} = 8\text{ mA}$ and $I_{DSS(max)} = 16\text{ mA}$.
11. Determine the position of the operating points for the voltage divider bias circuit of Fig.8-88. The JFET used in this circuit has $V_{P(min)} = -3\text{ V}$ and $V_{P(max)} = -5\text{ V}$, and $I_{DSS(min)} = 8\text{ mA}$ and $I_{DSS(max)} = 12\text{ mA}$.

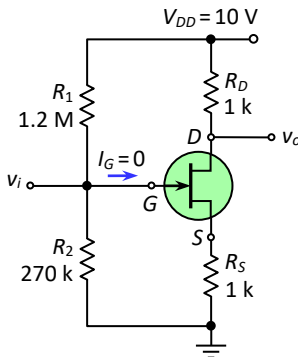


Fig.8-87: Voltage divider bias Exe no.

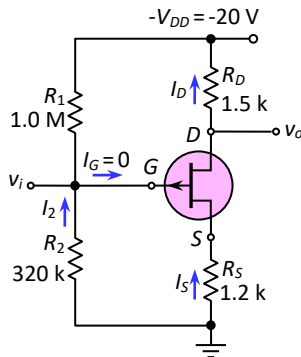


Fig.8-88: Voltage divider bias Exe no.

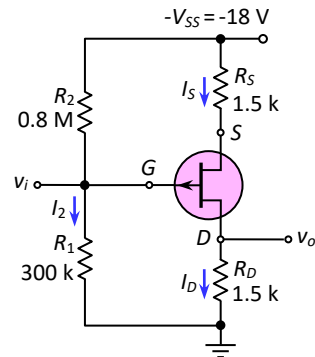
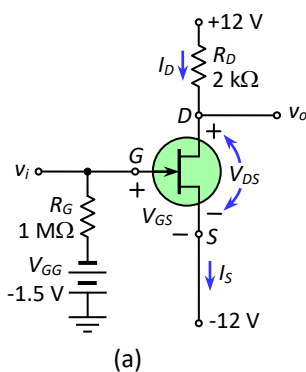
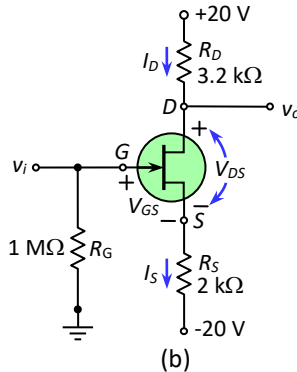


Fig.8-89: Voltage divider bias Exe no.

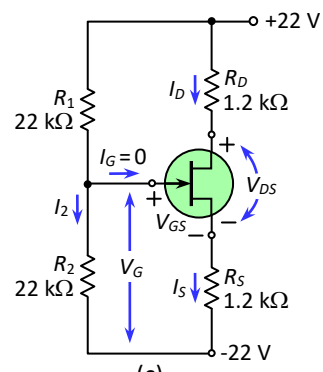
12. Repeat Exercise no. 10 considering each of the following changes individually: i) considering $R_D = 2.7\text{ kΩ}$, and ii) $R_S = 240\text{ Ω}$, iii) $V_P = -10\text{ V}$, and iv) $I_{DSS} = 50\text{ mA}$.
13. Using graphical method, determine the operating point of the circuit of Fig.8-89. Assume, $V_P = -3\text{ V}$, and $I_{DSS} = 12\text{ mA}$.
14. Fig.8-90 shows three biasing circuits for JFET with split power supply. Analyze them and calculate the values of I_D , I_S , I_G , V_G , V_{GS} , and V_{DS} . Assume, $V_P = -5\text{ V}$, and $I_{DSS} = 16\text{ mA}$.



(a)



(b)



(c)

Fig.8-90: JFET biasing circuit using split power supply: (a) Fixed bias, (b) Self bias, and (c) Voltage divider bias

15. The maximum and minimum transfer characteristics of a JFET are shown in Fig.8-91. Using these characteristics design a gate bias circuits for N-channel JFETs. Assume $V_{DD} = 16\text{ V}$.

16. The maximum and minimum transfer characteristics of a JFET are shown in Fig.8-91. Using these characteristics design self-bias circuits for N-channel JFETs. Assume $V_{DD} = 20\text{ V}$.

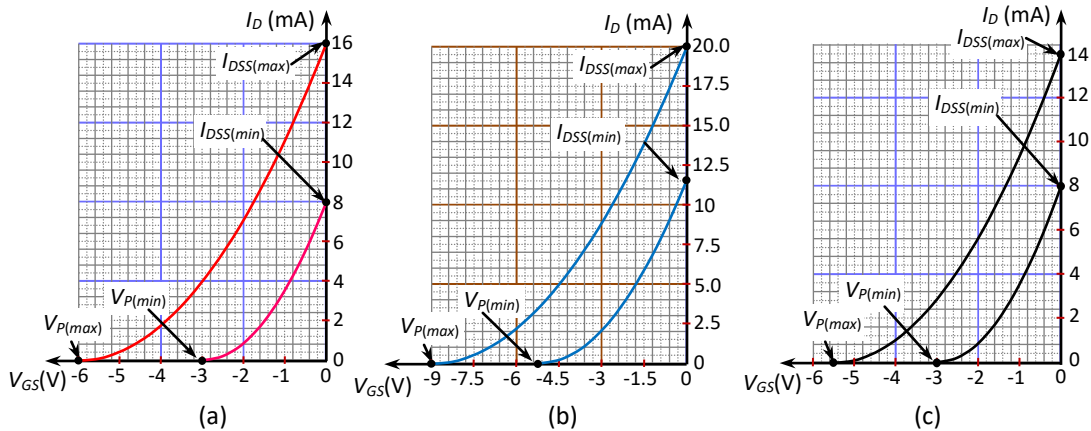


Fig.8-91: Maximum and minimum transfer characteristics for Exercise JFET biasing circuit using split power supply: (a) Fixed bias,

17. The values of maximum and minimum transfer characteristics of a JFET are given as: $V_{P(min)} = -4\text{ V}$ and $V_{P(max)} = -5\text{ V}$, and $I_{DSS(min)} = 10\text{ mA}$ and $I_{DSS(max)} = 12\text{ mA}$. Using these characteristics, design voltage divider biasing circuit for this FET.
18. A P-channel JFET has $V_P = 5\text{ V}$, and $I_{DSS} = 10\text{ mA}$. Using this JFET design a voltage divider biasing circuit for a common source amplifier. Assume, $V_{DD} = 20\text{ V}$, and $I_{D(max)} = 5\text{ mA}$. Use analytical method.
19. Assuming $V_P = -4\text{ V}$, and $I_{DSS} = 12\text{ mA}$, determine the Q-point of the common-drain amplifier of Fig.8-92 using analytical method.
20. Fig.8-93 shows two Common Gate biasing circuits. Assuming $V_P = -4\text{ V}$, and $I_{DSS} = 12\text{ mA}$, determine the Q-point of the amplifiers using analytical and graphical method. Also calculate the terminal currents and inter-terminals voltages.

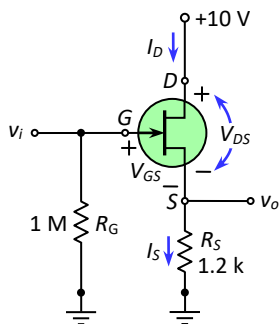


Fig.8-92: Common drain biasing circuit for Exercise 19

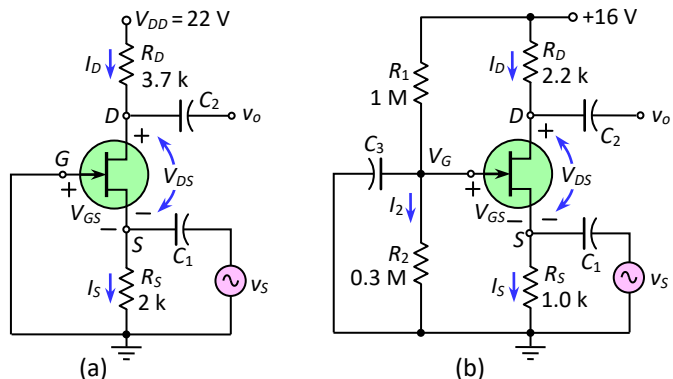


Fig.8-93: Common gate circuits: (a) Self bias, and (b) Voltage divider bias

21. Design a voltage divider bias circuit for common drain JFET amplifier. Assume $V_p = -5\text{ V}$, and $I_{DSS} = 10\text{ mA}$, $V_{DD} = 12\text{ V}$.
22. Fig.8-94 shows three Common Source biasing circuits for depletion MOSFETs. Assuming $V_p = -5\text{ V}$, and $I_{DSS} = 10\text{ mA}$, determine the Q-points of the amplifiers using graphical method. Also calculate the values of I_D , I_S , I_G , V_G , V_{GS} , and V_{DS} .

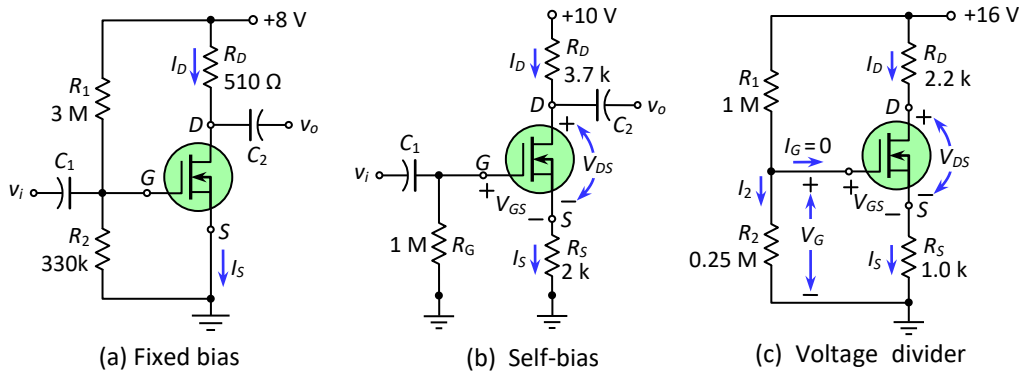


Fig.8-94: Different bias circuits of DMOSFET for Exercise 22

23. The self-bias circuit of Fig.8-95 uses a P-channel depletion MOSFET of $V_p = +6\text{ V}$, and $I_{DSS} = 16\text{ mA}$. Determine the operating point, and the levels of voltage and currents at three terminals of the circuit using analytical and graphical method.
24. Fig.8-96(a) shows a voltage divider bias circuit for P-channel DMOSFET and Fig.8-96(b) is its transfer characteristics. Determine the operating point using this transfer characteristic.

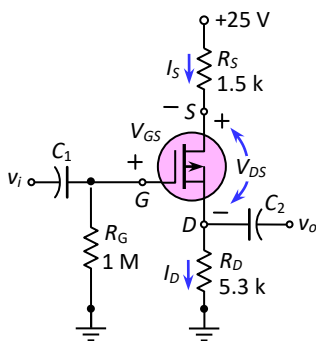


Fig.8-95: Self bias circuit of P-channel JFET for Exercise 23

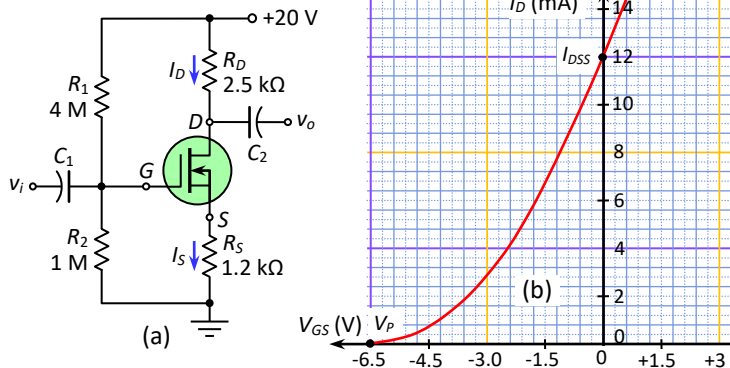


Fig.8-96: (a) Voltage divider bias circuit for Exercise 24, (b) Transfer characteristics for Exercise 24

25. The P-channel DMOSFET of the voltage divider biasing circuit of Fig.8-97(a) has the transfer characteristic curve as given in Fig.8-97(b). Determine the operating point of the circuit.

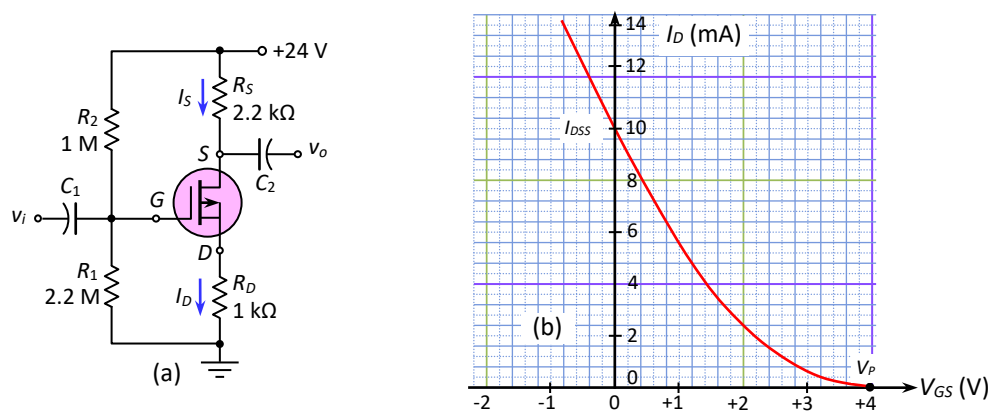


Fig.8-97: (a) Voltage divider bias circuit for DMOSFET, (b) Its transfer characteristics

26. The transfer characteristic curves of the EMOSFET used in the circuits of Fig.8-98, is given in Fig.8-99. Determine the operating points of the circuits using graphical method. Also calculate the terminal currents and inter-terminal voltages.

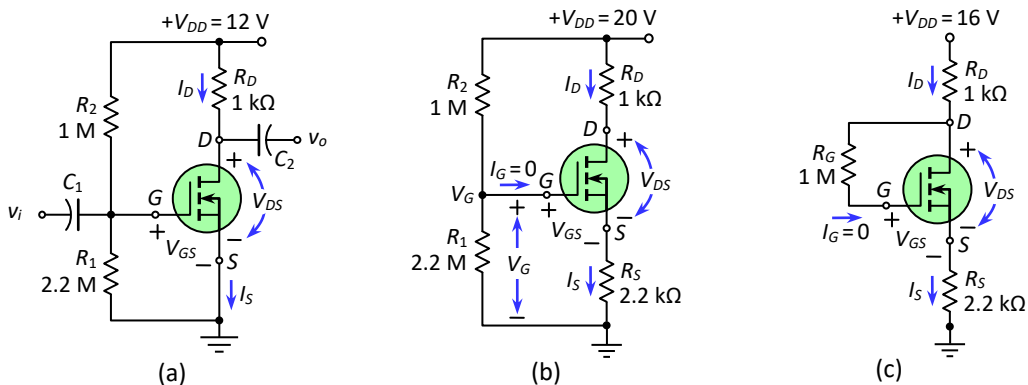


Fig.8-98: Different bias circuits of EMOSFET for Exercise 26: (a) Fixed bias, (b) Voltage divider bias, and (c) Drain feedback circuit

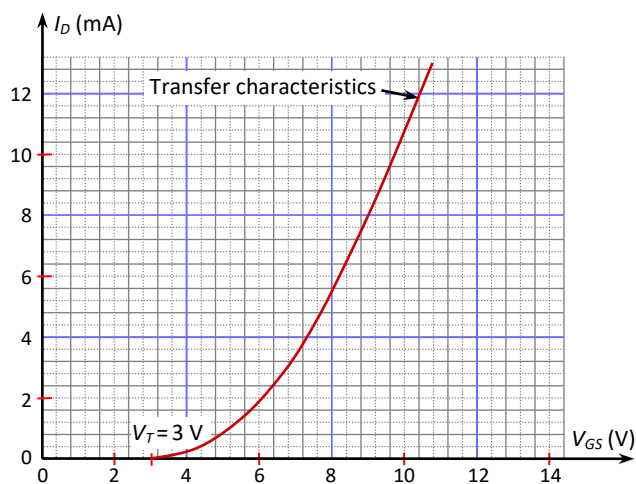


Fig.8-99: Transfer characteristics for Exercise 26

27. Using universal characteristic curve, solve the problem of Example 8-4. For better understanding, the problem is mentioned here again.
28. Using the universal graph, determine the Q -point of the circuit of Exercise no. 21, 22, 24, 25.
29. Using the universal graph, determine the Q -point of the circuit of Exercise no. 21, 22, 24, 25.
30. Using the universal graph, determine the Q -point of the circuit of Fig.8-100.

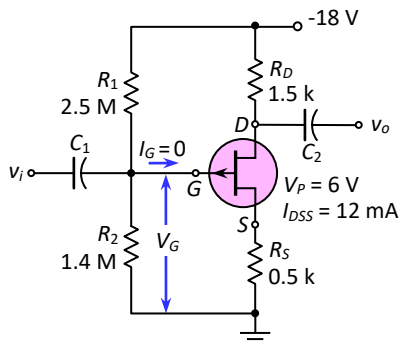


Fig.8-100: Voltage divider biasing circuit for P-channel JFET

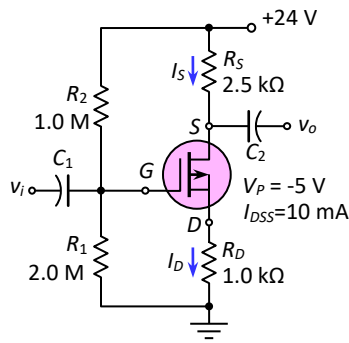


Fig.8-101: Voltage divider biasing circuit for P-channel DMOSFET

31. Using the universal graph, determine the Q -point of the circuit of Fig.8-101
32. The JFET used in the following circuit (Fig.8-102), has $V_P = 6$ V, and $I_{DSS} = 10$ mA. Graphically determine the operating point of the biasing circuit.

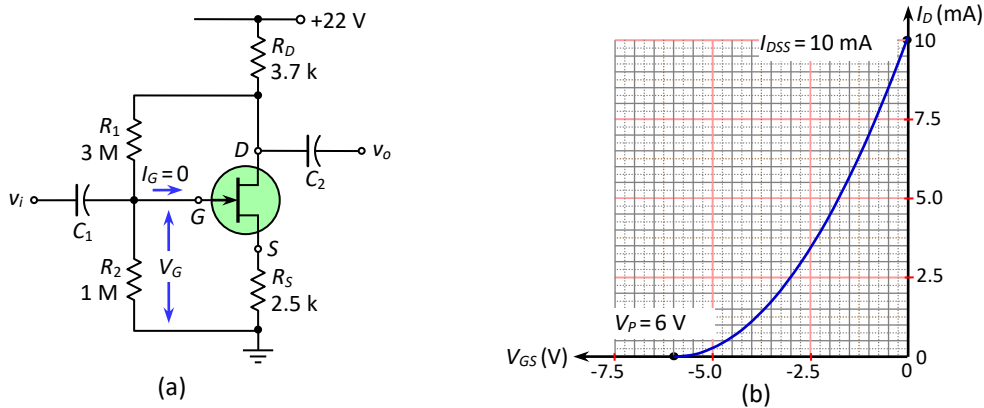


Fig.8-102: Voltage divider bias circuits with drain feedback and transfer characteristics for Exercise no. 32 for (a) JFET, and (b) EMOSFET

33. For the BIFET amplifier of Fig.8-103, determine the operating point. Assume, the FET has $V_P = -6\text{ V}$, and $I_{DSS} = 10\text{ mA}$.

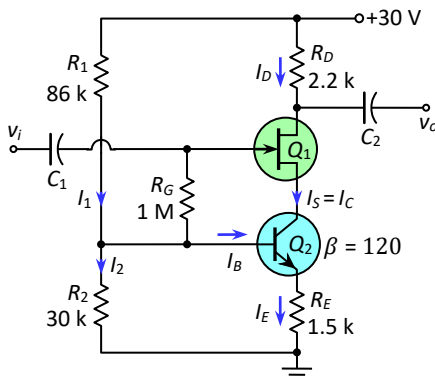


Fig.8-103: Biasing circuit with constant current source for Exercise no. 33

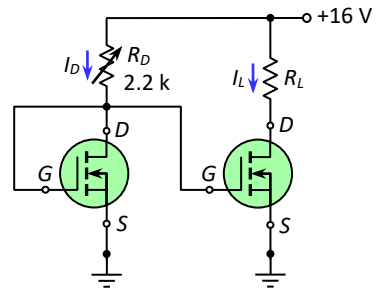


Fig.8-104: Constant current sink or current mirror circuit

34. Assume the EMOSFETs used in Fig.8-60 has $V_T = 2\text{ V}$, and $k = 0.15\text{ mA/V}^2$. Calculate the value of I_L .
35. Generally, a normal LED requires 10 mA to 20 mA to glow. The forward voltage drop of LED depends on color, for example, a blue LED needs 3.0 V to 3.6 V. Assuming $V_{DD} = 12\text{ V}$, design a switching circuit to drive a blue LED.
36. Design a switch to control (ON/OFF) a 24 V, 60 W DC load. The switch will be driven by a controlling circuit of 5 V output.

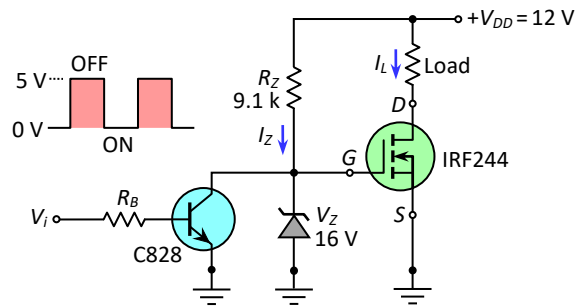


Fig.8-4: A switching circuit using MOSFET and Zener diode

37. The high power MOSFET switching circuit is shown in Fig.8-105. Determine the load current in ON and OFF state of the FET.

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